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(54) **METHODS AND COMPOSITIONS FOR THE
ACTIVATION OF GAMMA-DELTA T-CELLS**

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International Inc.**, Rockville, MD (US)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

The present invention relates generally to methods and
compositions for gene therapy and immunotherapy that
activate gamma delta T-cells, and in particular, can be used
in the treatment of various cancers and infectious diseases.

16 Claims, 15 Drawing Sheets

Specification includes a Sequence Listing.

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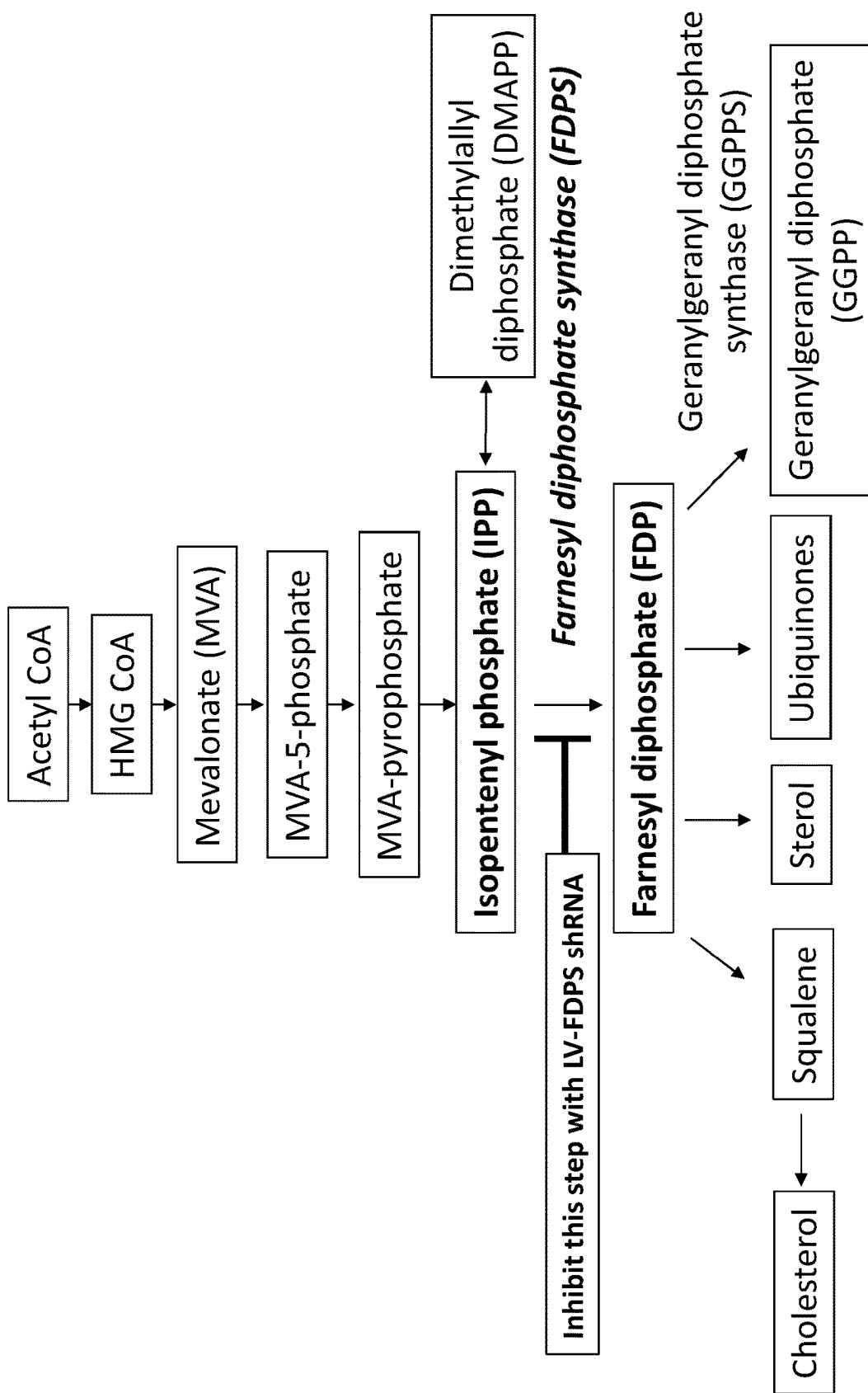


Figure 1

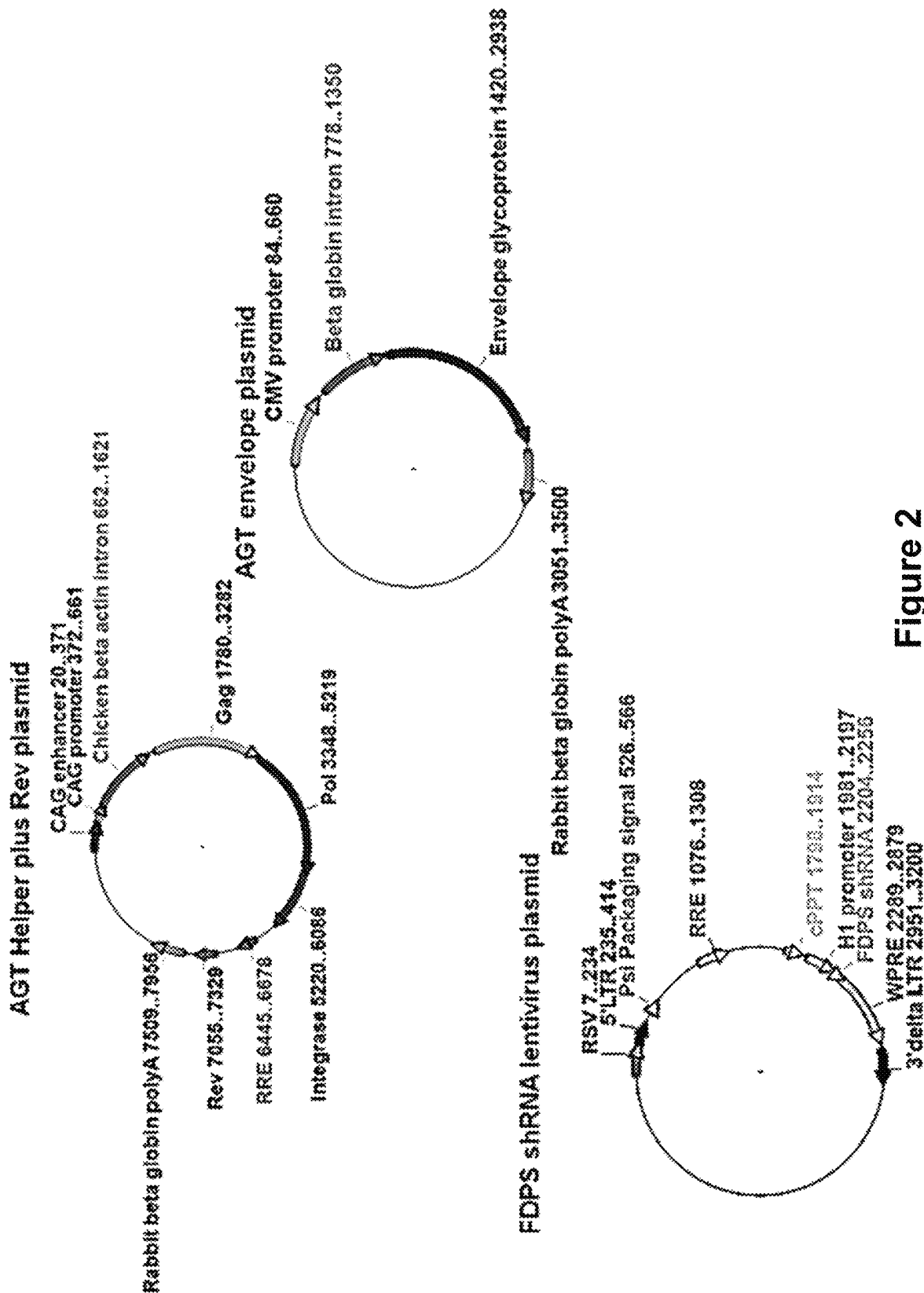


Figure 2

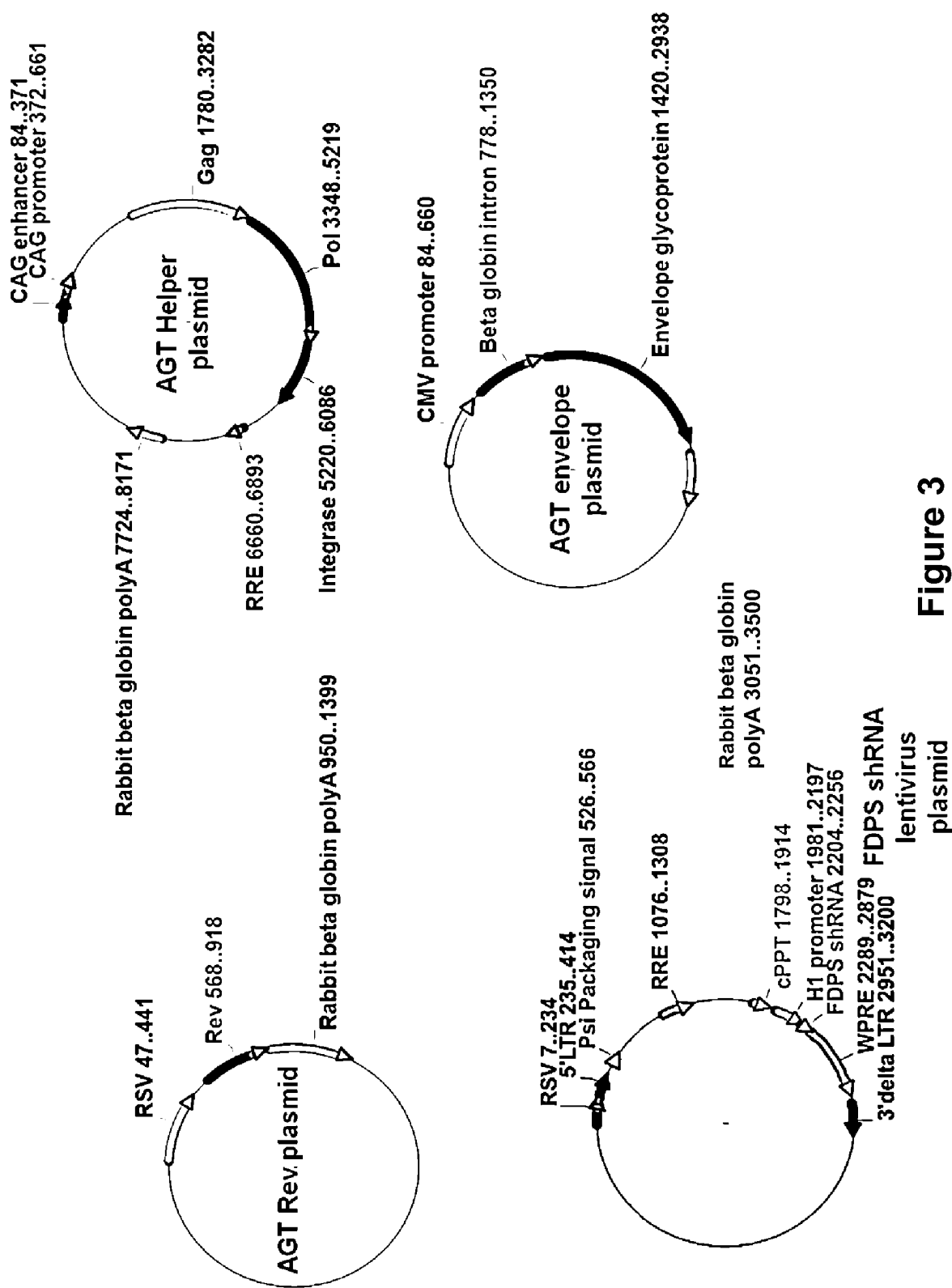


Figure 3

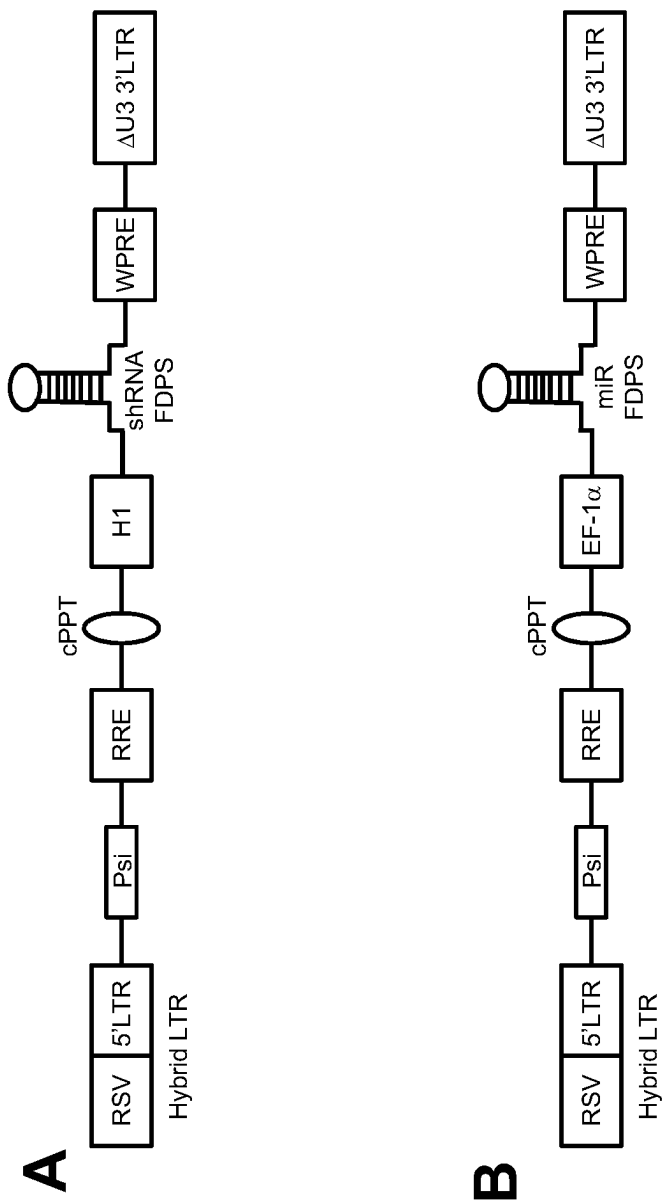


Figure 4

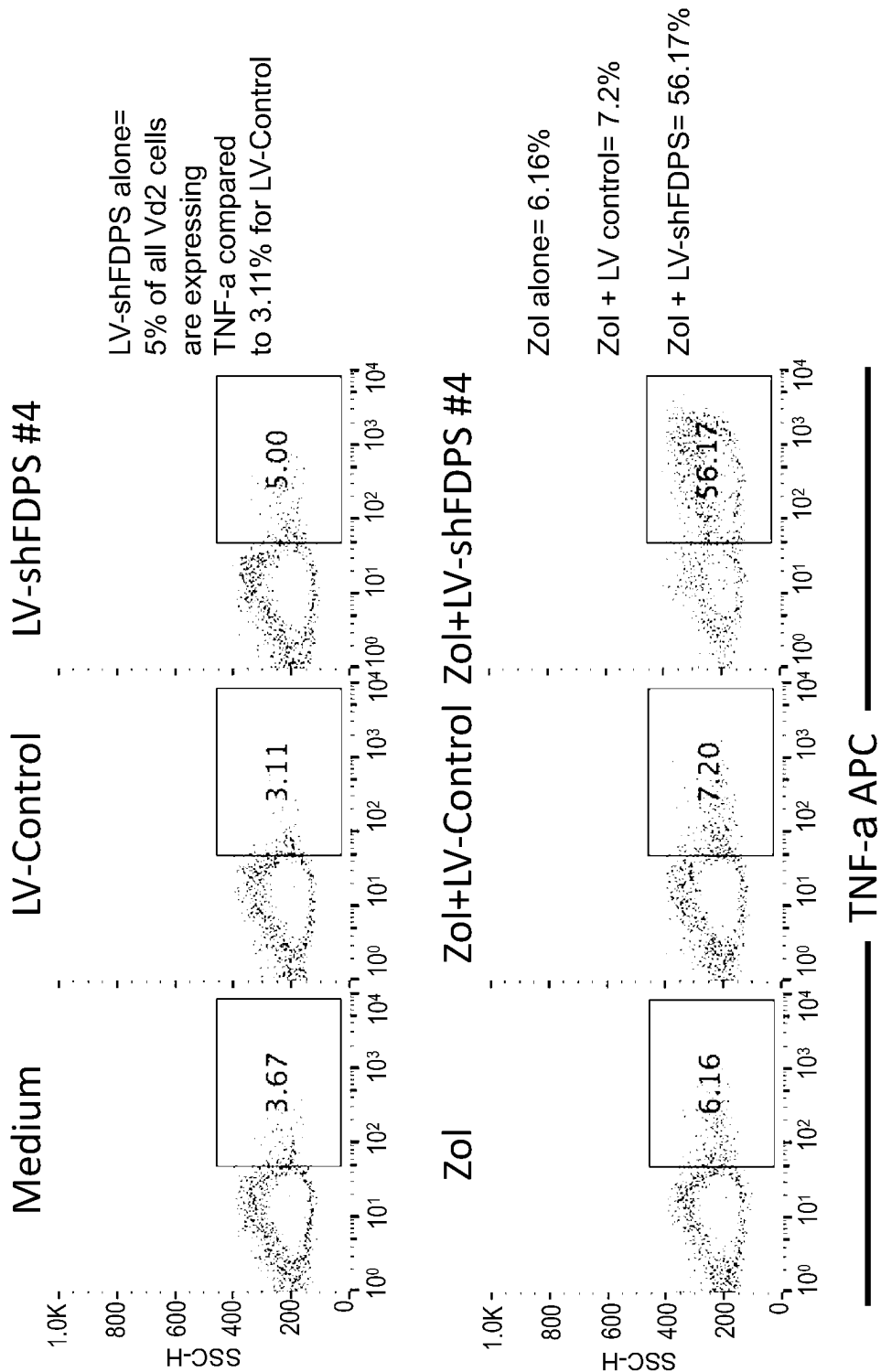


Figure 5

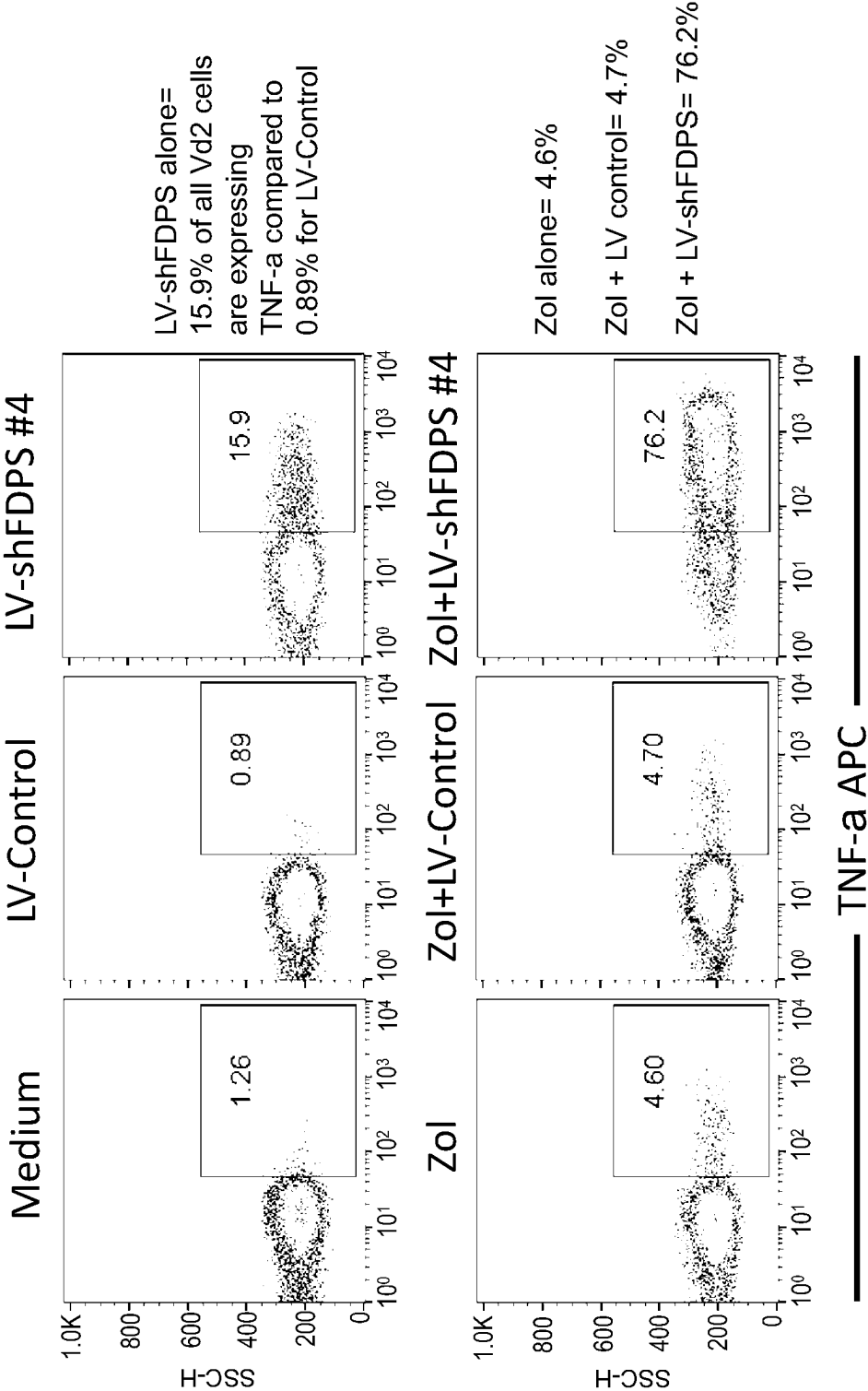


Figure 6

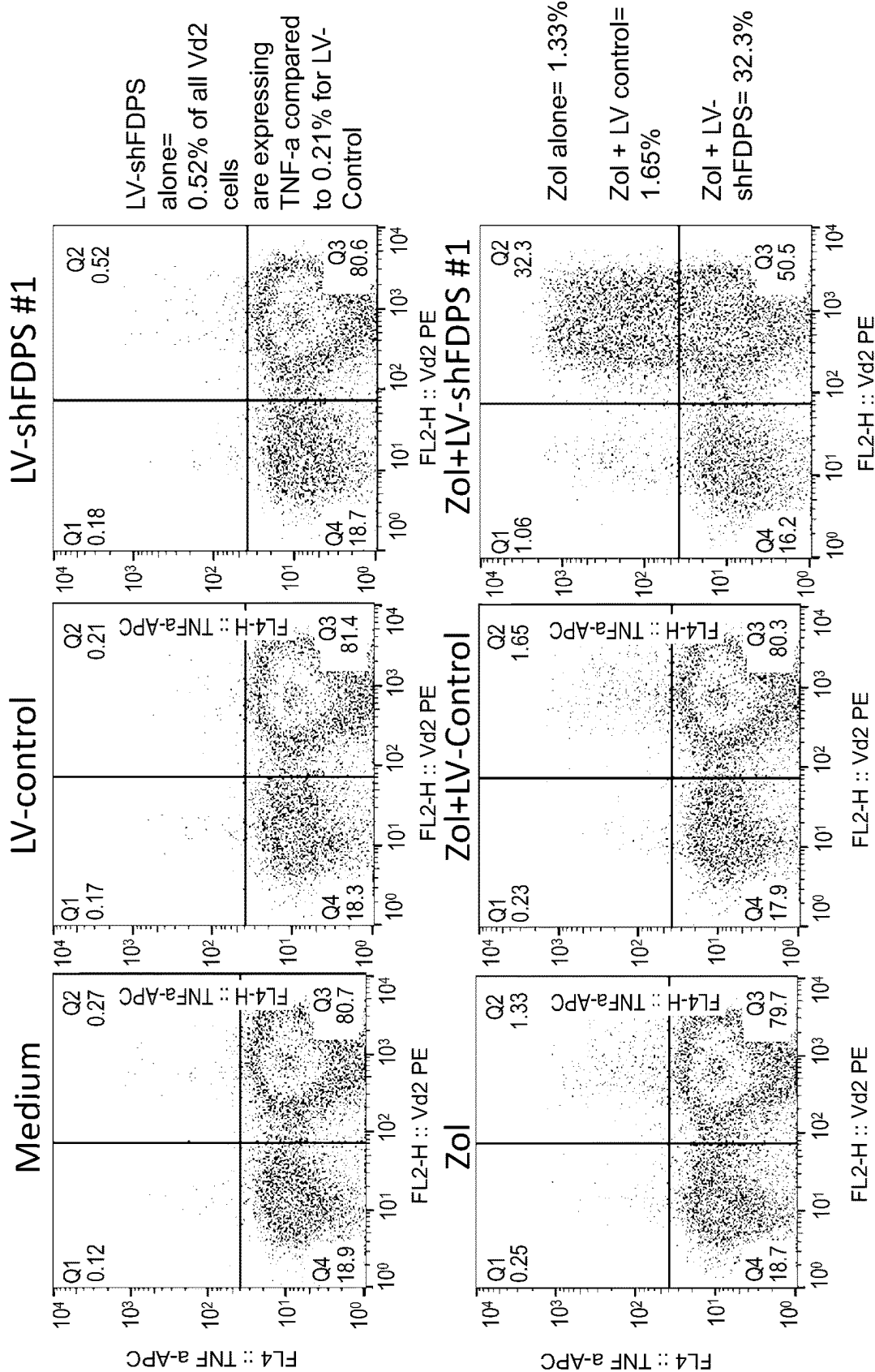


Figure 7

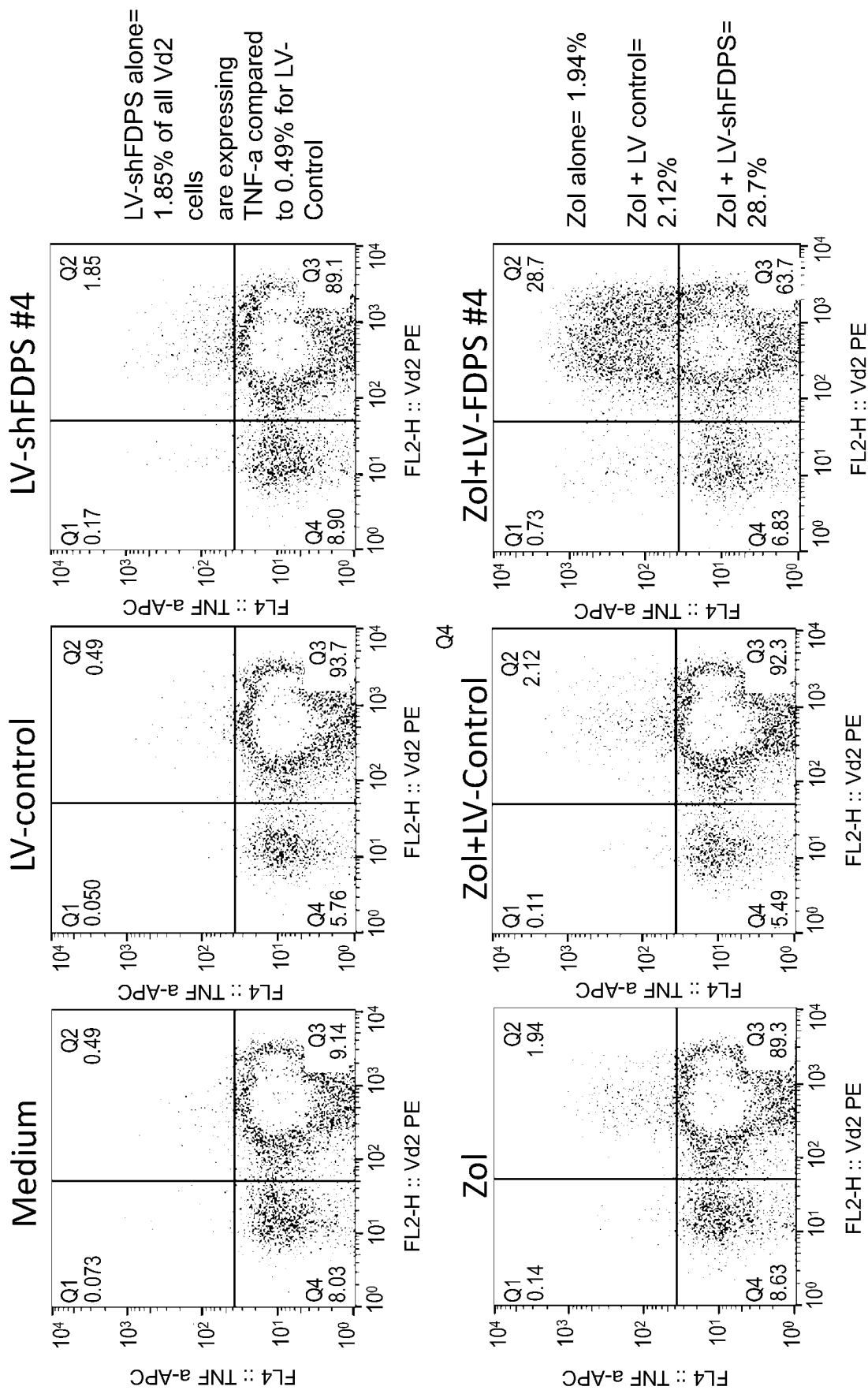


Figure 8

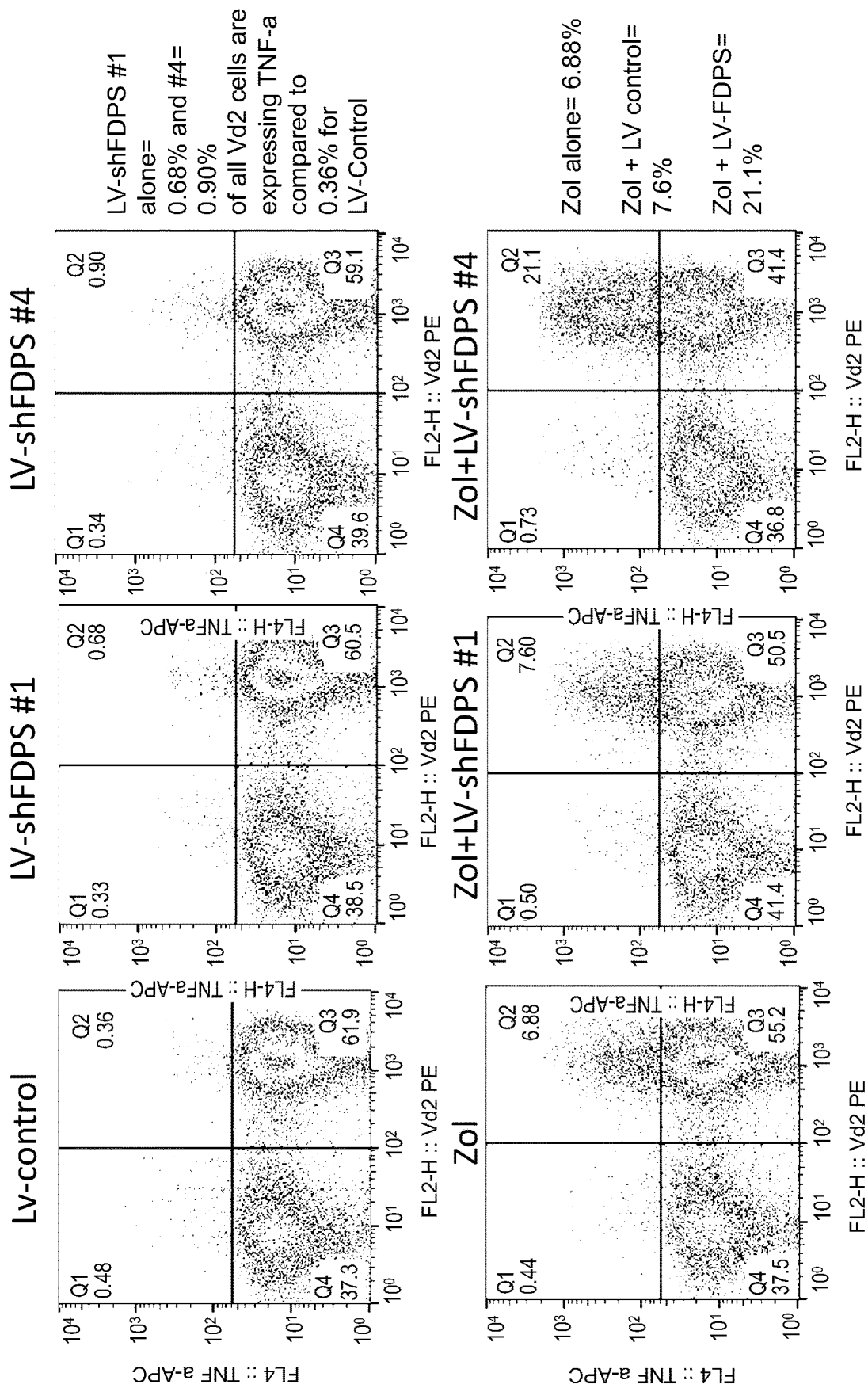


Figure 9

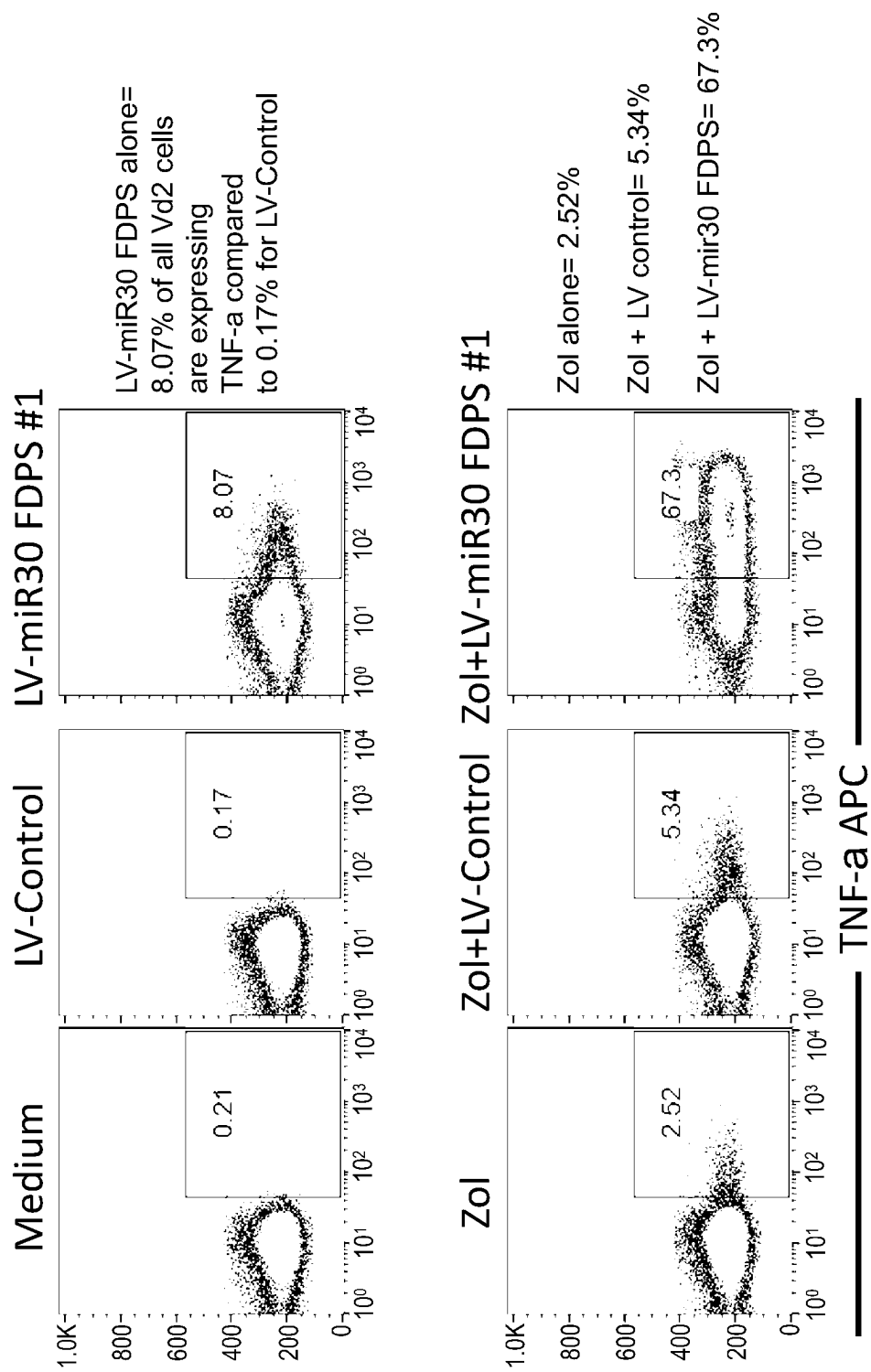


Figure 10

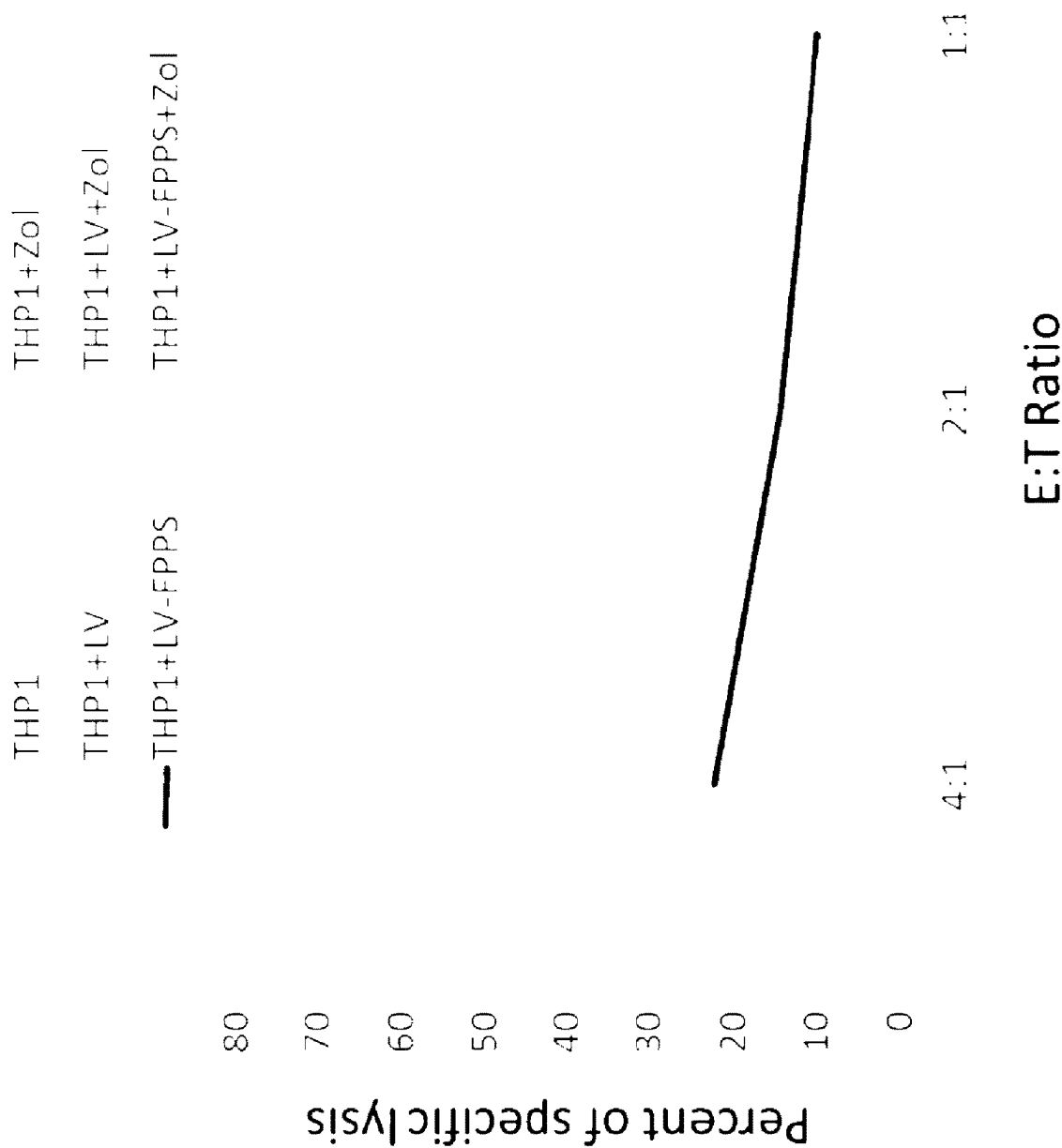


Figure 11

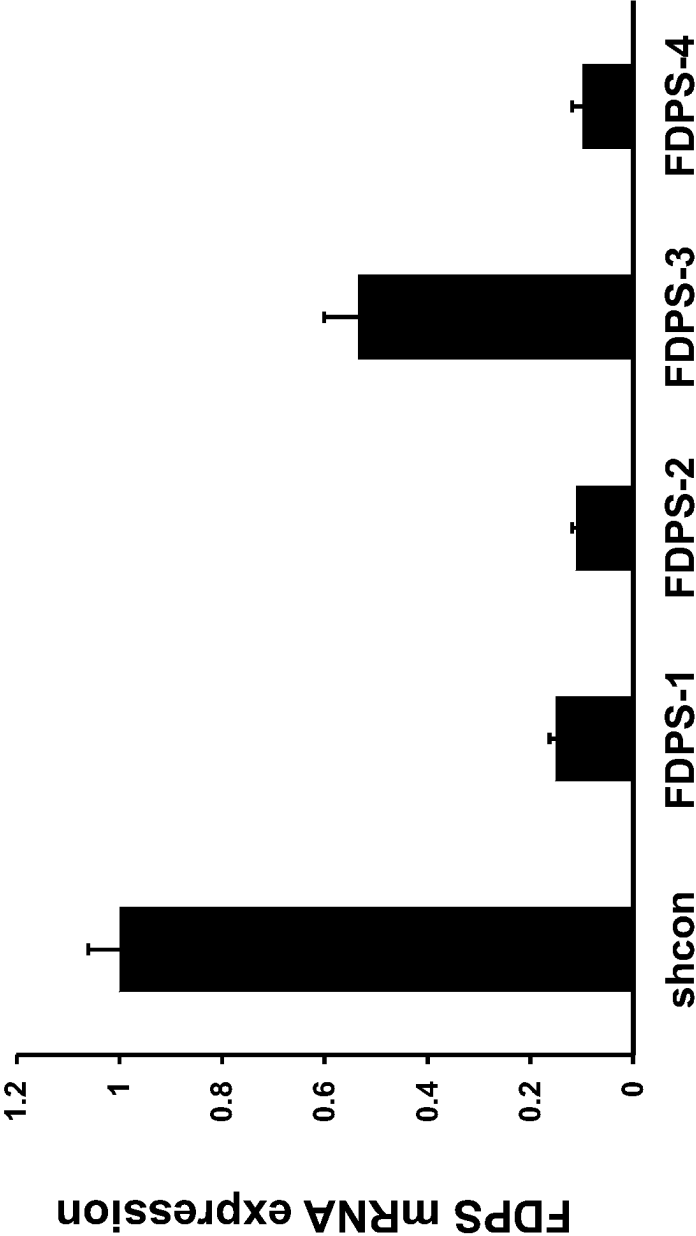


Figure 12

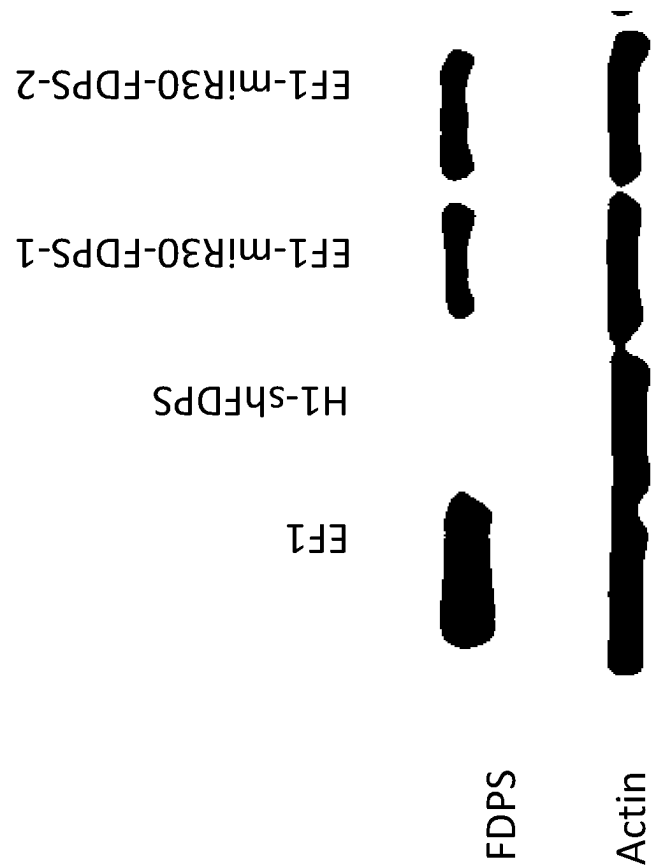


Figure 13

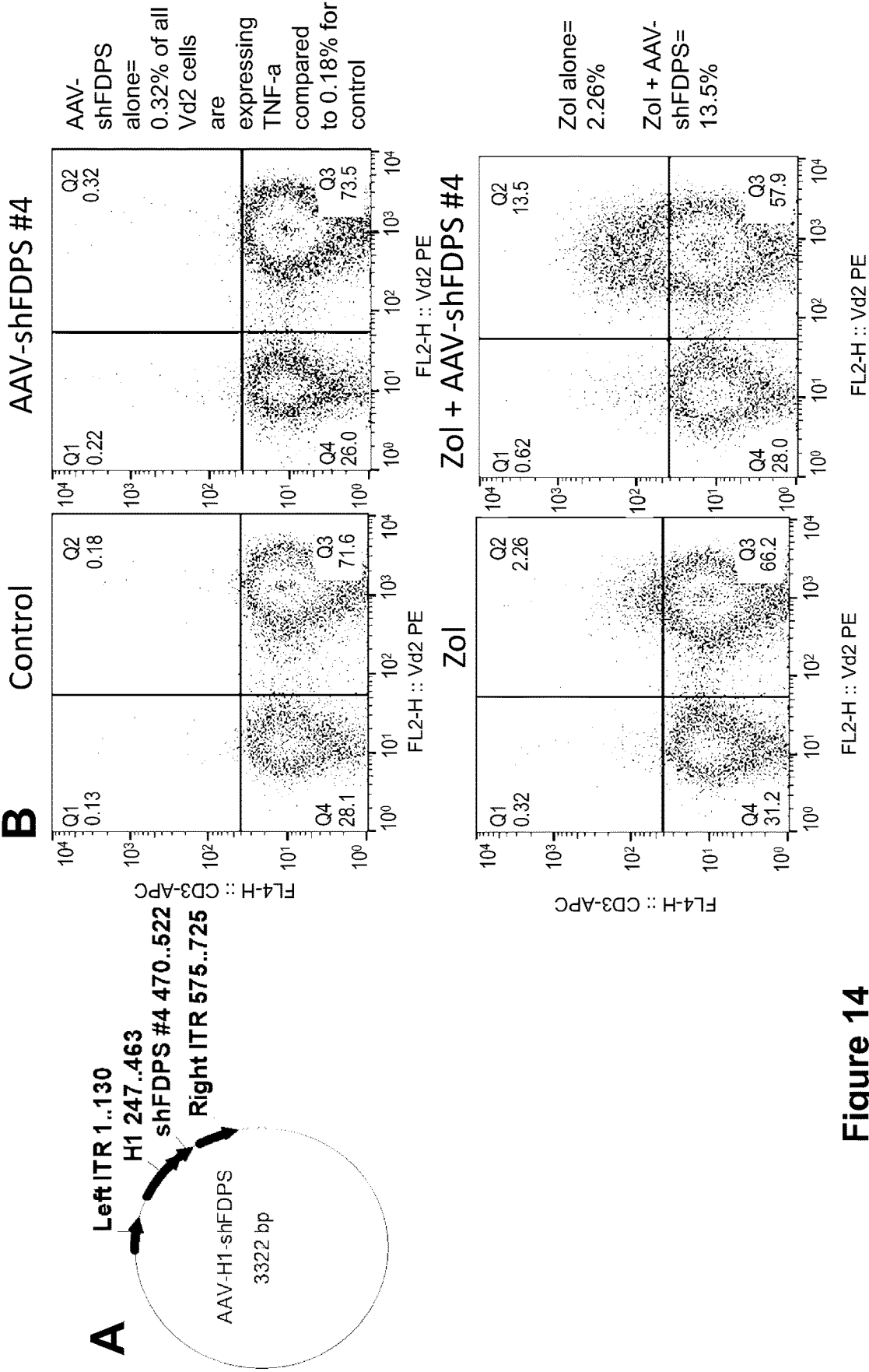


Figure 14

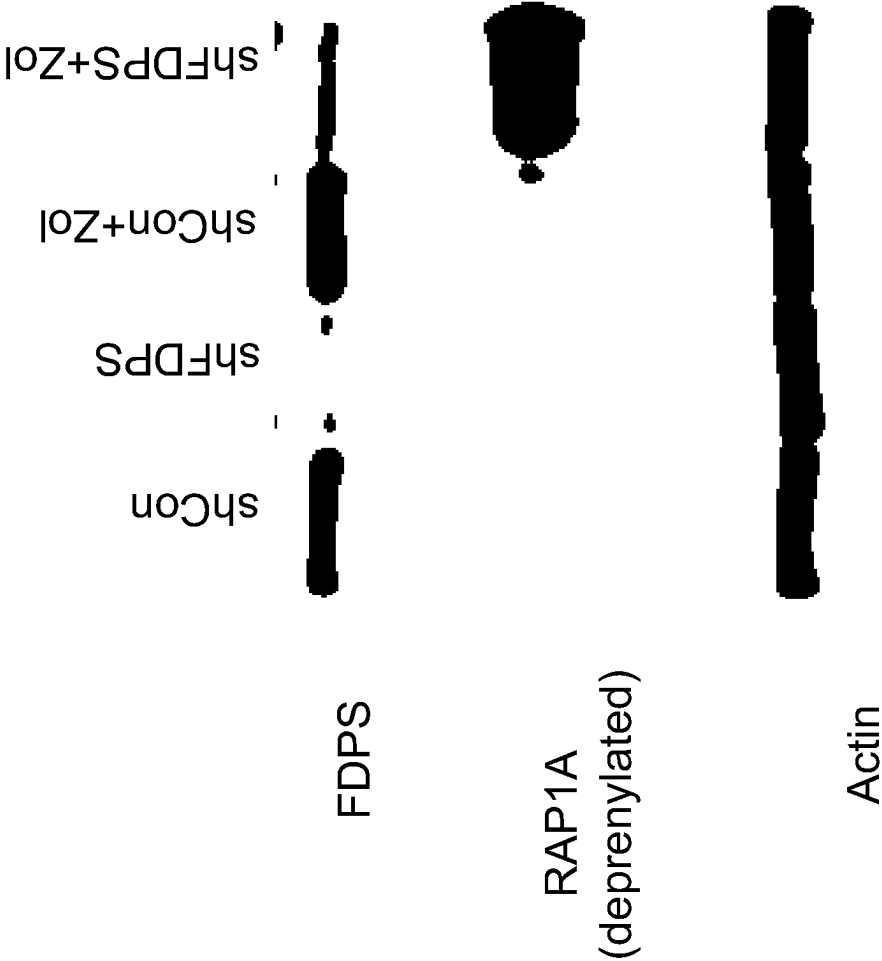


Figure 15

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METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/988,427 filed on Aug. 7, 2020 entitled "METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS", which is a continuation of U.S. patent application Ser. No. 16/530,908 filed on Aug. 2, 2019 entitled "METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS", which is a continuation of U.S. patent application Ser. No. 16/132,247 filed on Sep. 14, 2018 entitled "METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS," which is a continuation of U.S. patent application Ser. No. 15/904,131 filed on Feb. 23, 2018 entitled "METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS", which is a continuation in part of U.S. patent application Ser. No. 15/652,080 filed on Jul. 17, 2017 entitled "METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS", which is a continuation of International Application No. PCT/US17/13399 filed on Jan. 13, 2017 entitled "METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS", which claims priority to U.S. Provisional Patent Application No. 62/279,474 filed on Jan. 15, 2016 entitled "METHODS AND COMPOSITIONS FOR THE ACTIVATION OF GAMMA-DELTA T-CELLS", the disclosures of which are incorporated herein by reference.

SEQUENCE LISTING

A Sequence Listing is enclosed with this application and is incorporated herein by reference. The text file of the Sequence Listing is named 436313003138_SL.xml and the file size is 119 kilobytes.

FIELD OF THE INVENTION

The present disclosure relates generally to the fields of gene therapy and immunotherapy, specifically in relation to increased activation and effector cell function of gamma delta ("GD") T cells.

BACKGROUND

Human T cells are distinguished on the basis of T cell receptor structure. The major populations, including CD4+ and CD8+ subsets, express a receptor composed of alpha and beta chains. A smaller subset expresses T cell receptor made from gamma and delta chains. Gamma delta ("GD") T cells make up 3-10% of circulating lymphocytes, and a Vδ2+ subset makes up 75% of GD T cells in blood. Vδ2+ cells recognize non-peptide epitopes and do not require antigen presentation by major histocompatibility complexes ("MHC") or human leukocyte antigen ("HLA"). The majority of Vδ2+ T cells also express a Vγ9 chain and are stimulated by exposure to 5-carbon pyrophosphate compounds that are intermediates in mevalonate and non-mevalonate sterol/isoprenoid synthesis pathways. The response to isopentenyl pyrophosphate (5-carbon) is universal among healthy human beings.

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Another subset of GD T cells, Vδ1+, make up a much smaller percentage of the T cells circulating in the blood, but Vδ1+ cells are commonly found in the epithelial mucosa and the skin.

In general, GD T cells have several functions, including killing tumor cells and pathogen-infected cells. Stimulation through their unique T cell receptor ("TCRs") composed of two glycoprotein chains, γ and δ, improves the capacity for cellular cytotoxicity, cytokine secretion and other effector functions. The TCRs of GD T cells have unique specificities and the cells themselves occur in high clonal frequencies, thus allowing rapid innate-like responses to tumors and pathogens.

Bisphosphonate drugs and other inhibitors of farnesyl diphosphate synthase ("FDPS"), which are downstream from isopentenyl pyrophosphate ("IPP") in the mevalonate pathway (see, for e.g., FIG. 1), have been used to treat various diseases, including cancers, specifically those involving bone metastasis. Bisphosphonate drugs include, for example, Zometa® (Novartis) and Fosamax® (Merck).

Certain bisphosphonates have also been investigated for stimulation of GD T cells. This may be because when FDPS is inhibited in myeloid cells, IPP begins to accumulate and geranylgeranyl pyrophosphate ("GGPP"), a downstream product of FDPS that suppresses activation of the inflammasome pathway, is reduced. The reduction in GGPP removes an inhibitor of the caspase-dependent inflammasome pathway and allows secretion of mature cytokines including interleukin-beta and interleukin-18, the latter being especially important for gamma delta T cell activation.

Thus, when FDPS is blocked, the increased IPP and decreased GGPP combine to activate Vδ2+ T cells. Vδ2+ cells activated by IPP or bisphosphonates will proliferate rapidly, express a number of cytokines and chemokines, and can function to cytotoxically destroy tumor cells or cells infected with pathogenic microorganisms.

However, bisphosphonates are associated with inflammation and osteonecrosis, as well as having poor bioavailability due to their chemistry. Likewise, IPP has a very short half-life and is difficult to synthesize. Both types of compounds require systemic administration in an individual. Accordingly, both bisphosphonates in general, and IPP specifically, leave a great deal to be desired for therapeutic purposes involving activation of GD T cells.

SUMMARY OF THE INVENTION

In one aspect, a method of activating a GD T cell is provided. The method includes infecting, in the presence of the GD T cell, a target cell with a viral delivery system that encodes at least one genetic element. In embodiments, the at least one genetic element includes a small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In embodiments, the enzyme is FDPS. In embodiments, when the enzyme is inhibited in the target cell, the target cell subsequently activates the GD T cell. In embodiments, the target cell is a cancer cell or a cell that has been infected with an infectious agent. In a preferred embodiment, the activation of the GD T cell results in the GD T cell killing the cancer cell or the cell infected with an infectious agent. In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA. In further embodiments, the target cell is also contacted with a bisphosphonate drug. In embodiments, the bisphosphonate drug is zoledronic acid.

In another aspect, a method of treating cancer in a subject is provided. The method includes administering to the

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subject a therapeutically-effective amount of a viral delivery system that encodes at least one genetic element. In embodiments, the at least one genetic element includes a small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In further embodiments, when the enzyme is inhibited in a cancer cell in the presence of a GD T cell, the cancer cell activates the GD T cell, to thereby treat the cancer. In embodiments, the enzyme is FDPS. In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA. In further embodiments, the target cell is also contacted with a bisphosphonate drug. In embodiments, the bisphosphonate drug is zoledronic acid.

In another aspect, a method of treating an infectious disease in a subject is provided. The method includes administering to the subject a therapeutically-effective amount of a viral delivery system that encodes at least one genetic element. In embodiments, the at least one genetic element includes a small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In further embodiments, when the enzyme is inhibited in a cell that is infected with an infectious agent in the presence of a GD T cell, the infected cell activates the GD T cell, to thereby treat the infected cell, and the infectious disease. In embodiments, the enzyme is FDPS. In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA. In further embodiments, the target cell is also contacted with a bisphosphonate drug. In embodiments, the bisphosphonate drug is zoledronic acid.

In another aspect, the at least one encoded genetic element includes a shRNA having at least 80%, or at least 85%, or at least 90%, or at least 95% percent identity with GTCCTG-GAGTACAATGCCATTCTCGAGAATGGCATTGTACT CCAGGACTTTTT (SEQ ID NO: 1); GCAGGATTT CGTTTCAGCACTTCTCGAGAAGTGCTGAACGAA AT CCTGCTTTTT (SEQ ID NO: 2); GCCATGTACAT GGCAGGAATTCTCGAGAA TTCCTGCCATGTACAT GGCTTTTT (SEQ ID NO: 3); or GCAGAAGGAGGCTGA GAAAGTCTCGAGACTTTCTCAGCCTCCTTCT GCTT TTT (SEQ ID NO: 4). In a preferred embodiment, the shRNA includes GTCCTGGAGTACAATGCCATTCT CGAG AATGGCATTGTACTCCAGGACTTTTT (SEQ ID NO: 1); GCAGGATTTTCGTTCA GCACTTCTCGAG AAGTGCTGAACGAAATCCTGCTTTTT (SEQ ID NO: 2); GCCA TGTACATGGCAGGAATTCTCGAGAAT-TCCTGCCATGTACATGGCTTTTT (SEQ ID NO: 3); or GCAGAAGGAGGCTGAGAAAGTCTCGAGACTTTC TCAGCCTCCTT CTGCTTTTT (SEQ ID NO: 4).

In another aspect, the at least one encoded genetic element includes a microRNA having at least 80%, or at least 85%, or at least 90%, or at least 95% percent identity with AAGGTATATTGCTGTTGACAGTGAGCGACACTTTC TCAGCCTCCTTCTGCGTGAAGC CACAGATGGCA GAAGGAGGCTGAGAAAGTGCTGCCTACTGCC TCG GACTTCAAGGG GCT (SEQ ID NO: 5); AAGGTATAT-TGCTGTTGACAGTGAGCGACACT TTCTCAGCCT CC TTCTGCGTGAAGCCACAGATGGCAGAAGGGCTGA GAAAGTGCTGC CTACTGCCTCGACTTCAAGGG GCT (SEQ ID NO: 6); TGCTGTTGACAGTG AGCGAC TTTCTCAGCCTCCTTCTGCGTGAAGCCACAGATGG CAGAAGGAGGCTGAG AAAGTTGCCTACTGCC TC GGA (SEQ ID NO: 7); CCTGGAGGCTTGCTGAAG GCTGTATGCTGACTTTCTCAGCCTCCTTCTGCTTTT GGCCACTGACTGAGCAGAAGGG CTGAGAAAGT-CAGGACACAAGGCCTGTTACTAGCACTCA (SEQ ID NO: 8); CATCTCCATGGCTGTACCACCTTGTC- GGGACTTTCTCAGCCTCCTTCTGCGTGTGAA TCT-CATGGCAGAAGGAGGCGAGAAAGTCTGACATTT-

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TGGTATCTTTCATCTGACCA (SEQ ID NO: 9); or GGGCCTGGCTCGAGCAGGGGGCGAGGGATACTT TCT CAGCCTCCTTCTGCTGGTCCCCCTCCCCGCAG A AGGAGGCTGAGAAAGTCCCTCCCTC CCAATGA CC GCGTCTTCGTCG (SEQ ID NO: 10). In a preferred embodiment, the microRNA includes AAGGTATAT-TGCTGTTGACAGTGAGCGACACTTTCTCAGCCT CCTTCTGCGTGAAGCCACAGATGGCAGAAGGAG GCTGAGAAAGTGCTGCCTACTGC CTCGGACTTCA AGGGGCT (SEQ ID NO: 5); AAGGTATATTGCT GTTGACAGT GAGCGACACTTTCTCAGCCTCCTTC GCGTGAAGCCACAGATGGCAGAAGGGCTGA GAA AGTGCTGCCTACTGCCTCGGACTTCAAGGGGCT (SEQ ID NO: 6); TGCTG TTGACAGTGAGCG ACTTTCTCAGCCTCCTTCTGCGTGAAGCCACAGAT GGCAGAAGG AGGCTGAGAAAGTTGCCTACTGC CTCGGA (SEQ ID NO: 7); CCTGGAGGCT TGCT- GAAGGCTGTATGCTGACTTTCTCAGCCTCCTTCTG CTTTGGCCACTGACTGAG CAGAAGGGCTGAG AAAGTCAGGACACAAGGCCTGTTACTAGCACTCA (SEQ ID NO: 8); CATCTCCATGGCTGTACCACCTTGT CGGGACTTTCTCAGCCTCCTT CTGCGTGTGAAATCT C ATGGCAGAAGGAGGCGAGAAAGTCTGACATTT TGGTATCTT TCATCTGACCA (SEQ ID NO: 9); or GGGCCTGGCTCGAGCAGGGGGCGAGGG ATACTTT CTCAGCCTCCTTCTGCTGGTCCCCCTCCCCGCAGA AGGAGGCTGAGAAAGT CCTTCCCTCCCAATGAC CGGCTCTTCGTCG (SEQ ID NO: 10).

In another aspect, a viral vector comprising at least one encoded genetic element is provided. The at least one encoded genetic element includes a small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In embodiments, the enzyme involved in the mevalonate pathway is farnesyl diphosphate synthase (FDPS). In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA.

In another aspect, the at least one encoded genetic element includes a shRNA having at least 80%, or at least 85%, or at least 90%, or at least 95% percent identity with v In a preferred embodiment, the shRNA includes SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4.

In another aspect, the at least one encoded genetic element includes a microRNA having at least 80%, or at least 85%, or at least 90%, or at least 95% percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In a preferred embodiment, the microRNA includes SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10.

In embodiments, the viral vector is comprised of any vector that can effectively transduce the small RNA into a target cell. In embodiments, the viral vector is a lentiviral vector. In other embodiments, the viral vector is an adeno-associated virus vector.

In another aspect, the viral vector includes a second encoded genetic element. In embodiments, the second genetic element includes at least one cytokine or chemokine. In embodiments, the at least one cytokine is selected from the group consisting of: IL-18, TNF- α , interferon- γ , IL-1, IL-2, IL-15, IL-17, and IL-12. In embodiments, the at least one chemokine is a CC chemokine or a CXC chemokine. In further embodiments, the at least one chemokine is RANTES.

In another aspect, a lentiviral vector system for expressing a lentiviral particle is provided. The system includes a lentiviral vector, at least one envelope plasmid for expressing an envelope protein optimized for infecting a cell; and at

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least one helper plasmid for expressing gag, pol, and rev genes. When the lentiviral vector, the at least one envelope plasmid, and the at least one helper plasmid are transfected into a packaging cell, a lentiviral particle is produced by the packaging cell. In embodiments, the lentiviral particle is capable of infecting a targeting cell, and inhibiting an enzyme involved in the mevalonate pathway within the target cell. In embodiments, the enzyme involved in the mevalonate pathway is FDPS. In embodiments, the lentiviral vector system includes a first helper plasmid for expressing the gag and pol genes, and a second helper plasmid for expressing the rev gene. In embodiments, the envelope protein is preferably optimized for infecting a target cell. In embodiments, the target cell is a cancer cell. In other embodiments, the target cell is a cell that is infected with an infectious agent.

In another aspect a pharmaceutical combination is disclosed which includes a bisphosphonate compound; and a lentiviral particle produced by a packaging cell and capable of infecting a target cell. The lentiviral particle comprises an envelope protein capable of infecting the target cell, and: at least one encoded shRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded shRNA comprises a sequence having at least 80% percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4; or at least one encoded microRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded microRNA comprises a sequence having at least 80% percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10, wherein the pharmaceutical combination is at least one of fixed and non-fixed. In embodiments, the at least one encoded shRNA comprises a sequence having at least 85% or at least 90% or at least 95% percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4; or the at least one encoded microRNA comprises a sequence having at least 85% percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the at least one encoded shRNA comprises SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4 or the at least one encoded microRNA comprises SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the pharmaceutical composition comprises a fixed combination. In embodiments, the pharmaceutical composition comprises a non-fixed combination. In embodiments, the bisphosphonate drug comprises zoledronic acid. In embodiments, the bisphosphonate drug and the lentiviral particle are present in synergistically effective amounts. In embodiments, the target cell is one or more cancer cells that are present in a cancer selected from one or more of a carcinoma, a leukemia, a lymphoma, a sarcoma, a myeloma, a mesothelioma, a mixed type, or mixtures thereof. In embodiments, the target cell is one or more cancer cells that are present in a hepatocellular carcinoma. In embodiments, the target cell is capable of activating a gamma delta T cell following infection with the lentiviral particle. In embodiments, the enzyme is FDPS.

In another aspect, a method of treating a cancer in a subject using an immunotherapy-based composition is disclosed. The method includes administering a therapeutically-effective amount of a bisphosphonate drug to the subject; and administering a therapeutically-effective amount of the immunotherapy-based composition to the subject, wherein the immunotherapy-based composition comprises a lentiviral particle. The lentiviral particle com-

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prises an envelope protein capable of infecting one or more cancer cells, and at least one encoded shRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded shRNA comprises a sequence having at least 80% percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4 or at least one encoded microRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded microRNA comprises a sequence having at least 80% percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the at least one encoded shRNA comprises a sequence having at least 85% or at least 90% or at least 95% percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4, or at least one encoded microRNA comprises a sequence having at least 85% or at least 90% or at least 95% percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the at least one encoded shRNA comprises SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4. In embodiments, the at least one encoded microRNA comprises SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the one or more cancer cells are present in a cancer selected from one or more of a carcinoma, a leukemia, a lymphoma, a sarcoma, a myeloma, a mesothelioma, a mixed type, or mixtures thereof. In embodiments, the bisphosphonate drug comprises zoledronic acid. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered in a fixed combination. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered in a non-fixed combination. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered simultaneously. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered sequentially. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered in synergistically effective amounts. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered at a synergistically effective time interval. In embodiments, the one or more cancer cells are capable of activating a gamma delta T cell resident in the subject following infection of the one or more cancer cells with the immunotherapy-based composition. In embodiments, activating the gamma delta T cell comprises increasing tumor necrosis factor (TNF)- α expression by the gamma delta T cell. In embodiments, activating the gamma delta T cell comprises increasing expression and/or secretion of cytokines, chemokines, and/or cell death ligands including but not limited to FasL and TRAIL. In embodiments, the enzyme of the mevalonate pathway is farnesyl diphosphate synthase (FDPS).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 depicts an overview of the major steps in the mevalonate pathway for biosynthesis of steroids and isoprenoids.

FIG. 2 depicts an exemplary 3-vector lentiviral vector system in a circularized form.

FIG. 3 depicts an exemplary 4-vector lentiviral vector system in a circularized form.

FIG. 4 depicts: (A) a linear map of a lentiviral vector expressing a FDPS shRNA targeting sequence; and (B) a

linear map of a lentiviral vector expressing a synthetic microRNA with a FDPS targeting sequence.

FIG. 5 depicts data demonstrating activation of Vδ2+ T cells THP-1 leukemia cells with a lentivirus expressing FDPS shRNA #4 (SEQ ID NO: 4), as described herein.

FIG. 6 depicts data demonstrating activation of Vδ2+ T cells by THP-1 leukemia cells with a lentivirus expressing FDPS shRNA #4 (SEQ ID NO: 4), as described herein.

FIG. 7 depicts data demonstrating activation of Vδ2+ T cells by PC3 prostate carcinoma cells with a lentivirus expressing FDPS shRNA #1 (SEQ ID NO: 1), as described herein.

FIG. 8 depicts data demonstrating activation of Vδ2+ T cells by PC3 prostate carcinoma cells with a lentivirus expressing FDPS shRNA #4 (SEQ ID NO: 4), as described herein.

FIG. 9 depicts data demonstrating activation of Vδ2+ T cells by HepG2 carcinoma cells with a lentivirus expressing FDPS shRNA #1 (SEQ ID NO: 1) or FDPS shRNA #4 (SEQ ID NO: 4), as described herein.

FIG. 10 depicts data demonstrating activation of Vδ2+ T cells by THP-1 leukemia cells with a lentivirus expressing miR30 FDPS #1 (SEQ ID NO: 5), as described herein.

FIG. 11 depicts data demonstrating the percent of specific lysis versus an E:T ratio for a variety of experimental conditions, as described herein.

FIG. 12 depicts data demonstrating lentiviral-delivered shRNA-based RNA interference targeting the human FDPS gene.

FIG. 13 depicts data demonstrating lentiviral-delivered miR-based RNA interference targeting the human FDPS gene.

FIG. 14 depicts data demonstrating activation of Vδ2+ T cells by HepG2 carcinoma cells with an adeno-associated virus expressing FDPS shRNA #4 (SEQ ID NO: 4), as described herein.

FIG. 15 depicts immunoblot data demonstrating lack of RAP1 prenylation in the cells transduced with LV-shFDPS and treated with zoledronic acid.

DETAILED DESCRIPTION

Overview of Disclosure

The present disclosure relates to gene therapy constructs and delivery of the same to cells, resulting in suppression of Farnesyl diphosphate synthase ("FDPS"), which is necessary to convert isopentenyl phosphate (IPP) to farnesyl diphosphate (FDP), as shown, for example, in FIG. 1. In embodiments, one or more viral vectors are provided with microRNAs or short hairpin RNAs (shRNA) that target FDPS, thereby reducing expression levels of this enzyme. The viral vectors include lentiviral vectors and AAV vectors. A consequence of modulating expression of FDPS is to increase the accumulation of IPP, which is a stimulator of GD T cell proliferation and differentiation. Accordingly, the constructs provided herein are used to activate GD T cells, and are used to treat cancers and infectious diseases.

Definitions and Interpretation

Unless otherwise defined herein, scientific and technical terms used in connection with the present disclosure shall have the meanings that are commonly understood by those of ordinary skill in the art. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. Generally, nomenclature used in connection with, and techniques of, cell and tissue culture, molecular biology, immunology, microbiology, genetics and protein and nucleic acid chemistry and hybrid-

ization described herein are those well-known and commonly used in the art. The methods and techniques of the present disclosure are generally performed according to conventional methods well-known in the art and as described in various general and more specific references that are cited and discussed throughout the present specification unless otherwise indicated. See, e.g.: Sambrook J. & Russell D. *Molecular Cloning: A Laboratory Manual*, 3rd ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2000); Ausubel et al., *Short Protocols in Molecular Biology: A Compendium of Methods from Current Protocols in Molecular Biology*, Wiley, John & Sons, Inc. (2002); Harlow and Lane *Using Antibodies: A Laboratory Manual*; Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (1998); and Coligan et al., *Short Protocols in Protein Science*, Wiley, John & Sons, Inc. (2003). Any enzymatic reactions or purification techniques are performed according to manufacturer's specifications, as commonly accomplished in the art or as described herein. The nomenclature used in connection with, and the laboratory procedures and techniques of, analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well-known and commonly used in the art.

As used in the description and the appended claims, the singular forms "a", "an" and "the" are used interchangeably and intended to include the plural forms as well and fall within each meaning, unless the context clearly indicates otherwise. Also, as used herein, "and/or" refers to and encompasses any and all possible combinations of one or more of the listed items, as well as the lack of combinations when interpreted in the alternative ("or").

All numerical designations, e.g., pH, temperature, time, concentration, and molecular weight, including ranges, are approximations which are varied (+) or (-) by increments of 0.1. It is to be understood, although not always explicitly stated that all numerical designations are preceded by the term "about". The term "about" also includes the exact value "X" in addition to minor increments of "X" such as "X+0.1" or "X-0.1." It also is to be understood, although not always explicitly stated, that the reagents described herein are merely exemplary and that equivalents of such are known in the art.

As used herein, the term "about" will be understood by persons of ordinary skill in the art and will vary to some extent depending upon the context in which it is used. If there are uses of the term which are not clear to persons of ordinary skill in the art given the context in which it is used, "about" will mean up to plus or minus 10% of the particular term.

As used herein, the terms "administration of" or "administering" refer to providing an active agent to a subject in need of treatment in a form that can be introduced into that individual's body in a therapeutically useful form and therapeutically effective amount.

As used herein, the terms "bisphosphonates" and "bisphosphonate drugs" refer to therapeutic agents of various embodiments, and encompass any of aminobisphosphonates, diphosphonates, biphosphonic acids, and diphosphonic acids, as well as pharmaceutically acceptable salts and derivatives thereof. The use of a specific nomenclature in referring to bisphosphonates is not meant to limit the scope of the present invention, unless specifically indicated.

As used herein, the terms "co-administration" or "combined administration" or "combined use" or "combination therapy" or the like as utilized herein refer to administration of a therapeutic vector or a lentiviral particle and a bispho-

sphosphate drug or a therapeutic vector or a lentiviral particle and an antibody or a therapeutic vector or a lentiviral particle and a bisphosphonate drug and an antibody to a single subject in need thereof (e.g., a patient), and are intended to include treatment regimens in which the agents are not necessarily administered by the same route of administration and/or at the same time.

As used herein, the term “fixed combination,” refers to two or more active ingredients or components, including any of their respective compositions, formulations or drug forms, e.g., a therapeutic vector or a lentiviral particle and a bisphosphonate drug or any combination of these, that are administered essentially in combination to a patient, for example essentially simultaneously, in the form of a single entity or dosage or combined entities or dosages, e.g., in one tablet or in one capsule or in combined tablets or capsules or combined liquid forms.

As used herein, the term “non-fixed combination,” refers to two or more active ingredients or components, including any of their respective compositions, formulations or drug forms, e.g., a therapeutic vector or a lentiviral particle and a bisphosphonate drug or any combination of these, that are administered in combination to a patient as separate entities either simultaneously, concurrently or sequentially with no specific time limits, wherein such administration provides therapeutically effective levels of the active components in the patient. The non-fixed combination can be dosed independently of each other or by use of different fixed combinations e.g., simultaneously or at different time points. The active components may be administered as separate pharmaceutical dosage forms or pharmaceutical formulations that may be, for example, sold independently of each other, with or without label instructions concerning the possibility of a combined use. Such instructions may be provided in the package equipment, e.g., leaflet or the like, or in other information, e.g., provided to physicians and medical staff. A non-fixed combination, its respective active ingredients or components, including any of their respective compositions, formulations or drug forms, or the parts thereof, can be administered simultaneously or chronologically staggered, e.g., at different time points and with equal or different time intervals for any part of the administration. Such time intervals may be chosen such that the effect on the treated disease, when treated in combination, is more effective than would be obtained by use of only any one of the active components.

As used herein, the terms “combination,” “in combination” and “combination therapies,” may refer generally to any or both of the “fixed combination” and “non-fixed combination” definitions and embodiments described above.

As used herein, the transitional term “comprising,” when used to define compositions and methods, means that the compositions and methods include the recited elements, but does not exclude others. As used herein, “consisting essentially of,” when used to define compositions and methods, means that the composition and methods include additional elements, but only if those additional elements do not materially affect the basic and novel characteristics of the composition or methods. As used herein, “consisting of,” when used to define compositions and methods, means that the compositions and methods exclude more than trace elements of other ingredients for compositions and substantial method steps. Embodiments defined by each of these transitional terms are within the scope of this disclosure. For example, it is intended that the methods and compositions can include additional steps and components (comprising) or alternatively including steps and compositions of no signifi-

cance (consisting essentially of) or alternatively, intending only the stated method steps or compositions (consisting of).

As used herein, the terms “expression,” “expressed,” or “encodes” refer to a process by which polynucleotides are transcribed into mRNA and/or the process by which the transcribed mRNA is subsequently being translated into peptides, polypeptides, or proteins. Expression may include splicing of the mRNA in a eukaryotic cell or other forms of post-transcriptional modification or post-translational modification.

As used herein, the term “farnesyl diphosphate synthase” may also be referred to herein as FDPS, and may also be referred to herein as farnesyl pyrophosphate synthase or FPPS.

As used herein, the term “gamma delta T cell” may also be referred to herein as a $\gamma\delta$ T cell, or further as a GD T cell. The term “gamma delta T cell activation” refers to any measurable biological phenomenon associated with a gamma delta T cell that is representative of such T cell being activated. Non-limiting examples of such a biological phenomenon include an increase of cytokine production, changes in the qualitative or quantitative composition of cell surface proteins, an increase in T cell proliferation, and/or an increase in T cell effector function, such killing of a target cell or assisting another effector cell to kill a target cell.

As used herein, the terms “individual,” “subject,” and “patient” are used interchangeably herein, and refer to any individual mammal subject, e.g., bovine, canine, feline, equine, and/or human.

As used herein, the term “miRNA” refers to a microRNA, and also may be referred to herein as “miR”.

The term “packaging cell line” refers to any cell line that can be used to express a lentiviral particle.

As used herein, the term “homology” refers to the percentage number of amino acids, nucleic acids, or analogs thereof, that are identical or constitute conservative substitutions. Homology may be determined using sequence comparison programs such as GAP (Deveraux et al., 1984, Nucleic Acids Research 12, 387-395). In this way sequences of a similar or substantially different length to those cited herein could be compared by insertion of gaps into the alignment, such gaps being determined, for example, by the comparison algorithm used by GAP.

As used herein, the term “sequence identity,” which also may appear in the non-limiting context of “a sequence 50% identical to,” and “having at least 80%, or at least 85%, or at least 90%, or at least 95% identity with” a given sequence, as similar phrasings, as used herein, refers to the extent that sequences are identical on a nucleotide-by-nucleotide basis or an amino acid-by-amino acid basis over a window of comparison. Thus, a “percentage of sequence identity” may be calculated by comparing two optimally aligned sequences over the window of comparison, determining the number of positions at which the identical nucleic acid base (e.g., A, T, C, G, I) or the identical amino acid residue (e.g., Ala, Pro, Ser, Thr, Gly, Val, Leu, Ile, Phe, Tyr, Trp, Lys, Arg, His, Asp, Glu, Asn, Gln, Cys and Met) occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison (i.e., the window size), and multiplying the result by 100 to yield the percentage of sequence identity. Optimal alignment of sequences for aligning a comparison window may be conducted by computerized implementations of algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package Release 7.0, Genetics Computer Group, 575 Science Drive Madison, Wis., USA) or by inspection and the best align-

ment (i.e., resulting in the highest percentage homology over the comparison window) generated by any of the various methods selected. Reference also may be made to the BLAST family of programs as for example disclosed by Altschul et al., Nucl. Acids Res. 25:3389, 1997.

As used here, the term "percent identity," which may be used interchangeably with the term "sequence identity", in the context of two or more nucleic acid or polypeptide sequences, refer to two or more sequences or subsequences that have a specified percentage of nucleotides or amino acid residues that are the same, when compared and aligned for maximum correspondence, as measured using one of the sequence comparison algorithms described below (e.g., BLASTP and BLASTN or other algorithms available to persons of skill) or by visual inspection. Depending on the application, the "percent identity" can exist over a region of the sequence being compared, e.g., over a functional domain, or, alternatively, exist over the full length of the two sequences to be compared. For sequence comparison, typically one sequence acts as a reference sequence to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are input into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.

Optimal alignment of sequences for comparison can be conducted, e.g., by the local homology algorithm of Smith & Waterman, Adv. Appl. Math. 2:482 (1981), by the homology alignment algorithm of Needleman & Wunsch, J. Mol. Biol. 48:443 (1970), by the search for similarity method of Pearson & Lipman, Proc. Nat'l. Acad. Sci. USA 85:2444 (1988), by computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, Wis.), or by visual inspection (see generally Ausubel et al., *infra*).

Suitable algorithms for determining percent sequence identity include the BLAST algorithm, which is described in Altschul et al., J. Mol. Biol. 215:403-410 (1990). Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information website.

The percent identity between two nucleotide sequences can be determined using the GAP program in the GCG software package (available at <http://www.gcg.com>), using a NWSgapdna.CMP matrix and a gap weight of 40, 50, 60, 70, or 80 and a length weight of 1, 2, 3, 4, 5, or 6. The percent identity between two nucleotide or amino acid sequences can also be determined using the algorithm of E. Meyers and W. Miller (CABIOS, 4:11-17 (1989)) which has been incorporated into the ALIGN program (version 2.0), using a PAM120 weight residue table, a gap length penalty of 12 and a gap penalty of 4. In addition, the percent identity between two amino acid sequences can be determined using the Needleman and Wunsch (*J. Mol. Biol.* (48):444-453 (1970)) algorithm which has been incorporated into the GAP program in the GCG software package (available at <http://www.gcg.com>), using either a Blossum 62 matrix or a PAM250 matrix, and a gap weight of 16, 14, 12, 10, 8, 6, or 4 and a length weight of 1, 2, 3, 4, 5, or 6.

The nucleic acid and protein sequences of the present disclosure can further be used as a "query sequence" to perform a search against public databases to, for example, identify related sequences. Such searches can be performed

using the NBLAST and XBLAST programs (version 2.0) of Altschul, et al. (1990) *J. Mol. Biol.* 215:403-10. BLAST nucleotide searches can be performed with the NBLAST program, score=100, word length=12 to obtain nucleotide sequences homologous to the nucleic acid molecules provided in the disclosure. BLAST protein searches can be performed with the XBLAST program, score=50, word-length=3 to obtain amino acid sequences homologous to the protein molecules of the disclosure. To obtain gapped alignments for comparison purposes, Gapped BLAST can be utilized as described in Altschul et al., (1997) *Nucleic Acids Res.* 25(17):3389-3402. When utilizing BLAST and Gapped BLAST programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used. See <http://www.ncbi.nlm.nih.gov>.

As used herein, the term "pharmaceutically acceptable" refers to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues, organs, and/or bodily fluids of human beings and animals without excessive toxicity, irritation, allergic response, or other problems or complications commensurate with a reasonable benefit/risk ratio.

As used herein, the term "pharmaceutically acceptable carrier" refers to, and includes, any and all solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like that are physiologically compatible. The compositions can include a pharmaceutically acceptable salt, e.g., an acid addition salt or a base addition salt (see, e.g., Berge et al. (1977) *J Pharm Sci* 66:1-19).

As used herein, the term "pharmaceutically acceptable salt" refers to derivatives of compounds or other active ingredients, wherein the parent compound or active ingredient is modified by converting an existing acid or base moiety to its salt form. Non-limiting examples of pharmaceutically acceptable salts include, but are not limited to, mineral or organic acid salts of basic residues such as amines; alkali or organic salts of acidic residues such as carboxylic acids; alkali metal, alkaline metal, ammonium, and mono-, di-, tri-, or tetra-C1-C30-alkyl-substituted ammonium; and the like. The pharmaceutically acceptable salts of various embodiments include the conventional non-toxic salts of the compound or active ingredient formed, for example, from nontoxic inorganic or organic acids. Suitable organic acids are, e.g., carboxylic acids or sulfonic acids, such as acetic acid, succinic acid, fumaric acid or methane-sulfonic acid. The pharmaceutically acceptable salts herein can be synthesized from the parent compound or active ingredient which contains a basic or acidic moiety by conventional chemical methods. Generally, such salts can be prepared by reacting the free acid or base forms of these compounds with a stoichiometric amount of the appropriate base or acid in water or in an organic solvent, or in a mixture of the two; generally, nonaqueous media like ether, ethyl acetate, ethanol, isopropanol, or acetonitrile are preferred. Lists of suitable salts are found in Remington's Pharmaceutical Sciences, 17th ed., Mack Publishing Company, Easton, Pa., 1985, p. 1418 and Journal of Pharmaceutical Science, 66, 2 (1977), each of which is incorporated herein by reference in its entirety.

As used herein, the term "SEQ ID NO" is synonymous with the term "Sequence ID No."

As used herein, the term "small RNA" refers to non-coding RNA that are generally less than about 200 nucleotides or less in length and possess a silencing or interference function. In embodiments, the small RNA is about 175

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nucleotides or less, about 150 nucleotides or less, about 125 nucleotides or less, about 100 nucleotides or less, or about 75 nucleotides or less in length. Such RNAs include microRNA (miRNA), small interfering RNA (siRNA), double stranded RNA (dsRNA), and short hairpin RNA (shRNA). In embodiments, "small RNA" are capable of inhibiting or knocking-down gene expression of a target gene, generally through pathways that result in the inhibitions or destruction of the target gene mRNA.

As used herein, the term "therapeutically effective amount" refers to a sufficient quantity of the active agents of the present disclosure, in a suitable composition, and in a suitable dosage form to treat or prevent the symptoms, progression, or onset of the complications seen in patients suffering from a given ailment, injury, disease, or condition. The therapeutically effective amount will vary depending on the state of the patient's condition or its severity, and the age, weight, etc., of the subject to be treated. A therapeutically effective amount can vary, depending on any of a number of factors, including, e.g., the route of administration, the condition of the subject, as well as other factors understood by those in the art.

As used herein, the term "therapeutic vector" includes, without limitation, reference to a lentiviral vector or an AAV vector.

As used herein, the terms "treatment" and "treating" refer to the intended targeting of a disease state and combatting of it, i.e., ameliorating or preventing the disease state. A particular treatment thus will depend on the disease state to be targeted and the current or future state of medicinal therapies and therapeutic approaches. A treatment may have associated toxicities.

Desirable effects include, but are not limited to, preventing occurrence or recurrence of disease, alleviating symptoms, suppressing, diminishing or inhibiting any direct or indirect pathological consequences of the disease, ameliorating or palliating the disease state, and causing remission or improved prognosis.

Description of Aspects of the Disclosure

In one aspect, a method of activating a GDT cell is provided. The method includes infecting, in the presence of the GD T cell, a target cell with a viral delivery system encoding at least one genetic element. In embodiments, the at least one encoded genetic element includes a small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In embodiments, the enzyme is FDPS. In embodiments, when the enzyme is inhibited in the target cell, the target cell activates the GD T cell. In embodiments, the target cell is a cancer cell or a cell that has been infected with an infectious agent. In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA.

In embodiments, the at least one encoded genetic element includes a shRNA having at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95% or more percent identity with GTCCTGGAGTACAATGC-CATTCTCGAGAATGGCATTGTACTCCAGGACTTTTT (SEQ ID NO: 1); GCAGGATTTCGTTTCAG CACTTC TCGAGAAGTGCTGAACGAA ATCCTGCTTTTT (SEQ ID NO: 2); GCCATGTACATGGCAGGAATTCTC GAGAA TTCCTGCCATGTACATGGCTTTTT (SEQ ID NO: 3); or GCAGAAGGAGGCTGA GAAAGTCTCGA-GACTTTCTCAGCCTCCTTCTGCTTTTT (SEQ ID NO: 4). In a preferred embodiment, the shRNA includes GTCCTGGAGTACAATGCCATTCTCGAG AATGGCAT-

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TGTA CTCCAGGACTTTTT (SEQ ID NO: 1); GCAGGAT-TTCGTTCA GCACTTCTCGAGAAGTGCTGAACGA A ATCCTGCTTTTT (SEQ ID NO: 2); GCCA TGTA-CATGGCAGGAATTCTCGAGAATTCCCTGGCCATGTA-CATGGCTTTTT (SEQ ID NO: 3); or GCAGAAGG AGGCTGAGAAAGTCTCGAGACTTTCTCAGCCTCC TTCTGCTTTTT (SEQ ID NO: 4).

In another aspect, the at least one encoded genetic element includes a microRNA having at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95% or more percent identity with AAGGTATAT-TGCTGTTGACAGTGAGCGACACTTTCTCAGCCTC CTCTGCGTGAAGC CACAGATGGCAGAAGG AGG CTGAGAAAGTGCTGCCTACTGCCTCGGACTTCAA GGG GCT (SEQ ID NO: 5); AAGGTATATTGCTGTTG ACAGTGAGCGACACT TTCTCAGCCTCCTTCTGCGT-GAAGCCACAGATGGCAGAAGGGCTGAGAAAGTG CTGC TACTGCCTCGGACTTCAAGGGGCT (SEQ ID NO: 6); TGCTGTTGACAGTG AGCGACTTTCTCAG CCTCCTTCTGCGTGAAGCCACAGATGGCAGAA GG AGGCTGAG AAAGTTGCCTACTGCCTCGGA (SEQ ID NO: 7); CCTGGAGGCTTGCTGAAG GCTGT ATGCTGACTTTCTCAGCCTCCTTCTGCTTTTGGC-CACTGACTGAGCAGAAGGG CTGAGAAAGTCAGG ACACAAGGCCTGTTACTAGCACTA (SEQ ID NO: 8); CATCTCCATGGCTGTACCACCTTGTCGGGACTTTC TCAGCCTCCTTCTGCGTGTGAA TCTCATGG CAG AAGGAGGCGAGAAAGTCTGACATTTTGGTATCTTT-CATCTGACCA (SEQ ID NO: 9); or GGGCCTGGC TCGAGCAGGGGGCGAGGGATACTTTCT CAGCCT CCTTCTGCTGGTCCCCCTCCCCGAGAAGGAGGCT-GAGAAAGTCCTTCCCTC CCAATGACCGCGT CTTC GTCG (SEQ ID NO: 10). In a preferred embodiment, the microRNA includes AAGGTATATTGCTGTTGACAGT-GAGCGACACTTTCTCAGCCT CCTTCTGCGTGAAG CCACAGATGGCAGAAGGAGGCTGAGAAAGT CTG CTTACTGC CTCGGACTTCAAGGGGCT (SEQ ID NO: 5); AAGGTATATTGCTGTTGACAGT GAGCGACACT TTCTCAGCCTCCTTCTGCGTGAAGCCACAGAT GG CA GAAGGGCTGA GAAAGTGCTGCCTACTGCC TCGGACTTCAAGGGGCT (SEQ ID NO: 6); TGCTG TTGACAGTGAGCGACTTTCTCAGCCTCCTTCTG CG TGAAGCCACAGATGGCAGAAGG AGGCTGAGAA AGTTGCCTACTGCCTCGGA (SEQ ID NO: 7); CCTG-GAGGCT TGCTGAAGGCTGTATGCTGACTTTCTC AG CC TCCTTCTGCTTTTGGCCACTGACTGAG CAGAA GGGCTGAGAAAGTCAGGACACAAGGCCTGTTACT AGCACTCA (SEQ ID NO: 8); CATCTCCATGGCTGTAC-CACCTTGTGCGGACTTTCTCAGCCTCCTTCTGCTG CC TGGAATCTCATGGCAGAAGGAGGCGAGAGAAAG TCTGACATTTTGGTATCTT TCATCTGACCA (SEQ ID NO: 9); or GGGCCTGGCTCGAGCAGGGGGCGAGGG ATACTTTCTCAGCCTCCTTCTGCTGGTCCCCCTCCC CGCAGAAGGAGGCTGAGAAAGT CCTTCCCTCC-CAATGACCGCGTCTTCGTCG (SEQ ID NO: 10).

In another aspect, the target cell is also contacted with a bisphosphonate drug. In a preferred embodiment, the bisphosphonate drug is zoledronic acid. The bisphosphonate drug may be a pharmaceutically acceptable salt, hydrate or a solvate thereof.

In another aspect, a method of treating cancer in a subject is provided. The method includes administering to the subject a therapeutically-effective amount of a viral delivery system encoding at least one genetic element. In embodiments, the at least one encoded genetic element includes a

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small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In further embodiments, when the enzyme is inhibited in a cancer cell in the presence of a GD T cell, the cancer cell activates the GD T cell, to thereby treat the cancer. In embodiments, the enzyme is FDPS. In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA.

In another aspect, a method of treating an infectious disease in a subject is provided. The method includes administering to the subject a therapeutically-effective amount of a viral delivery system encoding at least one genetic element. In embodiments, the at least one encoded genetic element includes a small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In further embodiments, when the enzyme is inhibited in a cell that is infected with an infectious agent and is in the presence of a GD T cell, the infected cell activates the GD T cell, to thereby treat the infected cell, and the infectious disease. In embodiments, the enzyme is FDPS. In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA.

In embodiments, the at least one encoded genetic element includes a shRNA having at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95% or more percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4. In a preferred embodiment, the shRNA includes SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4.

In other embodiments, the at least one encoded genetic element includes a microRNA having at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95% or more percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In a preferred embodiment, the microRNA includes SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10.

In another aspect, a viral vector comprising at least one encoded genetic element is provided. The at least one encoded genetic element includes a small RNA capable of inhibiting production of an enzyme involved in the mevalonate pathway. In embodiments, the enzyme involved in the mevalonate pathway is farnesyl diphosphate synthase (FDPS). In embodiments, the at least one encoded genetic element includes a microRNA or a shRNA.

In another aspect, the at least one encoded genetic element includes a shRNA having at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95% or more percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4. In a preferred embodiment, the shRNA includes SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4.

In another aspect, the at least one encoded genetic element includes a microRNA having at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95% or more percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In a preferred embodiment, the microRNA includes SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10.

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In embodiments, the viral vector includes any vector that can effectively transduce the small RNA. In embodiments, the viral vector is a lentiviral vector. In other embodiments, the viral vector is an adeno-associated virus (AAV) vector.

In another aspect, the viral vector includes a second encoded genetic element. In embodiments, the second genetic element includes at least one cytokine or chemokine. In embodiments, the at least one cytokine is selected from the group consisting of: IL-18, TNF- α , interferon- γ , IL-1, IL-2, IL-15, IL-17, and IL-12. In embodiments, the at least one chemokine is a CC chemokine, CXC chemokine, a CX3 chemokine or a XC chemokine. In a further embodiment, the at least one chemokine is the CC chemokine, RANTES.

In another aspect, a lentiviral vector system for expressing a lentiviral particle is provided. The system includes a lentiviral vector, at least one envelope plasmid for expressing an envelope protein optimized for infecting a cell; and at least one helper plasmid for expressing gag, pol, and rev genes. When the lentiviral vector, the at least one envelope plasmid, and the at least one helper plasmid are transfected into a packaging cell, a lentiviral particle is produced by the packaging cell. In embodiments, the lentiviral particle is capable of infecting a targeting cell, and inhibiting an enzyme involved in the mevalonate pathway within the target cell. In embodiments, the enzyme involved in the mevalonate pathway is FDPS. In embodiments, the lentiviral vector system includes a first helper plasmid for expressing the gag and pol genes, and a second helper plasmid for expressing the rev gene. In embodiments, the envelope protein is preferably optimized for infecting a target cell. In embodiments, the target cell is a cancer cell. In other embodiments, the target cell is a cell that is infected with an infectious disease.

In another aspect a pharmaceutical combination is disclosed which includes a bisphosphonate compound; and a lentiviral particle produced by a packaging cell and capable of infecting a target cell. The lentiviral particle comprises an envelope protein capable of infecting the target cell, and: at least one encoded shRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded shRNA comprises a sequence having at least 80%, or at least 81%, or at least 82%, or at least 83%, or at least 84%, or at least 85%, or at least 86%, or at least 87%, or at least 88%, or at least 89%, or at least 90%, or at least 91%, or at least 92%, or at least 93%, or at least 94%, or at least 95%, or at least 96%, or at least 97%, or at least 98% or at least 99% percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4; or at least one encoded microRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded microRNA comprises a sequence having at least 80%, or at least 81%, or at least 82%, or at least 83%, or at least 84%, or at least 85%, or at least 86%, or at least 87%, or at least 88%, or at least 89%, or at least 90%, or at least 91%, or at least 92%, or at least 93%, or at least 94%, or at least 95%, or at least 96%, or at least 97%, or at least 98% or at least 99% percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10, wherein the pharmaceutical combination is at least one of fixed and non-fixed. In embodiments, the at least one encoded shRNA comprises SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4 or the at least one encoded microRNA comprises SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the pharmaceutical composition comprises a fixed combination. In embodiments, the pharmaceutical composition comprises a non-

fixed combination. In embodiments, the bisphosphonate drug comprises zoledronic acid. In embodiments, the bisphosphonate drug and the lentiviral particle are present in synergistically effective amounts. In embodiments, the target cell is one or more cancer cells that are present in a cancer selected from one or more of a carcinoma, a leukemia, a lymphoma, a sarcoma, a myeloma, a mesothelioma, a mixed type, or mixtures thereof. In embodiments, the target cell is one or more cancer cells that are present in a hepatocellular carcinoma. In embodiments, the target cell is capable of activating a gamma delta T cell following infection with the lentiviral particle. In embodiments, the enzyme is FDPS.

In another aspect, a method of treating a cancer in a subject using an immunotherapy-based composition is disclosed. The method includes administering a therapeutically-effective amount of a bisphosphonate drug to the subject; and administering a therapeutically-effective amount of the immunotherapy-based composition to the subject, wherein the immunotherapy-based composition comprises a lentiviral particle. The lentiviral particle comprises an envelope protein capable of infecting one or more cancer cells, and at least one encoded shRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded shRNA comprises a sequence having at least 80%, or at least 81%, or at least 82%, or at least 83%, or at least 84%, or at least 85%, or at least 86%, or at least 87%, or at least 88%, or at least 89%, or at least 90%, or at least 91%, or at least 92%, or at least 93%, or at least 94%, or at least 95%, or at least 96%, or at least 97%, or at least 98% or at least 99% percent identity with SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4 or at least one encoded microRNA capable of inhibiting production of an enzyme of the mevalonate pathway, wherein the at least one encoded microRNA comprises a sequence having at least 80%, or at least 81%, or at least 82%, or at least 83%, or at least 84%, or at least 85%, or at least 86%, or at least 87%, or at least 88%, or at least 89%, or at least 90%, or at least 91%, or at least 92%, or at least 93%, or at least 94%, or at least 95%, or at least 96%, or at least 97%, or at least 98% or at least 99% percent identity with SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the at least one encoded shRNA comprises SEQ ID NO: 1; SEQ ID NO: 2; SEQ ID NO: 3; or SEQ ID NO: 4. In embodiments, the at least one encoded microRNA comprises SEQ ID NO: 5; SEQ ID NO: 6; SEQ ID NO: 7; SEQ ID NO: 8; SEQ ID NO: 9; or SEQ ID NO: 10. In embodiments, the one or more cancer cells are present in a cancer selected from one or more of a carcinoma, a leukemia, a lymphoma, a sarcoma, a myeloma, a mesothelioma, a mixed type, or mixtures thereof. In embodiments, the bisphosphonate drug comprises zoledronic acid. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered in a fixed combination. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered in a non-fixed combination. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered simultaneously. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered sequentially. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered in synergistically effective amounts. In embodiments, the bisphosphonate drug and the immunotherapy-based composition are administered at a synergistically effective time interval. In embodiments, the one or more cancer cells are

capable of activating a gamma delta T cell resident in the subject following infection of the one or more cancer cells with the immunotherapy-based composition. In embodiments, activating the gamma delta T cell comprises increasing tumor necrosis factor (TNF)- α expression by the gamma delta T cell. In embodiments, the enzyme of the mevalonate pathway is farnesyl diphosphate synthase (FDPS).

Cancer

The compositions and methods provided herein are used to treat cancer. A cell, tissue, or target may be a cancer cell, a cancerous tissue, harbor cancerous tissue, or be a subject or patient diagnosed or at risk of developing a disease or condition. In certain aspects, a cell may be an epithelial, an endothelial, a mesothelial, a glial, a stromal, or a mucosal cell. The cancer cell population can include, but is not limited to a brain, a neuronal, a blood, an endometrial, a meninges, an esophageal, a lung, a cardiovascular, a liver, a lymphoid, a breast, a bone, a connective tissue, a fat, a retinal, a thyroid, a glandular, an adrenal, a pancreatic, a stomach, an intestinal, a kidney, a bladder, a colon, a prostate, a uterine, an ovarian, a cervical, a testicular, a splenic, a skin, a smooth muscle, a cardiac muscle, or a striated muscle cell, and can also include a cancer cell population from any of the foregoing, and can be associated with one or more of carcinomas, sarcomas, myelomas, leukemias, lymphomas, mixed types or mixtures of the foregoing. In still a further aspect cancer includes, but is not limited to astrocytoma, acute myeloid leukemia, anaplastic large cell lymphoma, acute lymphoblastic leukemia, angiosarcoma, B-cell lymphoma, Burkitt's lymphoma, breast carcinoma, bladder carcinoma, carcinoma of the head and neck, cervical carcinoma, chronic lymphoblastic leukemia, chronic myeloid leukemia, colorectal carcinoma, endometrial carcinoma, esophageal squamous cell carcinoma, Ewing's sarcoma, fibrosarcoma, glioma, glioblastoma, gastrinoma, gastric carcinoma, hepatoblastoma, hepatocellular carcinoma, Kaposi's sarcoma, Hodgkin lymphoma, laryngeal squamous cell carcinoma, larynx carcinoma, leukemia, leiomyosarcoma, lipoma, liposarcoma, melanoma, mantle cell lymphoma, medulloblastoma, mesothelioma, myxofibrosarcoma, myeloid leukemia, mucosa-associated lymphoid tissue B cell lymphoma, multiple myeloma, high-risk myelodysplastic syndrome, nasopharyngeal carcinoma, neuroblastoma, neurofibroma, high-grade non-Hodgkin lymphoma, non-Hodgkin lymphoma, lung carcinoma, non-small cell lung carcinoma, ovarian carcinoma, esophageal carcinoma, osteosarcoma, pancreatic carcinoma, pheochromocytoma, prostate carcinoma, renal cell carcinoma, retinoblastoma, rhabdomyosarcoma, salivary gland tumor, schwannoma, small cell lung cancer, squamous cell carcinoma of the head and neck, testicular tumor, thyroid carcinoma, urothelial carcinoma, and Wilms tumor.

The compositions and methods provided herein are also used to treat NSCLC (non-small cell lung cancer), pediatric malignancies, cervical and other tumors caused or promoted by human papilloma virus (HPV), melanoma, Barrett's esophagus (pre-malignant syndrome), adrenal and skin cancers and auto immune, neoplastic cutaneous diseases.

Infectious Diseases

The compositions and methods disclosed herein can be used to treat infectious diseases. The term "infectious disease" includes any disease that is caused by an infectious agent. An "infectious agent" includes any exogenous pathogen including, without limitation, bacteria, fungi, viruses, mycoplasma, and parasites. Infectious agents that may be treated with compositions provided for in this disclosure include any art-recognized infectious organisms that cause

pathogenesis in an animal, including such organisms as bacteria that are gram-negative or gram-positive cocci or bacilli, DNA and RNA viruses, including, but not limited to, DNA viruses such as papilloma viruses, parvoviruses, adenoviruses, herpesviruses and vaccinia viruses, and RNA viruses, such as arenaviruses, coronaviruses, rhinoviruses, respiratory syncytial viruses, influenza viruses, picornaviruses, paramyxoviruses, reoviruses, retroviruses, and rhabdoviruses. Examples of fungi that may be treated with the compositions and methods of the disclosure include fungi that grow as molds or are yeast-like, including, for example, fungi that cause diseases such as ringworm, histoplasmosis, blastomycosis, aspergillosis, cryptococcosis, sporotrichosis, coccidioidomycosis, paracoccidio-idomycosis, and candidiasis. Compositions and methods provided for herein may be utilized to treat parasitic infections including, but not limited to, infections caused by somatic tapeworms, blood flukes, tissue roundworms, ameba, and *Plasmodium*, *Trypanosoma*, *Leishmania*, and *Toxoplasma* species.

Methods of GD T Cell Activation

Provided herein are compositions and methods for activating GD T cells in an individual, as well as methods for treating tumors and infectious diseases. For instance, in embodiments, the compositions and methods provided herein can be used in methods to treat all known cancers because activated GD T cells comprise a natural mechanism for immune surveillance of tumors (See for e.g.: Pauza et al. 2014 *Frontiers in Immunol.* 5:687). Likewise, in embodiments, the compositions and methods provided herein can be used to treat infectious diseases, including but not limited to flavivirus, influenza virus, human retrovirus, mycobacteria, plasmodia and a variety of other viral, fungal and bacterial infections. (See for e.g.: Pauza and Cairo, 2015 *Cell Immunol.* 296(1)).

In general, a vector system is administered to an individual to transfect or transduce a target cell population with the disclosed constructs for decreasing expression of FDPS and, in other embodiments, increasing expression of chemokines or cytokines. Administration and transfection/transduction can occur in vivo or ex vivo, with the transfected cells later administered back into the subject in the latter scenario.

Administration of the disclosed vectors and transfection or transduction of the disclosed constructs into a subject's cells result in decreased expression of FDPS, increased expression of cytokines or chemokines, accumulation of IPP and in many cases, reduced growth rates for genetically modified tumor cells. All of these features work together to activate and co-localize GD T cells to the site of a tumor or infection.

The disclosed methods can also increase the capacity of NK cells to recognize and destroy tumor cells and/or infected cells. Crosstalk between GD T cells and NK cells is an important aspect of regulating the immune and inflammatory responses. Further, GD T cells are known to trigger dendritic cell maturation, recruit B cells and macrophages, and participate in a variety of cytolytic activities, such as secretion of interferon- γ and TNF- α .

In embodiments, the disclosed compositions and methods provided herein comprise a form of gene therapy for activating GD T cells at the site of tumor or infectious disease pathology. In an aspect, the compositions and methods provided herein activate GD T cells and support their proliferation, differentiation, and functional capacities by promoting the production of specific cytokines needed for cytolytic activity capable of killing cancer cells or treating infectious diseases.

In embodiments the gene therapy sequences (e.g., FDPS shRNAs) are carried by therapeutic vectors, including but not limited to viral vectors such as lentiviruses or adeno-associated viruses, although other viral vectors can also be suitable. Gene therapy constructs may also be delivered in the form of DNA or RNA, including but not limited to plasmid forms. In embodiments, the disclosed gene therapy constructs may also be delivered in the form of protein-nucleic acid complexes or lipid nucleic acid complexes and mixtures of these formulations. For instance, a protein-nucleic acid complex can comprise nucleic acids of interest in a complex with cationic peptides such as lysine and arginine. Lipid-nucleic acids complexes can comprise lipid emulsions, micelles, liposomes, and/or mixtures of neutral and cationic lipids such as DOTMA, DOSPA, DOTAP, and DMRIE.

In embodiments, therapeutic vectors may comprise a single construct or at least two, at least three, at least four, or at least five different constructs. When more than one construct is present in a vector the constructs may be identical, or they may be different. For instance, the constructs may vary in terms of their promoters, the presence or absence of integrating elements, and/or their sequences. In some embodiments, a therapeutic vector will comprise at least one construct that encodes a small RNA capable of knocking down the expression of FDPS. In embodiments, the therapeutic vector will also encode a specific cytokine(s) and/or chemokine(s), including but not limited to TNF- α , interferon- γ , IL-1, IL-2, IL-15, IL-17, IL-18 or IL-12. In some embodiments, a single construct may encode both small RNAs capable of knocking down the expression of FDPS and specific cytokines or chemokines, including but not limited to TNF- α , interferon- γ , IL-1, IL-2, IL-15, IL-17, IL-18 or IL-12.

In embodiments, viral vectors may introduce nucleic acid constructs that become integrated into the host chromosome. Alternately, transient delivery vectors may be used to prevent chromosomal integration and limit the lifespan of gene therapy constructs.

In embodiments, the disclosed constructs and vectors comprise short hairpin RNA ("shRNA"), micro RNA ("miRNA"), or siRNA capable of reducing or knocking down expression of FDPS and/or geranyl pyrophosphate synthase ("GPPS") and/or farnesyl transferase ("FT") genes. By down regulating these genes, which control steroid and isoprenoid synthesis, isopentenyl pyrophosphate ("IPP") levels are elevated. Elevation and accumulation of IPP is a known mechanism for increasing GD T cells activation. Further, down regulation of these pyrophosphate synthase genes removes an important negative regulator of inflammation function that in turn results in increased expression of cytokines that are important for GD T cell activation and effector cell function.

In embodiments, the disclosed constructs are regulated by specific promoters that are capable of producing interleukin-2 and/or interleukin-15 to sustain GD T cell proliferation. In addition, the disclosed constructs may be regulated by specific promoters that are capable of producing interleukin-1 beta and/or interleukin-18 and/or interferon-gamma required for GD T cell differentiation and acquisition of all effector cell function. Desirable effector cell functions include the capacity for direct cytotoxic cell killing of tumors and/or infected cells, secretion of beneficial cytokines and/or chemokines, increased expression of NK receptors required to recognize cancerous or infected cells, and increased expression of Fc receptors needed to bind target-

ing antibodies in order to co-localize GD T cells with cancerous or infected cell targets.

In embodiments, the disclosed methods activate GD T cells, resulting in the indirect effect of increasing the capacity for NK cells to attack and destroy cancerous cells, tumors, or infected cells. The activation of NK cells requires GD T cells that are stimulated to proliferate and differentiate, and to express 4-1BBL costimulatory ligand needed to engage the 4-1BB costimulatory receptor on NK cells. This form of crosstalk is known as an important mechanism for activating NK cells and is achieved here through the action of the disclosed methods and compositions.

In another aspect, crosstalk between GD T cells and NK cells is an important mechanism for eliminating inflammatory dendritic cells that accumulate in diseased tissues. Alone, neither GD T cells nor NK cells are capable of destroying dendritic cells, but once the aforementioned crosstalk interactions have occurred, NK cells are altered to become cytotoxic against inflammatory dendritic cells. This immuno-regulatory mechanism depends on strong activation and proliferation of GD T cells.

In embodiments, the disclosed methods for activation of GD T cells further comprise a step of suppressing pathologic inflammatory responses that may include cellular proliferation leading to atherosclerosis, chronic immune activation that stimulates tumor growth, autoimmune diseases including psoriasis and other presentations in the epidermis, inflammatory diseases of the central nervous system, and arthritis and other diseases of unregulated immune responses.

In embodiments, therapeutic vectors are administered in combination with bisphosphonate drugs. In various embodiments, such combinations achieve synergistic, positive or heightened activation of gamma delta T cells. Such positive activation may allow alternate, modified or reduced doses of bisphosphonates and may decrease adverse reactions to bisphosphonates including acute inflammatory responses and chronic diseases. Combinations of therapeutic vectors with bisphosphonates may be together or separate, with or without instructions for combined use or to combination products. The therapeutic vectors and/or bisphosphonates may be administered entirely separately and may be formulated in entirely distinct pharmaceutical dosage forms. The therapeutic vectors and/or bisphosphonates may be sold independently of each other, with or without label instructions concerning the possibility of a combined use. Such instructions also may be provided in the package equipment, e.g., leaflet or the like, or in other information, e.g., provided to physicians and medical staff (e.g., oral communications, communications in writing or the like). Such labels or other instructions can refer to either a fixed combination in one dosage unit form, or a non-fixed combination as a kit of parts for the combined administration where the therapeutic vector may be administered independently of the bisphosphonate drug, at the same time, or separately within time intervals. In various embodiments, the combination exhibits a cooperative or joint effect, or a decrease in toxicity or complications of treatment. In one embodiment the effect of the combination is synergistic. A synergistic effect is achieved when the active ingredients used together is greater than the sum of the effects that results from using the compounds separately. A synergistic effect may be attained when the active ingredients are: (1) co-formulated and administered or delivered simultaneously in a combined, unit dosage formulation; (2) delivered by alternation or in parallel as separate formulations; or (3) by some other regimen. When delivered in alternation therapy, a synergis-

tic effect may be attained when the compounds are administered or delivered sequentially, e.g., by different injections in separate syringes. In general, during alternation therapy, an effective dosage of each active ingredient is administered sequentially, i.e., serially, whereas in combination therapy, effective dosages of two or more active ingredients are administered together, albeit subject to potential variances in timing as detailed herein.

The combinations herein may be manufactured and/or formulated by the same or different manufacturers. The active ingredients may be brought together into a combination therapy: (i) prior to release of the combination product to physicians (e.g., in the case of a kit comprising the compound of the disclosure and the other therapeutic agent); (ii) by the treating physician (or under the guidance of a physician) shortly before administration; (iii) in the actual patient, e.g., during sequential administration of the active ingredients disclosed herein.

In embodiments, a therapeutically effective amount of each of the combinations may be administered simultaneously or sequentially and in any order, and the components may be administered together or separate. For example, the method of treating a proliferative disease according to the disclosure may comprise (i) administration of a first agent such as a therapeutic vector that forms part of a lentiviral particle, and (ii) administration of a second agent such as a bisphosphonate drug in free or pharmaceutically acceptable salt form, simultaneously or sequentially in any order, in jointly therapeutically effective amounts, preferably in cooperative, jointly effective, and/or synergistically effective, amounts, e.g., in daily or intermittent dosages corresponding to the amounts described herein. The combinations may be administered separately at different times during the course of therapy or concurrently in divided or single drug forms. Furthermore, the term "administering" also encompasses the use of a pro-drug of a combination partner that converts in vivo to the combination partner as such. The instant disclosure is therefore to be understood as embracing all such regimens of simultaneous or alternating treatment and the term "administering" is to be interpreted accordingly.

In embodiments, agents (i) and (ii) can be administered using any pharmaceutically acceptable method, such as intranasal, buccal, sublingual, oral, rectal, ocular, parenteral (intravenously, intradermally, intramuscularly, subcutaneously, intraperitoneally), pulmonary, intravaginal, locally administered, topically administered, topically administered after scarification, mucosally administered, via an aerosol, in semi-solid media such as agarose or gelatin, or via a buccal or nasal spray formulation, and/or in solid media such as granules or powders including inert excipients. For example, a therapeutic vector and/or bisphosphonate drug may be administered intravenously. Further, agents (i) and (ii) can be formulated into any pharmaceutically acceptable dosage form, such as a solid dosage form, tablet, pill, lozenge, capsule, liquid dispersion, gel, aerosol, pulmonary aerosol, nasal aerosol, ointment, cream, semi-solid dosage form, a solution, an emulsion, and a suspension. For example, a bisphosphonate drug may be formulated into a tablet and administered orally.

A combination therapy according to the disclosure can besides or in addition be administered especially for cancer therapy in combination with chemotherapy, radiotherapy, immunotherapy, surgical intervention, or a combination of these. Long-term therapy is equally possible as is adjuvant therapy in the context of other treatment strategies, as described above. In embodiments, a combination therapy can also include immune adjuvants (e.g., Toll-like receptor

ligands), immune stimulating toxins, or stimulatory protozoans or stimulatory bacilli (e.g., bacille Calmette-Guerin), cancer therapeutic drugs, cell-based therapies (gamma delta T cell or other cell types known to be in use or under evaluation for tumor therapy and may also include natural or genetically-engineered cells and cells cultured under) ionizing radiation or surgery. Other possible treatments are therapy to maintain the patient's status after tumor regression, or even chemo-preventive therapy, for example in patients at risk.

Constructs for GD T Cell Activation

Inhibition of FDPS results in IPP accumulation, resulting in activation of Vδ2+GD T cells and expression of IL-18, which is also important in activating GD T cells. Inhibition of farnesyl transferase results in decreased prenylation of proteins. The disclosed constructs can be transfected or transduced into specific target cells, like tumor cells or infected cells, where they can express RNA sequences (i.e., siRNA, shRNA or microRNA) that will inhibit translation of FDPS as well as encode and express cytotoxic cytokines or chemokines.

Disclosed herein are constructs for decreasing expression of FDPS and/or FT, increasing expression of cytokines, and increasing expression of chemokines including RANTES. For instance, in some embodiments the constructs may encode for interferon-gamma, IL-1, IL-2, IL-15, IL-17, IL-18 or IL-12.

Expression of cytokines and chemokines, like those listed above, will result in localized cytotoxic destruction of tumor cells or cells infected with pathogenic organisms. Accordingly, expression of such constructs by a tumor cell or an infected cell will result in the unwanted cells assisting in its own destruction.

Likewise, if the disclosed constructs are expressed in a tumor cell or infected cell, decreasing the expression of FDPS and FT will result in activation and recruitment of GD T cells to the tumor site of site of cell infection. Increasing expression of RANTES will further attract GD T cells to intended tissue location. Because GD T cells can kill a broad range of tumors of epithelial origin as well as many leukemias and lymphomas, and are further able to produce high levels of the anti-tumor cytokine, IFNγ, recruitment of GD T cells to the site of a tumor can be a particularly effective means of inducing anti-tumor immunity.

Decreased expression of FDPS can be achieved via shRNA, microRNA, siRNA, or other means known in the art. For instance, shRNAs according to SEQ ID NOS: 1, 2, 3, or 4, or variants thereof can be used in the disclosed constructs and methods, although this example is not limiting. The coding regions for RNAs to decrease expression of FDPS and FT and the coding regions of cytokine and chemokines may be in the same construct or on different constructs.

The classical approach for the production of recombinant polypeptides or gene regulatory molecules including small RNA is the use of stable expression constructs. These constructs are based upon chromosomal integration of a transduced expression plasmid (or at least a portion thereof) into the genome of the host cell, short-duration plasmid transfection, or non-integrating viral vectors also with limited half-life. The sites of gene integration are generally random, and the number and ratio of genes integrating at any particular site are often unpredictable; likewise, non-integrating plasmids or viral vectors also generate nuclear DNA but these species usually lack sequences required for DNA replication and continuous maintenance. Thus, constructs

that rely on chromosomal integration result in permanent maintenance of the recombinant gene that may exceed the therapeutic interval.

An alternative to stable expression constructs for gene expression are transient expression constructs. The expression of the latter gene expression construct is based on non-integrated plasmids, and hence the expression is typically lost as the cell undergoes division or the plasmid vectors are destroyed by endogenous nucleases.

The disclosed constructs are preferably episomal constructs that are transiently expressed. Episomal constructs are degraded or diluted over time such that they do not make permanent changes to a subject's genome, nor are they incorporated into the chromosome of a target cell. The process of episomal replication typically incorporates both host cell replication machinery and viral trans-acting factors.

Avoiding chromosomal integration reduces certain barriers to in vivo gene delivery. However, even integration-defective constructs can have a background frequency of integration, and any DNA molecule can find rare homologs to recombine with host sequences; but these rates of integration are exceptionally rare and generally not clinically significant.

Thus, in some embodiments, the disclosed vectors support active gene and/or small RNA delivery over a period of about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 11, or about 12 weeks. In some embodiments, the disclosed vectors support active gene and/or small RNA delivery over a period of about 1 month, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 12 months, or longer. Any combination of these time periods can also be used in the methods of the invention, e.g., 1 month and 1 week, or 3 months and 2 weeks.

However, in some embodiments, the constructs comprise integrating elements that depend on a retroviral integrase gene, such that the construct becomes integrated into the subject's chromosome. Retrotransposition and transposition are additional examples of mechanisms whereby mobile genetic elements become integrated or inserted into the chromosome. Plasmids may become integrated into the chromosome by recombination, and gene editing technologies including CRISPR and TALEN utilize guide RNA sequences and alter chromosomal loci by gene conversion mechanisms.

Constructs may comprise specific promoters for expressing cytokines involved in the maintenance of GD T cells (i.e., IL-2, IL-7, IL-17, and IL-15). For example, promoters that may be incorporated into the disclosed constructs include but are not limited to TATA-box promoters, CpG-box promoters, CCAAT-box promoters, TTGACA-box promoters, BRE-box promoters, INR-box promoters, AT-based promoters, CG-based promoters, ATCG-compact promoters, ATCG-balanced promoters, ATCG-middle promoters, ATCG-less promoters, AT-less promoters, CG-less promoters, AT-spike promoters, and CG-spike promoters. See Gagnic and Ionescu-Tirgoviste, Eukaryotic genomes may exhibit up to 10 generic classes of gene promoters, BMC GENOMICS 13:512 (2012).

Therapeutic Vectors

The construct can be delivered via known transfection and/or transduction vectors, including but not limited to lentiviral vectors, gamma-retroviruses, adeno-associated virus, poxvirus, herpesvirus vectors, protein and/or lipid complexes, liposomes, micelles, and the like.

Viral vectors can be preferentially targeted to cell types that are useful for the disclosed methods (i.e., tumor cells or

myeloid cells). Viral vectors can be used to transduce genes into target cells owing to specific virus envelope-host cell receptor interactions and viral mechanisms for gene expression. As a result, viral vectors have been used as vehicles for the transfer of genes into many different cell types including whole embryos, fertilized eggs, isolated tissue samples, tissue targets in situ, and cultured cell lines. The ability to introduce and express foreign genes in a cell is useful for the study of gene expression, and the elucidation of cell lineages as well as providing the potential for therapeutic interventions such as gene therapy, somatic cell reprogramming of induced pluripotent stem cells, and various types of immunotherapy. Viral components from viruses like Papovaviridae (e.g. bovine papillomavirus or BPV) or Herpesviridae (e.g. Epstein Barr Virus or EBV) or Hepadnaviridae (e.g. Hepatitis B Virus or HBV) or pox vectors including vaccinia may be used in the disclosed vectors.

Lentiviral vectors are a preferred type of vector for the disclosed compositions and methods, although the disclosure is not specifically limited to lentiviral vectors. Lentivirus is a genus of viruses that can deliver a significant amount of viral nucleic acid into a host cell. Lentiviruses are characterized as having a unique ability to infect/transduce non-dividing cells, and following transduction, lentiviruses integrate their nucleic acid into the host cell's chromosomes.

Infectious lentiviruses have three main genes coding for the virulence proteins gag, pol, and env, and two regulatory genes including tat and rev. Depending on the specific serotype and virus, there may be additional accessory genes that code for proteins involved in regulation, synthesis, and/or processing viral nucleic acids and other replicative functions.

Moreover, lentiviruses contain long terminal repeat (LTR) regions, which may be approximately 600 nt long. LTRs may be segmented into U3, R, and U5 regions. LTRs can mediate integration of retroviral DNA into the host chromosome via the action of integrase. Alternatively, without functioning integrase, the LTRs may be used to circularize the viral nucleic acid.

Viral proteins involved in early stages of lentivirus replication include reverse transcriptase and integrase. Reverse transcriptase is the virally encoded, RNA-dependent DNA polymerase. The enzyme uses a viral RNA genome as a template for the synthesis of a complementary DNA copy. Reverse transcriptase also has RNaseH activity for destruction of the RNA-template. Integrase binds both the viral cDNA generated by reverse transcriptase and the host DNA. Integrase processes the LTR before inserting the viral genome into the host DNA. Tat acts as a trans-activator during transcription to enhance initiation and elongation. The rev responsive element acts post-transcriptionally, regulating mRNA splicing and transport to the cytoplasm.

Viral vectors, in general, comprise glycoproteins and the various glycoproteins may provide specific affinities. For instance, VSV-G peptides can increase transfection into myeloid cells. Alternatively, viral vectors can also have targeting moieties, such as antibodies, attached to their shell peptides. Targeting antibodies can be specific for antigens that are overexpressed on a tumor, for instance, like HER-2, PSA, CEA, M2-PK, and CA19-9.

Other viral vector specificities are also known in the art and can be used to target particular populations of cells. For example, poxvirus vectors target to macrophages and dendritic cells.

Lentiviral Vector System

A lentiviral virion (particle) is expressed by a vector system encoding the necessary viral proteins to produce a

virion (viral particle). There is at least one vector containing a nucleic acid sequence encoding the lentiviral pol proteins necessary for reverse transcription and integration, operably linked to a promoter. In another embodiment, the pol proteins are expressed by multiple vectors. There is also a vector containing a nucleic acid sequence encoding the lentiviral gag proteins necessary for forming a viral capsid operably linked to a promoter. In an embodiment, this gag nucleic acid sequence is on a separate vector than at least some of the pol nucleic acid sequence. In another embodiment, the gag nucleic acid is on a separate vector from all the pol nucleic acid sequences that encode pol proteins.

Numerous modifications can be made to the vectors, which are used to create the particles to further minimize the chance of obtaining wild type revertants. These include, but are not limited to deletions of the U3 region of the LTR, tat deletions and matrix (MA) deletions.

The gag, pol and env vector(s) do not contain nucleotides from the lentiviral genome that package lentiviral RNA, referred to as the lentiviral packaging sequence.

The vector(s) forming the particle preferably do not contain a nucleic acid sequence from the lentiviral genome that expresses an envelope protein. Preferably, a separate vector that contains a nucleic acid sequence encoding an envelope protein operably linked to a promoter is used. This env vector also does not contain a lentiviral packaging sequence. In one embodiment the env nucleic acid sequence encodes a lentiviral envelope protein.

In another embodiment the envelope protein is not from the lentivirus, but from a different virus. The resultant particle is referred to as a pseudotyped particle. By appropriate selection of envelopes one can "infect" virtually any cell. For example, one can use an env gene that encodes an envelope protein that targets an endocytic compartment such as that of the influenza virus, VSV-G, alpha viruses (Semliki forest virus, Sindbis virus), arenaviruses (lymphocytic choriomeningitis virus), flaviviruses (tick-borne encephalitis virus, Dengue virus, hepatitis C virus, GB virus), rhabdoviruses (vesicular stomatitis virus, rabies virus), paramyxoviruses (mumps or measles) and orthomyxoviruses (influenza virus). Other envelopes that can preferably be used include those from Moloney Leukemia Virus such as MLV-E, MLV-A and GALV. These latter envelopes are particularly preferred where the host cell is a primary cell. Other envelope proteins can be selected depending upon the desired host cell. For example, targeting specific receptors such as a dopamine receptor can be used for brain delivery. Another target can be vascular endothelium. These cells can be targeted using a filovirus envelope. For example, the GP of Ebola, which by post-transcriptional modification become the GP, and GP₂ glycoproteins. In another embodiment, one can use different lentiviral capsids with a pseudotyped envelope (for example, FIV or SHIV [U.S. Pat. No. 5,654, 195]). A SHIV pseudotyped vector can readily be used in animal models such as monkeys.

As detailed herein, a lentiviral vector system typically includes at least one helper plasmid comprising at least one of a gag, pol, or rev gene. Each of the gag, pol and rev genes may be provided on individual plasmids, or one or more genes may be provided together on the same plasmid. In one embodiment, the gag, pol, and rev genes are provided on the same plasmid (e.g., FIG. 2). In another embodiment, the gag and pol genes are provided on a first plasmid and the rev gene is provided on a second plasmid (e.g., FIG. 3). Accordingly, both 3-vector and 4-vector systems can be used to produce a lentivirus as described in the Examples section and elsewhere herein. The therapeutic vector, the envelope

plasmid and at least one helper plasmid are transfected into a packaging cell line. A non-limiting example of a packaging cell line is the 293T/17 HEK cell line. When the therapeutic vector, the envelope plasmid, and at least one helper plasmid are transfected into the packaging cell line, a lentiviral particle is ultimately produced.

In another aspect, a lentiviral vector system for expressing a lentiviral particle is disclosed. The system includes a lentiviral vector as described herein; an envelope plasmid for expressing an envelope protein optimized for infecting a cell; and at least one helper plasmid for expressing gag, pol, and rev genes, wherein when the lentiviral vector, the envelope plasmid, and the at least one helper plasmid are transfected into a packaging cell line, a lentiviral particle is produced by the packaging cell line, wherein the lentiviral particle is capable of inhibiting production of chemokine receptor CCR5 or targeting an HIV RNA sequence.

In another aspect, and as detailed in FIG. 2, the lentiviral vector, which is also referred to herein as a therapeutic vector, can include the following elements: hybrid 5' long terminal repeat (RSV/5' LTR) (SEQ ID NOS: 11-12), Psi sequence (RNA packaging site) (SEQ ID NO: 13), RRE (Rev-response element) (SEQ ID NO: 14), cPPT (polypurine tract) (SEQ ID NO: 15), H1 promoter (SEQ ID NO: 16), FDPS shRNA (SEQ ID NOS: 1, 2, 3, 4), Woodchuck Post-Transcriptional Regulatory Element (WPPE) (SEQ ID NO: 17), and 3' Delta LTR (SEQ ID NO: 18). In another aspect, sequence variation, by way of substitution, deletion, addition, or mutation can be used to modify the sequences references herein.

In another aspect, and as detailed herein, a helper plasmid has been designed to include the following elements: CAG promoter (SEQ ID NO: 19); HIV component gag (SEQ ID NO: 20); HIV component pol (SEQ ID NO: 21); HIV Int (SEQ ID NO: 22); HIV RRE (SEQ ID NO: 23); and HIV Rev (SEQ ID NO: 24). In another aspect, the helper plasmid may be modified to include a first helper plasmid for expressing the gag and pol genes, and a second and separate plasmid for expressing the rev gene. In another aspect, sequence variation, by way of substitution, deletion, addition, or mutation can be used to modify the sequences references herein.

In another aspect, and as detailed herein, an envelope plasmid has been designed to include the following elements being from left to right: RNA polymerase II promoter (CMV) (SEQ ID NO: 25) and vesicular stomatitis virus G glycoprotein (VSV-G) (SEQ ID NO: 26). In another aspect, sequence variation, by way of substitution, deletion, addition, or mutation can be used to modify the sequences references herein.

In another aspect, the plasmids used for lentiviral packaging can be modified with similar elements and the intron sequences could potentially be removed without loss of vector function. For example, the following elements can replace similar elements in the plasmids that comprise the packaging system: Elongation Factor-1 (EF-1), phosphoglycerate kinase (PGK), and ubiquitin C (UbC) promoters can replace the CMV or CAG promoter. SV40 poly A and bGH poly A can replace the rabbit beta globin poly A. The HIV sequences in the helper plasmid can be constructed from different HIV strains or clades. The VSV-G glycoprotein can be substituted with membrane glycoproteins from feline endogenous virus (RD114), gibbon ape leukemia virus (GALV), Rabies (FUG), lymphocytic choriomeningitis virus (LCMV), influenza A fowl plague virus (FPV), Ross River alphavirus (RRV), murine leukemia virus 10A1 (MLV), or Ebola virus (EboV).

Of note, lentiviral packaging systems can be acquired commercially (e.g., Lenti-vpak packaging kit from OriGene Technologies, Inc., Rockville, Md.), and can also be designed as described herein. Moreover, it is within the skill of a person skilled in the art to substitute or modify aspects of a lentiviral packaging system to improve any number of relevant factors, including the production efficiency of a lentiviral particle.

Doses and Dosage Forms

The disclosed vectors allow for short, medium, or long-term expression of genes or sequences of interest and episomal maintenance of the disclosed vectors. Accordingly, dosing regimens may vary based upon the condition being treated and the method of administration.

In one embodiment, transduction vectors may be administered to a subject in need in varying doses. Specifically, a subject may be administered about $\geq 10^6$ infectious doses (where 1 dose is needed on average to transduce 1 target cell). More specifically, a subject may be administered about $\geq 10^7$, about $\geq 10^8$, about $\geq 10^9$, or about $\geq 10^{10}$ infectious doses, or any number of doses in-between these values. Upper limits of transduction vector dosing will be determined for each disease indication and will depend on toxicity/safety profiles for each individual product or product lot.

Additionally, a vector of the present disclosure may be administered periodically, such as once or twice a day, or any other suitable time period. For example, vectors may be administered to a subject in need once a week, once every other week, once every three weeks, once a month, every other month, every three months, every six months, every nine months, once a year, every eighteen months, every two years, every thirty months, or every three years.

In one embodiment, the disclosed vectors are administered as a pharmaceutical composition. In some embodiments, the pharmaceutical composition comprising the disclosed vectors can be formulated in a wide variety of dosage forms, including but not limited to nasal, pulmonary, oral, topical, or parenteral dosage forms for clinical application. Each of the dosage forms can comprise various solubilizing agents, disintegrating agents, surfactants, fillers, thickeners, binders, diluents such as wetting agents or other pharmaceutically acceptable excipients. The pharmaceutical composition comprising a vector can also be formulated for injection, insufflation, infusion, or intradermal exposure. For instance, an injectable formulation may comprise the disclosed vectors in an aqueous or non-aqueous solution at a suitable pH and tonicity.

The disclosed vectors may be administered to a subject via direct injection into a tumor site or at a site of infection. In some embodiments, the vectors can be administered systemically. In some embodiments, the vectors can be administered via guided cannulation to tissues immediately surrounding the sites of tumor or infection.

The disclosed vector compositions can be administered using any pharmaceutically acceptable method, such as intranasal, buccal, sublingual, oral, rectal, ocular, parenteral (intravenously, intradermally, intramuscularly, subcutaneously, intraperitoneally), pulmonary, intravaginal, locally administered, topically administered, topically administered after scarification, mucosally administered, via an aerosol, in semi-solid media such as agarose or gelatin, or via a buccal or nasal spray formulation.

Further, the disclosed vector compositions can be formulated into any pharmaceutically acceptable dosage form, such as a solid dosage form, tablet, pill, lozenge, capsule, liquid dispersion, gel, aerosol, pulmonary aerosol, nasal

aerosol, ointment, cream, semi-solid dosage form, a solution, an emulsion, and a suspension. Further, the composition may be a controlled release formulation, sustained release formulation, immediate release formulation, or any combination thereof. Further, the composition may be a transdermal delivery system.

In some embodiments, the pharmaceutical composition comprising a vector can be formulated in a solid dosage form for oral administration, and the solid dosage form can be powders, granules, capsules, tablets or pills. In some embodiments, the solid dosage form can include one or more excipients such as calcium carbonate, starch, sucrose, lactose, microcrystalline cellulose or gelatin. In addition, the solid dosage form can include, in addition to the excipients, a lubricant such as talc or magnesium stearate. In some embodiments, the oral dosage form can be immediate release, or a modified release form. Modified release dosage forms include controlled or extended release, enteric release, and the like. The excipients used in the modified release dosage forms are commonly known to a person of ordinary skill in the art.

In a further embodiment, the pharmaceutical composition comprising a vector can be formulated as a sublingual or buccal dosage form. Such dosage forms comprise sublingual tablets or solution compositions that are administered under the tongue and buccal tablets that are placed between the cheek and gum.

In some embodiments, the pharmaceutical composition comprising a vector can be formulated as a nasal dosage form. Such dosage forms of the present invention comprise solution, suspension, and gel compositions for nasal delivery.

In some embodiments, the pharmaceutical composition comprising a vector can be formulated in a liquid dosage form for oral administration, such as suspensions, emulsions or syrups. In some embodiments, the liquid dosage form can include, in addition to commonly used simple diluents such as water and liquid paraffin, various excipients such as humectants, sweeteners, aromatics or preservatives. In particular embodiments, the composition comprising vectors can be formulated to be suitable for administration to a pediatric patient.

In some embodiment, the pharmaceutical composition can be formulated in a dosage form for parenteral administration, such as sterile aqueous solutions, suspensions, emulsions, non-aqueous solutions or suppositories. In some embodiments, the solutions or suspensions can include propylene glycol, polyethylene glycol, vegetable oils such as olive oil or injectable esters such as ethyl oleate.

The dosage of the pharmaceutical composition can vary depending on the patient's weight, age, gender, administration time and mode, excretion rate, and the severity of disease.

In some embodiments, the treatment of cancer is accomplished by guided direct injection of the disclosed vector constructs into tumors, using needle, or intravascular cannulation. In some embodiments, the disclosed vectors are administered into the cerebrospinal fluid, blood or lymphatic circulation by venous or arterial cannulation or injection, intradermal delivery, intramuscular delivery or injection into a draining organ near the site of disease.

The following examples are given to illustrate the present invention. It should be understood, however, that the invention is not to be limited to the specific conditions or details

described in these examples. All printed publications referenced herein are specifically incorporated by reference.

EXAMPLES

Example 1: Development of a Lentiviral Vector System

A lentiviral vector system was developed as summarized in FIG. 4 (circularized form). Lentiviral particles were produced in 293T/17 HEK cells (purchased from American Type Culture Collection, Manassas, Va.) following transfection with the therapeutic vector, the envelope plasmid, and the helper plasmid. The transfection of 293T/17 HEK cells, which produced functional viral particles, employed the reagent Poly(ethylenimine) (PEI) to increase the efficiency of plasmid DNA uptake. The plasmids and DNA were initially added separately in culture medium without serum in a ratio of 3:1 (mass ratio of PEI to DNA). After 2-3 days, cell medium was collected and lentiviral particles were purified by high-speed centrifugation and/or filtration followed by anion-exchange chromatography. The concentration of lentiviral particles can be expressed in terms of transducing units/ml (TU/ml). The determination of TU was accomplished by measuring HIV p24 levels in culture fluids (p24 protein is incorporated into lentiviral particles), measuring the number of viral DNA copies per cell by quantitative PCR, or by infecting cells and using light (if the vectors encode luciferase or fluorescent protein markers).

As mentioned above, a 3-vector system (i.e., a 2-vector lentiviral packaging system) was designed for the production of lentiviral particles. A schematic of the 3-vector system is shown in FIG. 2. Briefly, and with reference to FIG. 2, the top-most vector is a helper plasmid, which, in this case, includes Rev. The vector appearing in the middle of FIG. 2 is the envelope plasmid. The bottom-most vector is the therapeutic vector, as described herein.

Referring more specifically to FIG. 2, the Helper plus Rev plasmid includes a CAG enhancer (SEQ ID NO: 27); a CAG promoter (SEQ ID NO: 19); a chicken beta actin intron (SEQ ID NO: 28); a HIV gag (SEQ ID NO: 20); a HIV Pol (SEQ ID NO: 21); a HIV Int (SEQ ID NO: 22); a HIV RRE (SEQ ID NO: 23); a HIV Rev (SEQ ID NO: 24); and a rabbit beta globin poly A (SEQ ID NO: 29).

The Envelope plasmid includes a CMV promoter (SEQ ID NO: 25); a beta globin intron (SEQ ID NO: 30); a VSV-G (SEQ ID NO: 28); and a rabbit beta globin poly A (SEQ ID NO: 31).

Synthesis of a 2-Vector Lentiviral Packaging System Including Helper (Plus Rev) and Envelope Plasmids. Materials and Methods:

Construction of the helper plasmid: The helper plasmid was constructed by initial PCR amplification of a DNA fragment from the pNL4-3 HIV plasmid (NIH Aids Reagent Program) containing Gag, Pol, and Integrase genes. Primers were designed to amplify the fragment with EcoRI and NotI restriction sites which could be used to insert at the same sites in the pCDNA3 plasmid (Invitrogen). The forward primer was (5'-TAAGCAGAATTC ATGAATTTGCCAG-GAAGAT-3') (SEQ ID NO: 32) and reverse primer was (5'-CCATACAATGAATGGACACTAGGCGGCCGCAC-GAAT-3') (SEQ ID NO: 33).

The sequence for the Gag, Pol, Integrase fragment was as follows:

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(SEQ ID NO: 34)

GAATTCATGAATTGCCAGGAAGATGGAACCAAAATGATAGGGGAAT

TGGAGGTTTTATCAAAGTAAGACAGTATGATCAGATACTCATAGAAATCT

GCGGACATAAAGCTATAGGTACAGTATTAGTAGGACCTACACCTGTCAAC

ATAATTGGAAGAAATCTGTTGACTCAGATTGGCTGCACCTTTAAATTTTCC

CATTAGTCTCTATTGAGACTGTACCAGTAAAATTAAAGCCAGGAATGGATG

GCCCCAAAGTTAAACAATGGCCATTGACAGAAGAAAAATAAAGCATT

GTAGAAATTTGTACAGAAATGGAAGGAAGAAAAATTTCAAAATTTGG

GCCTGAAATCCATACAATACTCCAGTATTTGCCATAAGAAAAAAGACA

GTACTAAATGGAGAAAATTAGTAGATTTAGAGAACTTAATAAGAGAAGT

CAAGATTTCTGGGAAGTTCAATTAGGAATACCACATCCTGCAGGGTTAA

ACAGAAAAATCAGTAACAGTACTGGATGTGGGCGATGCATATTTTTCAG

TTCCCTTAGATAAAGACTTCAGGAAGTATATGCATTTACCATACCTAGT

ATAAACAAATGAGACACAGGGATTAGATATCAGTACAATGTGCTTCCACA

GGGATGGAAGGATCACCAGCAATATTCAGTGTAGCATGACAAAAATCT

TAGAGCCTTTTAGAAAACAAATCCAGACATAGTCATCTATCAATACATG

GATGATTTGTATGTAGGATCTGACTTAGAAATAGGGCAGCATAGAACAAA

AATAGAGGAAGTGAAGACACATCTGTTGAGGTGGGGATTACCACACCAG

ACAAAAACATCAGAAAGAACCTCCATTCCCTTGGATGGGTTATGAATC

CATCTGATAAATGGACAGTACAGCCTATAGTGCTGCCAGAAAAGGACAG

CTGGACTGTCAATGACATACAGAAATTAGTGGGAAATGAATTGGGCAA

GTCAGATTTATGCAGGGATTAAAGTAAGCAATTATGTAACCTCTTAGG

GGAACCAAGCACTAACAGAAGTAGTACCATAACAGAAGAGCAGAGCT

AGAACTGGCAGAAAACAGGGAGATTCTAAAAGAACCGGTACATGGAGTGT

ATTATGACCCATCAAAGACTTAATAGCAGAAATACAGAAGCAGGGGCAA

GGCCAATGGACATATCAAAATTTATCAAGAGCCATTTAAAAATCTGAAAC

AGGAAAGTATGCAAGAATGAAGGGTGCCACACTAATGATGTGAAACAAT

TAACAGAGGCAGTACAAAAATAGCCACAGAAGCATAGTAATATGGGGA

AAGACTCTTAAATTTAAATTACCCATACAAAAGGAACATGGGAAGCATG

GTGGACAGAGTATTGGCAAGCCACCTGGATTCTGAGTGGGAGTTTGTCA

ATACCCCTCCCTTAGTGAAGTTATGGTACCAGTTAGAGAAGAACCATA

ATAGGAGCAGAAACTTTCTATGTAGATGGGCAGCCAATAGGGAACTAA

ATTAGGAAAAGCAGGATATGTAATGACAGAGGAAGACAAAAGTTGTCC

CCCTAACGGACACAACAAATCAGAAGACTGAGTTACAAGCAATTCATCTA

GCTTTGCAGGATTCGGGATTAGAAGTAAACATAGTGACAGACTCACAATA

TGCATTGGGAATCATTCAAGCACAAACCAGATAAGAGTGAATCAGAGTTAG

TCAGTCAAATAATAGAGCAGTTAATAAAAAAGGAAAAAGTCTACCTGGCA

TGGGTACCAGCACAAAGGAATTGGAGGAATGAACAAGTAGATAAAAT

GGTCAGTGCTGGAATCAGGAAAGTACTATTTTATAGTGAATAGATAAAG

CCCAAGAAGACATGAGAAATATCAGTAATTGGAGAGCAATGGCTAGT

GATTTTAACCTACCACCTGTAGTAGCAAAAGAAATAGTAGCCAGCTGTGA

32

-continued

TAAATGTCTAGCTAAAAGGGGAAGCCATGCATGGACAAGTAGACTGTAGCC

CAGGAATATGGCAGCTAGATTGTACACATTTAGAAGGAAAAGTTATCTTG

5 GTAGCAGTTTCATGTAGCCAGTGGATATATAGAAGCAGAAGTAATCCAGC

AGAGACAGGGCAAGAAAACAGCATACTTCTCTTAAAATTAGCAGGAAGAT

GGCCAGTAAAAACAGTACATACAGACAATGGCAGCAATTTACCAGTACT

10 ACAGTTAAGGCCCGCTGTTGGTGGGCGGGGATCAAGCAGGAATTTGGCAT

TCCCTACAATCCCCAAAGTCAAGGAGTAATAGAATCTATGAATAAAGAAT

TAAAGAAAATTATAGGACAGGTAAGAGATCAGGCTGAACATCTTAAGACA

15 GCAGTACAAATGGCAGTATTCATCCACAATTTTAAAGAAAAGGGGGAT

TGGGGGGTACAGTGCAGGGGAAAGAATAGTAGACATAATAGCAACAGACA

TACAACTAAAGAATTACAAAACAAATTACAAAAATTCAAAATTTTCGG

20 GTTTATTACAGGGACAGCAGAGATCCAGTTTGGAAAGGACCAGCAAAGCT

CCTCTGGAAGGTGAAGGGCAGTAGTAATACAAGATAATAGTGACATAA

AAGTAGTGCCAAGAAGAAAAGCAAAGATCATCAGGGATTATGGAACACG

ATGGCAGGTGATGATTGTGTGGCAAGTAGACAGGATGAGGATTAA

25 Next, a DNA fragment containing the Rev, RRE, and

rabbit beta globin poly A sequence with XbaI and XmaI

flanking restriction sites was synthesized by MWG Operon.

The DNA fragment was then inserted into the plasmid at the

30 XbaI and XmaI restriction sites The DNA sequence was as

follows:

(SEQ ID NO: 35)

35 TCTAGAATGGCAGGAAGAAGCGGAGACAGCGACGAAGAGCTCATCAGAAC

AGTCAGACTCATCAAGCTTCTCTATCAAGCAACCCACCTCCCAATCCCG

AGGGGACCCGACAGGCCGAAGGAATAGAAGAAGAAGTGGAGAGAGAGA

40 CAGAGACAGATCCATTTCGATTAGTGAACGGATCCTTGGCACTTATCTGGG

ACGATCTGCGGAGCCTGTGCCTCTCAGCTACCACCGCTTGAGAGACTTA

CTCTTGATTGTAACGAGGATTGTGGAACCTCTGGGACGCAGGGGGTGGGA

AGCCCTCAAATATTTGGTGAATCTCCTACAATATTGGAGTCAGAGCTAA

45 AGAATAGAGGAGCTTTGTTCTTGGGTCTTGGGAGCAGCAGGAAGCACT

ATGGGCGCAGCGTCAATGACGCTGACGGTACAGGCCAGACAATTATTGTC

TGGTATAGTGCAGCAGCAGAACAAATTTGCTGAGGGCTATTGAGCGCAAC

50 AGCATCTGTTGCAACTCACAGTCTGGGGCATCAAGCAGCTCCAGGCAAGA

ATCCTGGCTGTGAAAGATACCTAAAGGATCAACAGCTCCTAGATCTTTT

TCCCTCTGCCAAAAATTTAGGGACATCATGAAGCCCTTGAGCATCTGA

55 CTTCTGGCTAATAAAGGAAATTTATTTTCATTGCAATAGTGTGTTGGAAT

TTTTTGTGTCTCTCACTCGGAAGGACATATGGGAGGGCAATCATTTAAA

ACATCAGAAATGAGTATTTGGTTTAGAGTTTGGCAACATATGCCATATGCT

60 GGCTGCCATGAACAAAGGTGGCTATAAAGAGGTATCAGTATATGAAACA

GCCCCCTGCTGTCCATTCTTATCCATAGAAAAGCCTTGACTTGAGGTT

AGATTTTTTTTATATTTTGTGTTGTTATTTTTTCTTTTAACTCCCTA

65 AAATTTTCTTACATGTTTACTAGCCAGATTTTCTCTCTCTCTGACT

33

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ACTCCCAGTCATAGCTGTCCCTCTTCTCTTATGAAGATCCCTCGACCTGC
 AGCCCAAGCTTGGCGTAATCATGGTCATAGCTGTTCTGTGTGAAATTG
 TTATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTA
 AAGCCTGGGGTGCTTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGC
 TCACTGCCCGCTTTCAGTCGGGAAACCTGTCGTGCCAGCGGATCCGCAT
 CTCAATTAGTCAGCAACCATAGTCCCGCCCCTAACCTCCGCCATCCCGCC
 CCTAACTCCGCCAGTTCCGCCCATTTCTCCGCCCATGGCTGACTAATTT
 TTTTATTATGACAGAGCCGAGGCGCCTCGGCCTCTGAGCTATTCCAG
 AAGTAGTGAGGAGGCTTTTGGAGGCCTAGGCTTTTGCAAAAGCTAAC
 TTGTTTATTGACAGCTTATAATGGTTACAAATAAGCAATAGCATCACAAA
 TTTCACAAATAAGCATTTTTTCTACTGCATTCTAGTTGTGGTTTGTCCA
 AACTCATCAATGTATCTTATCAGCGGCCGCCCGGG

Finally, the CMV promoter of pCDNA3.1 was replaced with the CAG enhancer/promoter plus a chicken beta actin intron sequence. A DNA fragment containing the CAG enhancer/promoter/intron sequence with MluI and EcoRI flanking restriction sites was synthesized by MWG Operon. The DNA fragment was then inserted into the plasmid at the MluI and EcoRI restriction sites. The DNA sequence was as follows:

(SEQ ID NO: 36)

ACGCGTTAGTTATTAATAGTAATCAATTACGGGTCATTAGTTTCATAGCC
 CATATATGGAGTTCCCGTTACATAAATTACGGTAAATGGCCCGCTGGC
 TGACCGCCCAACGACCCCCGCCATTGACGTCAATAATGACGTATGTTCC
 CATAGTAACGCCAATAGGACTTTCATTGACGTCAATGGGTGACTATT
 TACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGT
 ACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCTGGCATTATGC
 CCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTAT
 TAGTCATCGCTATTACCATGGTTCGAGGTGAGCCCCACGTTCTGCTTCAC
 TCTCCCCATCTCCCCCCCCCTCCCAACCCCAATTTTGTATTTATTTATT
 TTTAATTATTTTGTGACGATGGGGGCGGGGGGGGGGGCGCGCGCC
 AGGCGGGGCGGGGCGGGGCGAGGGGCGGGGCGAGGCGAGAGGTG
 CGGCGGCAGCCAATCAGAGCGGCGCGCTCCGAAAGTTTCTTTTATGGCG
 AGGCGGCGGCGGCGGCGGCTTATAAAAAGCGAAGCGCGGCGGGCGGG
 AGTCGCTGCGTTGCTTCGCCCGGTGCCCGCTCCGCGCGCCTCGCGCC
 GCCCCCGCGCTCTGACTGACCGGTTACTCCACAGGTGAGCGGGCGG
 GACGCGCCTTCTCCTCCGGGTGTAATTAGCGCTTGGTTAATGACGGCT
 CGTTTCTTTTCTGTGGCTGCGTGAAAGCCTTAAAGGGCTCCGGGAGGGCC
 CTTTGTGCGGGGGGAGCGGCTCGGGGGTGCCTGCGTGTGTGTGCGT
 GGGGAGCGCCGCTGCGGCCCGCTGCCCCGCGCTGTGAGCGCTGCGG
 GCGCGGCGCGGGGCTTTGTGCGCTCCGCGTGTGCGGAGGGGAGCGCGG
 CGGGGGCGGTGCCCGCGTGCAGGGGGGTGCGAGGGGAACAAAGGCTG
 CGTGCAGGGTGTGCGTGGGGGGTGAGCAGGGGGTGTGGCGCGCGG

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TCGGGCTGTAACCCCCCTGCACCCCCCTCCCGAGTTGCTGAGCACGG
 CCCGGCTTCGGGTGCGGGGCTCCGTGCGGGCGTGCGCGGGGCTCGCCG
 5 TGCCGGGCGGGGGTGGCGGCAGGTGGGGTGCCGGGCGGGGCGGGCCG
 CCTCGGGCGGGGAGGGCTCGGGGAGGGGCGCGGCGGCCCGGAGCGCC
 GCGGCTGTCGAGGCGCGGCGAGCCGAGCCATTGCTTTTATGGTAATC
 10 GTGCGAGAGGGCGCAGGACTTCCTTTGTCCCAAATCTGGCGGAGCCGAA
 ATCTGGGAGGCGCGCCGCACCCCCCTCTAGCGGGCGGGGCGAAGCGGTG
 CGGCGCGCGCAGGAAGAAATGGGCGGGGAGGGCTTCGTGCGTCGCCG
 15 GCGCGCGTCCCTTCTCCATCTCCAGCCTCGGGGTGCGCGAGGGGACG
 GCTGCTTCGGGGGGGACGGGGCAGGGCGGGGTTCGGCTTCTGGCGTGTG
 ACCGGCGGAATTC

Construction of the VSV-G Envelope Plasmid:

The vesicular stomatitis Indiana virus glycoprotein (VSV-G) sequence was synthesized by MWG Operon with flanking EcoRI restriction sites. The DNA fragment was then inserted into the pCDNA3.1 plasmid (Invitrogen) at the EcoRI restriction site and the correct orientation was determined by sequencing using a CMV specific primer. The DNA sequence was as follows:

(SEQ ID NO: 37)

GAATTCATGAAGTGCCTTTTGTACTTAGCCTTTTATTATTGGGGTGAA
 TTGCAAGTTCACCATAGTTTTCACACAACCAAAAGGAACTGGAAAA
 ATGTTCTTCTAATTACCATATTGCCCCGTCAGCTCAGATTTAAATTGG
 35 CATAATGACTTAATAGGCACAGCCTTACAAGTCAAAATGCCCAAGAGTCA
 CAAGGCTATTCAAGCAGACGGTTGGATGTGTCATGCTTCCAAATGGGTCA
 CTACTTGTGATTTCCGCTGGTATGGACCGAAGTATATAACATTCATC
 CGATCCTTCACTCCATCTGTAGAACAAATGCAAGGAAAGCATTGAACAAAC
 40 GAAACAAGGAACCTGGCTGAATCCAGGCTTCCCTCCTCAAAGTTGTGGAT
 ATGCAACTGTGACGATGCCGAAGCAGTGATTGTCCAGGTGACTCCTCAC
 CATGTGCTGGTTGATGAATACACAGGAGAATGGGTTGATTACAGTTTAT
 45 CAACGGAATAATGACGAATTACATATGCCCACTGTCCATAACTCTACAA
 CCTGGCATTCTGACTATAAGGTCAAAGGCTATGTGATTCTAACCTCATT
 TCCATGGACATCACCTTCTTCTCAGAGGACGAGAGCTATCATCCTGGG
 50 AAAGGAGGGCACAGGTTTCAAGTAATACTTTGCTTATGAACTGGAG
 GCAAGGCTGCAAAATGCAATACTGCAAGCATTGGGGAGTCAGACTCCCA
 TCAGGTGTCTGGTTCGAGATGGCTGATAAGGATCTCTTGTGTCAGCCAG
 55 ATTCCTGAATGCCAGAGGGTCAAGTATCTCTGCTCCATCTCAGACCT
 CAGTGGATGTAAGTCTAATTCAGGACGTTGAGAGGATCTTGGATTATTCC
 CTCTGCCAAGAACTGGAGCAAAATCAGAGCGGGTCTTCAATCTCTCC
 60 AGTGGATCTCAGCTATCTTGCTCCTAAAAACCGAAGCGGTCCTGCTT
 TCACCATAATCAATGGTACCCTAAAAATACTTTGAGACCAGATACATCAGA
 GTCGATATTGCTGCTCAATCCTCTCAAGAATGGTCGAATGATCAGTGG
 65 AACTACCACAGAAAGGGAACGTGGGATGACTGGGCACCATATGAAGACG

35

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TGGAAATTGGACCCAATGGAGTTCTGAGGACCAGTTCAGGATATAAGTTT
 CTTTATACATGATTGGACATGGTATGTTGGACTCCGATCTTCATCTTAG
 CTCAAAGGCTCAGGTGTTTGAACATCCTCACATTCAAGACGCTGCTTCGC
 AACTTCCTGATGATGAGAGTTTATTTTTTGGTGATACTGGGCTATCCAAA
 AATCCAATCGAGCTTGTAGAAGTTGGTTCAGTAGTTGGAAAAGCTCTAT
 TGCCCTCTTTTTCTTTATCATAGGGTTAATCATTGGACTATCTTGGTTC
 TCCGAGTTGGTATCCATCTTTGCATTAAATTAAAGCACACCAAGAAAAGA
 CAGATTATACAGACATAGAGATGAGAATTC

A 4-vector system (i.e., a 3-vector lentiviral packaging system) has also been designed and produced using the methods and materials described herein. A schematic of the 4-vector system is shown in FIG. 3. Briefly, and with reference to FIG. 3, the top-most vector is a helper plasmid, which, in this case, does not include Rev. The vector second from the top is a separate Rev plasmid. The vector second from the bottom is the envelope plasmid. The bottom-most vector is the previously described therapeutic vector.

Referring, in part, to FIG. 2, the Helper plasmid includes a CAG enhancer (SEQ ID NO: 27); a CAG promoter (SEQ ID NO: 19); a chicken beta actin intron (SEQ ID NO: 28); a HIV gag (SEQ ID NO: 20); a HIV Pol (SEQ ID NO: 21); a HIV Int (SEQ ID NO: 22); a HIV RRE (SEQ ID NO: 23); and a rabbit beta globin poly A (SEQ ID NO: 29).

The Rev plasmid includes a RSV promoter and a HIV Rev (SEQ ID NO: 38); and a rabbit beta globin poly A (SEQ ID NO: 29).

The Envelope plasmid includes a CMV promoter (SEQ ID NO: 25); a beta globin intron (SEQ ID NO: 30); a VSV-G (SEQ ID NO: 28); and a rabbit beta globin poly A (SEQ ID NO: 29).

Synthesis of a 3-Vector Lentiviral Packaging System Including Helper, Rev, and Envelope Plasmids.

Materials and Methods:

Construction of the Helper Plasmid without Rev:

The Helper plasmid without Rev was constructed by inserting a DNA fragment containing the RRE and rabbit beta globin poly A sequence. This sequence was synthesized by MWG Operon with flanking XbaI and XmaI restriction sites. The RRE/rabbit poly A beta globin sequence was then inserted into the Helper plasmid at the XbaI and XmaI restriction sites. The DNA sequence is as follows:

(SEQ ID NO: 65)
 TCTAGAAGGAGCTTTGTTCTTGGGTTCTTGGGAGCAGCAGGAAGCACTA
 TGGGCGCAGCGTCAATGACGCTGACGGTACAGGCCAGACAATTATTGTCT
 GGTATAGTGCAGCAGCAGAACAATTGCTGAGGGCTATTGAGGCGCAACA
 GCATCTGTGCAACTCACAGTCTGGGGCATCAAGCAGCTCCAGGCAAGAA
 TCCTGGCTGTGGAAAGATACCTAAAGGATCAACAGCTCCTAGATCTTTTT
 CCCTCTGCCAAAAATTATGGGGACATCATGAAGCCCTTGAGCATCTGAC
 TTCTGGCTAATAAAGGAAATTTATTTTCATTGCAATAGTGTGTTGGAATT
 TTTTGTGTCTCTCACTCGGAAGGACATATGGGAGGGCAAATCATTAAAA
 CATCAGAATGAGTATTGGTTTAGAGTTTGGCAACATATGCCATATGCTG
 GCTGCCATGAACAAGGTGGCTATAAAGAGGTCACTAGTATATGAACAG

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CCCCCTGCTGCCATTCTTATTCATAGAAAAGCCTTGACTTGAGGTTA
 GATTTTTTTTATATTTTGTGTTATTTTTTTCTTTAAACATCCCTAA
 5 AATTTTCCTTACATGTTTACTAGCCAGATTTTTCTCCTCTCCTGACTA
 CTCCAGTCATAGCTGTCCCTCTTCTTATGAAGATCCCTCGACCTGCA
 GCCCAAGCTTGGCGTAATCATGGTCATAGCTGTTTCTGTGTGAAATTGT
 10 TATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTA
 AGCCTGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCT
 CACTGCCCGCTTTCCAGTCGGGAAACCTGTGTCGACGCGGATCCGCATC
 15 TCAATTAGTCAGCAACCATAGTCCCGCCCCCTAAGTCCGCCCATCCGCC
 CTAAGTCCGCCAGTTCCGCCCATTTCTCGCCCCATGGCTGACTAATTTT
 TTTTATTTATGCAGAGGCCGAGGCCCTCGGCCCTCTGAGCTATTCCAGA
 AGTAGTGAGGAGGCTTTTTTGGAGGCTAGGCTTTTGCAAAAAGCTAACT
 20 TGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCACAAAT
 TTCACAAATAAAGCATTTTTTTTCACTGCATCTAGTTGTGGTTTGTCCAA
 ACTCATCAATGTATCTTATCACCCGGG

Construction of the Rev Plasmid:

The RSV promoter and HIV Rev sequence was synthesized as a single DNA fragment by MWG Operon with flanking MfeI and XbaI restriction sites. The DNA fragment was then inserted into the pCDNA3.1 plasmid (Invitrogen) at the MfeI and XbaI restriction sites in which the CMV promoter is replaced with the RSV promoter. The DNA sequence was as follows:

(SEQ ID NO: 38)
 CAATTGCGATGTACGGGCCAGATATACGCGTATCTGAGGGGACTAGGGTG
 TGTTTAGCGGAAAAGCGGGGCTTCGGTTGTACGCGTTAGGAGTCCCTC
 AGGATATAGTAGTTTCGCTTTTGCATAGGGAGGGGAAATGTAGTCTTAT
 40 GCAATACACTGTAGTCTTGCAACATGGTAACGATGAGTTAGCAACATGC
 CTTACAAGGAGAGAAAAAGCACCGTGCATGCCGATTGGTGGAGTAAGGT
 GGTACGATCGTGCCCTTATTAGGAAGGCAACAGACAGGCTGACATGGATT
 45 GGACGAACCACTGAATCCGCATTGCAGAGATAATTGTATTAAAGTGCCCT
 AGCTCGATACAATAAACGCCATTTGACCATTACCCACATTGGTGTGCACC
 TCCAAGCTCGAGCTCGTTTGTAGTAACCGTCAGATCGCTGGAGACGCCAT
 CCACGCTGTTTTTGACCTCCATAGAAAGACACGGGACCGATCCAGCCTCCC
 CTCGAAGCTAGCGATTAGGCATCTCCTATGGCAGGAAGAAGCGGAGACAG
 CGACGAAGAACTCCTCAAGGCAGTCAGACTCATCAAGTTTCTCTATCAAA
 55 GCAACCCACCTCCCAATCCCGAGGGGACCCGACAGGCCCGAAGGAATAGA
 AGAAGAAGGTGGAGAGAGAGACAGAGACAGATCCATTGATTAGTGAACG
 GATCCTTAGCACTTATCTGGGACGATCTGCGGAGCCTGTGCCTCTCAGC
 60 TACCACCGCTTGAGAGACTTACTCTTGATTGTAACGAGGATTGTGGAAC
 TCTGGGACGCGAGGGGTGGGAAGCCCTCAATATTGGTGAATCTCTAC
 AATATTGGAGTCAGGAGCTAAAGAATAGTCTAGA

The plasmids for the 2-vector and 3-vector packaging systems could be modified with similar elements and the intron sequences could potentially be removed without loss

of vector function. For example, the following elements could replace similar elements in the 2-vector and 3-vector packaging system:

Promoters: Elongation Factor-1 (EF-1) (SEQ ID NO: 39), phosphoglycerate kinase (PGK) (SEQ ID NO: 40), and ubiquitin C (UbC) (SEQ ID NO: 41) can replace the CMV (SEQ ID NO: 25) or CAG promoter (SEQ ID NO: 19). These sequences can also be further varied by addition, substitution, deletion or mutation.

Poly A sequences: SV40 poly A (SEQ ID NO: 42) and bGH poly A (SEQ ID NO: 43) can replace the rabbit beta globin poly A (SEQ ID NO: 29). These sequences can also be further varied by addition, substitution, deletion or mutation.

HIV Gag, Pol, and Integrase sequences: The HIV sequences in the Helper plasmid can be constructed from different HIV strains or clades. For example, HIV Gag (SEQ ID NO: 20); HIV Pol (SEQ ID NO: 21); and HIV Int (SEQ ID NO: 22) from the Bal strain can be interchanged with the gag, pol, and int sequences contained in the helper/helper plus Rev plasmids as outlined herein. These sequences can also be further varied by addition, substitution, deletion or mutation.

Envelope: The VSV-G glycoprotein can be substituted with membrane glycoproteins from feline endogenous virus (RD114) (SEQ ID NO: 44), gibbon ape leukemia virus (GALV) (SEQ ID NO: 45), Rabies (FUG) (SEQ ID NO: 46), lymphocytic choriomeningitis virus (LCMV) (SEQ ID NO: 47), influenza A fowl plague virus (FPV) (SEQ ID NO: 48), Ross River alphavirus (RRV) (SEQ ID NO: 49), murine leukemia virus 10A1 (MLV) (SEQ ID NO: 50), or Ebola virus (EboV) (SEQ ID NO: 51). Sequences for these envelopes are identified in the sequence portion herein. Further, these sequences can also be further varied by addition, substitution, deletion or mutation.

In summary, the 3-vector versus 4-vector systems can be compared and contrasted, in part, as follows. The 3-vector lentiviral vector system contains: 1. Helper plasmid: HIV Gag, Pol, Integrase, and Rev/Tat; 2. Envelope plasmid: VSV-G/FUG envelope; and 3. Therapeutic vector: RSV 5'LTR, Psi Packaging Signal, Gag fragment, RRE, Env fragment, cPPT, WPRE, and 3' delta LTR. The 4-vector lentiviral vector system contains: 1. Helper plasmid: HIV Gag, Pol, and Integrase; 2. Rev plasmid: Rev; 3. Envelope plasmid: VSV-G/FUG envelope; and 4. Therapeutic vector: RSV 5'LTR, Psi Packaging Signal, Gag fragment, RRE, Env fragment, cPPT, WPRE, and 3' delta LTR. Sequences corresponding with the above elements are identified in the sequence listings portion herein.

Example 2: Development of a Lentiviral Vector that Expresses FDPS

The purpose of this Example was to develop an FDPS lentivirus vector.

Inhibitory RNA Design: The sequence of *Homo sapiens* Farnesyl diphosphate synthase (FDPS) (NM_002004.3) mRNA was used to search for potential siRNA or shRNA candidates to knockdown FDPS levels in human cells. Potential RNA interference sequences were chosen from candidates selected by siRNA or shRNA design programs such as from GPP Web Portal hosted by the Broad Institute (<http://portals.broadinstitute.org/gpp/public/>) or the BLOCK-iT RNAi Designer from Thermo Scientific (<https://rnaidesigner.thermofisher.com/rnaiexpress/>). Individual selected shRNA sequences were inserted into a lentiviral vector immediately 3 prime to a RNA polymerase III

promoter such as H1 (SEQ ID NO: 16), U6 (SEQ ID NO: 52), or 7SK (SEQ ID NO: 53) to regulate shRNA expression. These lentivirus shRNA constructs were used to transduce cells and measure the change in specific mRNA levels. The shRNA most potent for reducing mRNA levels were embedded individually within a microRNA backbone to allow for expression by either the EF-1alpha or CMV RNA polymerase II promoters. The microRNA backbone was selected from mirbase.org. RNA sequences were also synthesized as synthetic siRNA oligonucleotides and introduced directly into cells without using a lentiviral vector.

Vector Construction: For FDPS shRNA, oligonucleotide sequences containing BamHI and EcoRI restriction sites were synthesized by Eurofins MWG Operon. Overlapping sense and antisense oligonucleotide sequences were mixed and annealed during cooling from 70 degrees Celsius to room temperature. The lentiviral vector was digested with the restriction enzymes BamHI and EcoRI for one hour at 37 degrees Celsius. The digested lentiviral vector was purified by agarose gel electrophoresis and extracted from the gel using a DNA gel extraction kit from Thermo Scientific. The DNA concentrations were determined and vector to oligo (3:1 ratio) were mixed, allowed to anneal, and ligated. The ligation reaction was performed with T4 DNA ligase for 30 minutes at room temperature. 2.5 microliters of the ligation mix were added to 25 microliters of STBL3 competent bacterial cells. Transformation was achieved after heat-shock at 42 degrees Celsius. Bacterial cells were spread on agar plates containing ampicillin and drug-resistant colonies (indicating the presence of ampicillin-resistance plasmids) were recovered and expanded in LB broth. To check for insertion of the oligo sequences, plasmid DNA was extracted from harvested bacteria cultures with the Thermo Scientific DNA mini prep kit. Insertion of shRNA sequences in the lentiviral vector was verified by DNA sequencing using a specific primer for the promoter used to regulate shRNA expression. Using the following target sequences, exemplary shRNA sequences were determined to knock-down FDPS:

(FDPS target sequence #1; SEQ ID NO: 54)
GTCCTGGAGTACAATGCCATT;

(FDPS shRNA sequence #1; SEQ ID NO: 1)
GTCCTGGAGTACAATGCCATTCTCGAGAATGGCATTGTACTCCAGGACTT
TTT;

(FDPS target sequence #2; SEQ ID NO: 55)
GCAGGATTTCGTTTCAGCACTT;

(FDPS shRNA sequence #2; SEQ ID NO: 2)
GCAGGATTTCGTTTCAGCACTTCTCGAGAAGTGCTGAACGAAATCCTGCTT
TTT;

(FDPS target sequence #3; SEQ ID NO: 56)
GCCATGTACATGGCAGGAATT;

(FDPS shRNA sequence #3; SEQ ID NO: 3)
GCCATGTACATGGCAGGAATTCTCGAGAATTCCTGCCATGTACATGGCTT
TTT;

(FDPS target sequence #4; SEQ ID NO: 57)
GCAGAAGGAGGCTGAGAAAGT;
and

(FDPS shRNA sequence #4; SEQ ID NO: 4)
GCAGAAGGAGGCTGAGAAAGTCTCGAGACTTTCTCAGCCTCCTTCTGCTT
TTT.

shRNA sequences were then assembled into a synthetic microRNA (miR) under control of the EF-1 alpha promoter. Briefly, a miR hairpin sequences, such as miR30, miR21, or miR185 as detailed below, was obtained from mirbase.org. The 19-22mer shRNA target sequence was used to construct the synthetic miR sequence. The miR sequence was arranged as an anti-sense-target-sequence-hairpin loop sequence (specific for each microRNA)-sense target sequence.

The following miR sequences were developed:

```
(miR30 FDPS sequence #1; SEQ ID NO: 5)
AAGGTATATTGCTGTTGACAGTGAGCGACACTTTCTCAGCCTCCTTCTGC
GTGAAGCCACAGATGGCAGAAAGGAGGCTGAGAAAGTGCCTACTGCCT
CGGACTTCAAGGGGCT

(miR30 FDPS sequence #2; SEQ ID NO: 6)
AAGGTATATTGCTGTTGACAGTGAGCGACACTTTCTCAGCCTCCTTCTGC
GTGAAGCCACAGATGGCAGAAAGGCTGAGAAAGTGCCTACTGCCTCG
GACTTCAAGGGGCT

(miR30 FDPS sequence #3; SEQ ID NO: 7)
TGCTGTTGACAGTGAGCGACTTTCTCAGCCTCCTTCTGCGTGAAGCCACA
GATGGCAGAAAGGAGGCTGAGAAAGTGCCTACTGCCTCGGA

(miR155 FDPS sequence #1; SEQ ID NO: 8)
CCTGGAGGCTTGCTGAAGGCTGTATGCTGACTTTCTCAGCCTCCTTCTGC
TTTTGGCCACTGACTGAGCAGAAAGGCTGAGAAAGTCAGGACACAAGGCC
TGTTACTAGCACTCA

(miR21 FDPS sequence #1; SEQ ID NO: 9)
CATCTCCATGGCTGTACCACCTTGTGCGGACTTTCTCAGCCTCCTTCTGC
CTGTTGAATCTCATGGCAGAAAGGAGCGAGAAAGTCTGACATTTTGGTAT
CTTTCATCTGACCA

(miR185 FDPS sequence #1; SEQ ID NO: 10)
GGGCCTGGCTCGAGCAGGGGGCGAGGGATACTTTCTCAGCCTCCTTCTGC
TGGTCCCCTCCCGCAGAAAGGAGGCTGAGAAAGTCCCTCCCTCCCAATGA
CCGCGTCTTCGTCG
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Example 3—Knock-Down of FDPS for 3 Days in THP1 Monocytic Leukemia by shRNA #4

This Example illustrates that knock-down of FDPS in THP1 monocytic leukemia cells by lentiviral (LV)-expressing FDPS shRNA #4 stimulates TNF- α expression in gamma delta T cells, as shown in FIG. 5.

THP1 cells (1×10^5 cells) were transduced with LV-control or LV-FDPS shRNA #4 for 3 days. Two days after transduction, cells were treated with or without 1 μ M zoledronic acid. After 24 hours, the transduced THP-1 cells were co-cultured with 5×10^5 PBMC cells and IL-2 in a round bottom 96 well plate for 4 hours. The PBMC cells were pre-stimulated with zoledronic acid and IL-2 for 11 days to expand V γ 9V δ 2 T cells. After staining for V γ 9V δ 2 and TNF- α using fluorophore-conjugated anti TCR-V δ 2 and anti-TNF- α antibody, cells were analyzed via flow cytometry. Live cells were gated, and V δ 2+ and TNF- α + cells were selected on a dot blot. The activated cytotoxic V γ 9V δ 2 T cells appeared in the upper right quadrant of flow cytograms. Without zoledronic acid, LV-control stimulated 3.1% of TNF- α expressing V γ 9V δ 2 T cells and LV-FDPS shRNA #4 stimulated 5%. With zoledronic acid treatment, LV-

control stimulated 7.2% of TNF- α expressing V γ 9V δ 2 T cells and LV-FDPS shRNA #4 stimulated 56.2%.

Example 4—Knock-Down of FDPS for 14 Days in THP1 Leukemia Cells by shRNA #4

This Example illustrates that Knock-down of FDPS for 14 days in THP1 leukemia cells by lentiviral (LV)-expressing FDPS shRNA #4 stimulates TNF- α expression in GD T cells, as shown in FIG. 6.

THP1 cells (1×10^5 cells) were transduced with LV-control or LV-FDPS shRNA #4 for 14 days. Two days after transduction, cells were treated with or without 1 μ M zoledronic acid. After 24 hours, the transduced THP-1 cells were co-cultured with 5×10^5 PBMC cells and IL-2 in a round bottom 96 well plate for 4 hours. The PBMC cells were pre-stimulated with zoledronic acid and IL-2 for 11 days to expand V γ 9V δ 2 T cells. After staining for V γ 9V δ 2 and TNF- α using fluorophore-conjugated anti TCR-V δ 2 and anti-TNF- α antibody, cells were analyzed via flow cytometry. Live cells were gated, and V δ 2+ and TNF- α + cells were selected on a dot blot. The activated cytotoxic V γ 9V δ 2 T cells appeared in the upper right quadrant of flow cytograms. Without zoledronic acid, LV-control stimulated 0.9% of TNF- α expressing V γ 9V δ 2 T cells and LV-FDPS shRNA #4 (SEQ ID NO: 4) stimulated 15.9%. With zoledronic acid treatment, LV-control stimulated 4.7% of TNF- α expressing V γ 9V δ 2 T cells and LV-FDPS shRNA #4 (SEQ ID NO: 4) stimulated 76.2%.

Example 5—Knock-Down of FDPS for 3 Days in PC3 Prostate Carcinoma Cells by shRNA #1

This Example illustrates that knock-down of FDPS for 3 days in PC3 prostate carcinoma cells by lentiviral (LV)-expressing FDPS shRNA #1 stimulates TNF- α expression in GD T cells, as shown in FIG. 7.

PC3 cells were transduced with LV-control or LV-FDPS shRNA #1 (SEQ ID NO: 1) for 3 days. Two days after transduction, cells were treated with or without 1 μ M zoledronic acid. After 24 hours, the transduced PC3 cells were co-cultured with 5×10^5 PBMC cells and IL-2 in a round bottom 96 well plate for 4 hours. The PBMC cells were pre-stimulated with zoledronic acid and IL-2 for 11 days to expand V γ 9V δ 2 T cells. After staining for V γ 9V δ 2 and TNF- α using fluorophore-conjugated anti TCR-V δ 2 and anti-TNF- α antibody, cells were analyzed via flow cytometry. Live cells were gated, and V δ 2+ and TNF- α + cells were selected on a dot blot. The activated cytotoxic V γ 9V δ 2 T cells appeared in the upper right quadrant of flow cytograms. Without zoledronic acid, LV-control stimulated 0.2% of TNF- α expressing V γ 9V δ 2 T cells and LV-FDPS shRNA #1 stimulated 0.5%. With zoledronic acid treatment, LV-control stimulated 1.7% of TNF- α expressing V γ 9V δ 2 T cells and LV-FDPS shRNA #1 (SEQ ID NO: 1) stimulated 32.2%.

Example 6—Knock-Down of FDPS for 3 Days in PC3 Prostate Carcinoma Cells by shRNA #4

This Example illustrates that Knock-down of FDPS for 3 days in PC3 prostate carcinoma cells by lentiviral (LV)-expressing FDPS shRNA #4 stimulates TNF- α expression in GD T cells, as shown in FIG. 8.

PC3 cells were transduced with LV-control or LV-FDPS shRNA #4 (SEQ ID NO: 4) for 3 days. Two days after transduction, cells were treated with or without 1 μ M zoledronic acid. After 24 hours, the transduced PC3 cells were co-cultured with 5×10^5 PBMC cells and IL-2 in a round bottom 96 well plate for 4 hours. The PBMC cells were pre-stimulated with zoledronic acid and IL-2 for 11

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days to expand Vγ9Vδ2 T cells. After staining for Vγ9Vδ2 and TNF-α using fluorophore-conjugated anti TCR-Vδ2 and anti-TNF-α antibody, cells were analyzed via flow cytometry. Live cells were gated, and Vδ2+ and TNF-α+ cells were selected on a dot blot. The activated cytotoxic Vγ9Vδ2 T cells appeared in the upper right quadrant of flow cytograms. Without zoledronic acid, LV-control stimulated 0.5% of TNF-α expressing Vγ9Vδ2 T cells and LV-FDPS shRNA #4 (SEQ ID NO: 4) stimulated 1.9%. With zoledronic acid treatment, LV-control stimulated 2.1% of TNF-α expressing Vγ9Vδ2 T cells and LV-FDPS shRNA #4 stimulated 28.7%.

Example 7—Knock-Down of FDPS for 3 Days in
HepG2 Liver Carcinoma Cells by shRNA #1 and
#4

This Example illustrates that Knock-down of FDPS for 3 days in HepG2 liver carcinoma cells by lentiviral (LV)-expressing FDPS shRNA #1 (SEQ ID NO: 1) and shRNA #4 (SEQ ID NO: 4) stimulates TNF-α expression in GD T cells, as shown in FIG. 9.

HepG2 cells were transduced with LV-control, LV-FDPS shRNA #1 (SEQ ID NO: 1), or LV-FDPS shRNA #4 (SEQ ID NO: 4) for 3 days. Two days after transduction, cells were treated with or without 1 μM zoledronic acid. After 24 hours, the transduced HepG2 cells were co-cultured with 5×10⁵ PBMC cells and IL-2 in a round bottom 96 well plate for 4 hours. The PBMC cells were pre-stimulated with zoledronic acid and IL-2 for 11 days to expand Vγ9Vδ2 T cells. After staining for Vγ9Vδ2 and TNF-α using fluorophore-conjugated anti TCR-Vδ2 and anti-TNF-α antibody, cells were analyzed via flow cytometry. Live cells were gated, and Vδ2+ and TNF-α+ cells were selected on a dot blot. The activated cytotoxic Vγ9Vδ2 T cells appeared in the upper right quadrant of flow cytograms. Without zoledronic acid, LV-control stimulated 0.4% of TNF-α expressing Vγ9Vδ2 T cells and LV-FDPS shRNA #1 (SEQ ID NO: 1) and #4 (SEQ ID NO: 4) stimulated 0.7% and 0.9%, respectively. With zoledronic acid treatment, LV-control stimulated 6.9% of TNF-α expressing Vγ9Vδ2 T cells and LV-FDPS shRNA #1 and #4 stimulated 7.6% and 21.1%, respectively.

Example 8—Knock-Down of FDPS for 3 Days in
THP1 Leukemia by microRNA-30

This Example illustrates that Knock-down of FDPS for 3 days in THP1 leukemia cells by lentiviral (LV)-expressing FDPS-targeted synthetic microRNA-30 stimulates TNF-α expression in gamma delta T cells, as shown in FIG. 10.

THP1 cells (1×10⁵ cells) were transduced with LV-control or LV-miR30 FDPS #1 (SEQ ID NO: 5) for 3 days. Two days after transduction, cells were treated with or without 104 zoledronic acid. After 24 hours, the transduced THP-1 cells were co-cultured with 5×10⁵ PBMC cells and IL-2 in a round bottom 96 well plate for 4 hours. The PBMC cells were pre-stimulated with zoledronic acid and IL-2 for 11 days to expand Vγ9Vδ2 T cells. After staining for Vγ9Vδ2 and TNF-α using fluorophore-conjugated anti TCR-Vδ2 and anti-TNF-α antibody, cells were analyzed via flow cytometry. Live cells were gated, and Vδ2+ and TNF-α+ cells were selected on a dot blot. The activated cytotoxic Vγ9Vδ2 T cells appeared in the upper right quadrant of flow cytograms. Without zoledronic acid, LV-control stimulated 0.2% of TNF-α expressing Vγ9Vδ2 T cells and LV-miR30 FDPS stimulated 8.1%. With zoledronic acid treatment, LV-control stimulated 5.3% of TNF-α expressing Vγ9Vδ2 T cells and LV-miR30 FDPS #1 (SEQ ID NO: 5) stimulated 67.3%.

Example 9: E:T Ratios Resulting from Mixture of
THP-1 Cells, Cultured Human GD T Cells, and/or
Zometa (Zol)

This Example demonstrates results from mixing treated THP-1 monocytoid tumor cells with cultured human GD T cells, as shown in FIG. 11.

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The monocytoid cell line THP-1 was treated with control lentivirus vector (LV), LV suppressing farnesyl diphosphate synthase gene expression (LV-FDPS), zoledronic acid (Zol) or combinations. The legend, as shown in FIG. 11, was: lentiviral control vectors (LV-Control), lentiviral vectors expressing microRNA to down regulate FDPS (LV-FPPS), Zometa (Zol), Zometa plus lentiviral control (Zol+LV-Control), or Zometa plus lentiviral vectors expressing microRNA to down regulate FPPS (Zol+LV-FPPS).

Human GD T cells were cultured from an anonymous donor and added to treated THP-1 cells in 4:1, 2:1 or 1:1 ratios (GD T:THP-1) for 4 hours. Cell killing was measured by a fluorescence assay. When THP-1 cells were treated with a combination of LV-FDPS and Zol, cytotoxic T cell killing by GD T cells was increased greatly compared to either treatment alone. When LV-FDPS treatment alone was compared to Zol treatment alone, the LV-FDPS lead to greater killing but was >3-fold below tumor cell killing after combination treatment. The combined LV-FDPS plus Zol treatment caused nearly 70% tumor cell killing with 4:1 ratio; this was more than 3-fold higher than the second best treatment (LV-FDPS alone).

Example 10—Lentiviral-Delivered shRNA-Based
RNA Interference Targeting the Human Farnesyl
Diphosphate Synthase (FDPS) Gene

HepG2 human hepatocellular carcinoma cells were infected with lentiviral vectors containing the H1 promoter and either a non-targeting or four different FDPS shRNA sequences, as shown in FIG. 12. After 48 hours, RNA was extracted from the cells and converted to cDNA. Expression of FDPS cDNA was determined by quantitative PCR using SYBR Green and FDPS primers. FDPS expression was normalized to actin levels for each sample. FDPS-targeting lentiviral vectors containing the H1 promoter and either a non-targeting sequence

(SEQ ID NO: 58)

(5'-

GCCGCTTTGTAGGATAGAGCTCGAGCTCTATCCTACAAAGCGGCTTTTT-

3')

or one of four different FDPS shRNA sequences

(FDPS shRNA sequence #1; SEQ ID NO: 1)

GTCCTGGAGTACAATGCCATTCTCGAGAATGGCATTGTACTCCAGGACTT

TTT;

(FDPS shRNA sequence #2; SEQ ID NO: 2)

GCAGGATTTCGTTTCAGCACTTCTCGAGAAGTGCTGAACGAAATCCTGCTT

TTT;

(FDPS shRNA sequence #3; SEQ ID NO: 3)

GCCATGTACATGGCAGGAATTCTCGAGAATCCTGCCATGTACATGGCTT

TTT;
and

(FDPS shRNA sequence #4; SEQ ID NO: 4)

GCAGAAGGAGGCTGAGAAAGTCTCGAGACTTTCTCAGCCTCCTTCTGCTT

TTT

were produced in 293 T cells.

HepG2 human hepatocellular carcinoma cells were then infected with lentiviral vectors to determine the efficacy of FDPS knock-down. After 48 hours, RNA was extracted from the cells using the RNeasy RNA isolation kit (Qiagen) and

converted to cDNA with the SuperScript VILO cDNA synthesis kit (Thermo Scientific). Expression of FDPS cDNA was determined by quantitative PCR on an Applied Biosystems StepOne qPCR machine using a SYBR Green PCR mix (Thermo Scientific) and FDPS primers (Forward primer: 5'-AGGAATTGATGGCGAGAAGG-3' (SEQ ID NO: 59) and Reverse primer: 5'-CCCAAAGAGGT-CAAGGTAATCA-3' (SEQ ID NO: 60)). FDPS expression was normalized to actin levels for each sample using the actin primers (Forward primer: 5'-AGCGCGGCTA-CAGCTTCA-3' (SEQ ID NO: 61) and Reverse primer: 5'-GGCGACGTAGCACAGCTTCT-3' (SEQ ID NO: 62)). The relative FDPS RNA expression of the shCon sample is set at 100%. There was an 85% (FDPS sequence #1), 89% (FDPS sequence #2), 46% (FDPS sequence #3), and 98% (FDPS sequence #4) decrease in FDPS expression.

Example 11—Lentiviral-Delivered miR-Based
RNA Interference Targeting the Human Farnesyl
Diphosphate Synthase (FDPS) Gene

As shown in FIG. 13, HepG2 human hepatocellular carcinoma cells were infected with lentiviral vectors containing either the H1 promoter (SEQ ID NO: 16) the FDPS shRNA #4 (SEQ ID NO: 4) sequence or the EF-1a promoter (SEQ ID NO: 40) and miR30-based FDPS sequences. After 48 hours, cells were lysed and an immunoblot was performed using an anti-FDPS (Thermo Scientific) and an anti-actin (Sigma) antibody as a protein loading control.

More specifically, HepG2 human hepatocellular carcinoma cells were infected with lentiviral vectors containing either the H1 promoter (SEQ ID NO: 16) and the FDPS shRNA sequence

(FDPS shRNA sequence #4; SEQ ID NO: 4)
GCAGAAGGAGGCTGAGAAAGTCTGAGACTTTCTCAGCTCCTTCTGCTT
TTT
or the EF-1alpha promoter (SEQ ID NO: 39)
and
miR30-based FDPS sequences
(miR30 FDPS sequence #1; SEQ ID NO: 5)
AAGGTATATTGCTGTTGACAGTGAGCGACACTTTCTCAGCTCCTTCTGCG
GTGAAGCCACAGATGGCAGAAGGAGGCTGAGAAAGTGCTGCCTACTGCGT
CGGACTTCAAGGGGCT
and
(miR30 FDPS sequence #2; SEQ ID NO: 6)
AAGGTATATTGCTGTTGACAGTGAGCGACACTTTCTCAGCTCCTTCTGCG
TGAAGCCACAGATGGCAGAAGGAGGCTGAGAAAGTGCTGCCTACTGCGT
CTTCAAGGGGCT.

After 48 hours, cells were lysed with NP-40 lysis buffer and protein was quantified with the Bio-Rad protein assay reagent. Protein samples at 50 micrograms were electrophoresed on 4-12% Bis-Tris gels (Thermo Scientific) and transferred to PVDF membranes (EMD Millipore). An immunoblot was performed using an anti-FDPS (Thermo Scientific) and an anti-actin (Sigma) antibody as a protein loading control. Antibodies were bound with HRP-conjugated secondary antibodies and detected with a Licor c-DiGit Blot scanner using the Immobilon Western ECL reagent (EMD Millipore). The densitometry of the immunoblot bands were quantified with the NIH image software. The LV control with the EF-1 promoter was set at 100%.

There was a 68% (LV-shFDPS #4), 43% (LV-miR FDPS #1), and 38% (LV-miR FDPS #3) reduction of FDPS protein expression.

Example 12—Knock-Down of FDPS for 3 Days in
HepG2 Liver Carcinoma Cells by
Adeno-Associated Virus (AAV)-Expressing FDPS
shRNA #4

This Example illustrates that knock-down of FDPS for 3 days in HepG2 liver carcinoma cells by adeno-associated virus (AAV)-expressing FDPS shRNA #4 (SEQ ID NO: 4) stimulates TNF- α expression in GD T cells (FIG. 14, Panel B).

HepG2 cells were transduced with control or AAV-FDPS shRNA #4 (SEQ ID NO: 8) for 3 days. Two days after transduction, cells were treated with or without 1 μ M zoledronic acid. After 24 hours, the transduced HepG2 cells were co-cultured with 5×10^5 PBMC cells and IL-2 in a round bottom 96 well plate for 4 hours. The PBMC cells were pre-stimulated with zoledronic acid and IL-2 for 11 days to expand V γ 9V δ 2 T cells. After staining for V γ 9V δ 2 and TNF- α using fluorophore-conjugated anti TCR-V δ 2 and anti-TNF- α antibody, cells were analyzed via flow cytometry. Live cells were gated, and V δ 2+ and TNF- α + cells were selected on a dot blot. The activated cytotoxic V γ 9V δ 2 T cells appeared in the upper right quadrant of flow cytograms (FIG. 14, Panel B).

AAV Vector Construction. FDPS shRNA sequence #4 (SEQ ID NO: 4) was inserted into the pAAV plasmid (Cell Biolabs). FDPS oligonucleotide sequences containing BamHI and EcoRI restriction sites were synthesized by Eurofins MWG Operon. Overlapping sense and antisense oligonucleotide sequences were mixed and annealed during cooling from 70 degrees Celsius to room temperature. The pAAV was digested with the restriction enzymes BamHI and EcoRI for one hour at 37 degrees Celsius. The digested pAAV plasmid was purified by agarose gel electrophoresis and extracted from the gel using a DNA gel extraction kit from Thermo Scientific. The DNA concentrations were determined and vector to oligo (3:1 ratio) were mixed, allowed to anneal, and ligated. The ligation reaction was performed with T4 DNA ligase for 30 minutes at room temperature. 2.5 microliters of the ligation mix were added to 25 microliters of STBL3 competent bacterial cells. Transformation was achieved after heat-shock at 42 degrees Celsius. Bacterial cells were spread on agar plates containing ampicillin and drug-resistant colonies (indicating the presence of ampicillin-resistance plasmids) were recovered and expanded in LB broth. To check for insertion of the oligo sequences, plasmid DNA was extracted from harvested bacteria cultures with the Thermo Scientific DNA mini prep kit. Insertion of shRNA sequences in the pAAV plasmid was verified by DNA sequencing using a specific primer for the promoter used to regulate shRNA expression. An exemplary AAV vector with a H1 promoter (SEQ ID NO: 16), shFDPS sequence (e.g., SEQ ID NO: 4), Left Inverted Terminal Repeat (Left ITR; SEQ ID NO: 63), and Right Inverted Terminal Repeat (Right ITR; SEQ ID NO: 64) can be found in FIG. 14, Panel A).

Production of AAV particles. The AAV-FDPS shRNA plasmid was combined with the plasmids pAAV-RC2 (Cell Biolabs) and pHelper (Cell Biolabs). The pAAV-RC2 plasmid contains the Rep and AAV2 capsid genes and pHelper contains the adenovirus E2A, E4, and VA genes. To produce AAV particles, these plasmids were transfected in the ratio 1:1:1 (pAAV-shFDPS: pAAV-RC2: pHelper) into 293T

cells. For transfection of cells in 150 mm dishes (BD Falcon), 10 micrograms of each plasmid were added together in 1 ml of DMEM. In another tube, 60 microliters of the transfection reagent PEI (1 microgram/nil) (Polysciences) was added to 1 ml of DMEM. The two tubes were mixed together and allowed to incubate for 15 minutes. Then the transfection mixture was added to cells and the cells were collected after 3 days. The cells were lysed by freeze/thaw lysis in dry ice/isopropanol. Benzonase nuclease (Sigma) was added to the cell lysate for 30 minutes at 37 degrees Celsius. Cell debris were then pelleted by centrifugation at 4 degrees Celsius for 15 minutes at 12,000 rpm. The supernatant was collected and then added to target cells.

Example 13—Decreased RAP1 Prenylation in the Cells Transduced with LV-shFDPS and Treated with Zoledronic Acid

This Example illustrates that lentiviral-delivered shRNA targeting the human farnesyl diphosphate synthase (FDPS) gene and zoledronic acid synergize to inhibit farnesyl diphosphate production.

FDPS is an enzyme in the isoprenoid synthesis pathway that catalyzes the production of farnesyl diphosphate. Inhibiting the enzyme activity of FDPS by zoledronic acid or reduced protein expression by shRNA-mediated knock-down will result in reduced farnesyl diphosphate levels. Farnesylation of cellular proteins requires farnesyl diphosphate. RAP1A is a protein that is modified by farnesylation, which can be used as a biomarker for levels of cellular farnesyl diphosphate. An antibody that specifically recognizes reduced RAP1A farnesylation was used to measure FDPS activity after transduction with LV-shFDPS alone or in combination with zoledronic acid. HepG2 human hepatocellular carcinoma cells were infected with lentiviral vectors containing FDPS shRNA sequence #4. For the zoledronic acid treated cells, zoledronic acid (Sigma) was added for the last 24 hours. After 48 hours, cells were lysed with NP-40 lysis buffer and protein was quantified with the Bio-Rad protein assay reagent. Protein samples at 50 micrograms were electrophoresed on 4-12% Bis-Tris gels (Thermo Scientific) and transferred to PVDF membranes (EMD Millipore). As shown in FIG. 15, an immunoblot was performed using an anti-FDPS (Thermo Scientific), anti-RAP1A (Santa Cruz), and an anti-actin (Sigma) antibody as a protein loading control. Antibodies were bound with HRP-conjugated secondary antibodies and detected with a Licor c-DiGit Blot scanner using the Immobilon Western ECL reagent (EMD Millipore). An increase in the RAP1A band intensity correlates with reduced farnesylation. RAP1A defarnesylation occurred only in the cells transduced with LV-shFDPS and treated with zoledronic acid.

Example 14—Treatment of a Subject with Cancer

LV-FDPS is a Genetic Medicine Delivered by a Lentivirus Vector Via Local Administration to the Site of Late Stage, Non-Resectable Hepatocellular Carcinoma

A Phase I clinical trial will test safety and feasibility of delivering LV-FDPS to the site of hepatocellular carcinoma (HCC) using ultrasound guided cannulation of the liver in patients without concomitant radiotherapy or chemotherapy. It is rationally predicted that this study will result in the successful treatment of HCC. The study is an open label, 4x3 dose escalation (4 dose ranges, up to 3 subjects per

dose) to identify the maximum tolerable dose of LV-FDPS in patients 18 years or older with Stage III/IV non-resectable HCC.

LV-FDPS is a genetic therapy designed to reduce expression in tumor cells of the enzyme farnesyl diphosphate synthase. Experimental studies show that tumor cells modified by LV-FDPS induce the anti-tumor activity of human gamma delta T cells, including the capacity for tumor killing by cellular cytotoxicity.

- 5 Subjects with target lesions ≥ 1 cm in longest diameter (measured by helical CT) and ≤ 4.9 cm maximum diameter and meeting inclusion and exclusion criteria detailed below, are enrolled into the next available dosing category. A maximum of 3 subjects are recruited for each dosage group.
- 10 The dose is number of transducing units of LV-FDPS as described in the product release criteria, delivered via intra-hepatic cannulation in a single bolus with volume not to exceed 25 mL. The minimum dose is 1×10^9 transducing units and escalation is 10-fold to a next dose of 1×10^{10}
- 15 transducing units, the next dose is 1×10^{11} transducing units, and a maximum dose of 1×10^{12} transducing units based on reported experience with recombinant adenovirus therapy for HCC (Sangro et al., A phase I clinic trial of thymidine kinase-based gene therapy in advanced hepatocellular carcinoma, 2010, *Cancer Gene Ther.* 17:837-43). Subjects are enrolled, treated and evaluated for 3 months. All safety evaluations are completed for each group prior to enrolling and treating subjects at the next higher dose level. Enrollment and dose escalation continue until a maximum tolerable dose is achieved or the study is terminated.

Cannulation is via the left subclavian artery until tip of catheter is at the proper hepatic artery junction. Cannulation is guided by ultrasonography as described (Lin et al., Clinical effects of intra-arterial infusion chemotherapy with cisplatin, mitomycin C, leucovorin and 5-Fluorouracil for unresectable advanced hepatocellular carcinoma, 2004, *J. Chin. Med. Assoc.* 67:602-10).

Primary Outcome Measures

- 35 Safety: Systemic and locoregional adverse events are graded according to CTCAS and coded according to MedRA. The adverse events data for all subjects at a single dose range will be evaluated prior to dose escalation. The final safety assessment incorporates data from all dose ranges.

Secondary Outcome Measures

- 40 Lesion distribution and retention of LV-FDPS following locoregional administration and subsequent biopsy or necropsy to obtain tissues.
- Objective response rate (ORR) in target and measurable non-local lesions (if present) by physical analysis, medical imaging or biopsy during 3 months after treatment.
- 45 Levels of LV-FDPS in blood stream during 10 minutes, 30 minutes, 1 hour and 1 day after local injection.
- Changes in markers of hepatic function including ALP, ALT, ASAT, total bilirubin and GGT during 3 months after treatment.
- 50 Disease free survival beyond historical control (no LV-FDPS) patients in ad hoc analysis.

Inclusion Criteria

- 55 Greater than 18 years and including both males and females.
- Diagnosis confirmed by histology or cytology or based on currently accepted clinical standards of hepatocellular carcinoma of parenchyma cell origin that is not amenable, at the time of screening, to resection, transplant or other potentially curative therapies.

Treating physician determines that the lesion is amenable to locoregional targeted delivery.

Target lesion must represent measurable disease with a unidimensional longest diameter of ≥ 1.0 cm by computed tomography; the maximum longest diameter is ≤ 5.0 cm.

Karnofsky performance score 60-80% of ECOG values.

Life expectancy ≥ 12 weeks.

Hematopoietic function: WBC $\geq 2,500/\text{mm}^3$; ANC $\geq 1000/\text{mm}^3$; Hemoglobin ≥ 8 g/dL; Platelet count $\geq 50,000/\text{mm}^3$; Coagulation INR ≤ 1.3 .

AST and ALT < 5 times ULN; ALPS < 5 time ULN. Bilirubin ≤ 1.5 times ULV; Creatine ≤ 1.5 times ULN and eGFR ≥ 50 .

Thyroid function: Total T3 or free T3, total T4 or free T4 and THCAE Grade 2 abnormality.

Renal, cardiovascular and respiratory function adequate in the opinion of the attending physician.

Immunological function: Circulating Vgamma9Vdelta2+ T cells $\geq 30/\text{mm}^3$; no immunodeficiency disease.

Negative for HIV by serology and viral RNA test.

Written informed consent.

Exclusion Criteria

Target lesion contiguous with, encompasses or infiltrates blood vessel.

Primary HCC amenable to resection, transplantation or other potentially curative therapies.

Hepatic surgery or chemoembolization within the past 4 months.

Hepatic radiation or whole body radiation therapy within past 4 months.

Chemotherapy with 4 weeks or any use of nitrosourea, mitomycin C or cisplatin.

Current or within past 4 weeks receipt of bisphosphonate therapy

Investigational agents within 4 weeks or < 5 drug half-lives.

Impaired wound healing due to diabetes.

Significant psychiatric illness, alcohol dependence or illicit drug use.

Unwilling to comply with study protocols and reporting requirements.

Bisphosphonate treatment within past 4 months.

Presence of clinically significant cardiovascular, cerebrovascular (stroke), immunological (except hepatitis B or C virus infection, viral hepatitis or cirrhosis), endocrine or central nervous system disorders; current encephalopathy; variceal bleeding requiring hospitalization or transfusion within past 4 months.

History of HIV or acquired immune deficiency syndrome.

Current or prior treatment with antiretroviral medications.

Pregnant, lactating or refusal to adopt barrier or chemical contraceptive use throughout trial and follow-up interval.

LV-FDPS is a Genetic Medicine Delivered by a Lentivirus Vector Via Local Administration to the Site of Late Stage, Non-Resectable Hepatocellular Carcinoma—Adjunct Administration of Bisphosphonate

A Phase I clinical trial will test safety and feasibility of delivering LV-FDPS to the site of hepatocellular carcinoma (HCC) using ultrasound guided cannulation of the liver in patients with concomitant bisphosphonate chemotherapy. It is rationally predicted that this study will result in the successful treatment of HCC. The study is an open label, 4x3 dose escalation (4 dose ranges, up to 3 subjects per

dose) to identify the maximum tolerable dose of LV-FDPS in patients 18 years or older with Stage III/IV non-resectable HCC.

LV-FDPS is a genetic therapy designed to reduce expression in tumor cells of the enzyme farnesyl diphosphate synthase. Experimental studies show that tumor cells modified by LV-FDPS induce the anti-tumor activity of human gamma delta T cells, including the capacity for tumor killing by cellular cytotoxicity. Prior experimental studies also showed the potential for positive interactions of LV-FDPS and specific bisphosphonate drugs that may be prescribed in primary or metastatic diseases. For this study, subjects will receive dose escalating amounts of LV-FDPS with continuous standard of care dosing with Aredia® (pamidronate), Zometa® (zoledronic acid) or Actonel® (risedronate) according to physician advice and subject preference.

Subjects with target lesions ≥ 1 cm in longest diameter (measured by helical CT) and ≤ 4.9 cm maximum diameter and meeting inclusion and exclusion criteria detailed below, are enrolled and started on bisphosphonate therapy. 30 days later size of the target lesion is re-evaluated to ensure subjects still meet starting criteria for LV-FDPS. Subjects without objective clinical response on bisphosphonate are enrolled into the next available LV-FDPS dosing category. A maximum of 3 subjects are recruited for each dosage group and all continue on bisphosphonate for the study duration unless otherwise advised by the attending physician. The LV-FDPS dose is a number of transducing units of LV-FDPS as described in the product release criteria, delivered via intrahepatic cannulation in a single bolus with volume not to exceed 25 mL. The minimum dose is 1×10^9 transducing units and escalation is 10-fold to a next dose of 1×10^{10} transducing units, the next dose is 1×10^{11} transducing units, and a maximum dose of 1×10^{12} transducing units based on reported experience with recombinant adenovirus therapy for HCC (Sangro, et al., A phase I clinic trial of thymidine kinase-based gene therapy in advanced hepatocellular carcinoma, 2010, *Cancer Gene Ther.* 17:837-43). Subjects are enrolled, treated and evaluated for 3 months. All safety evaluations are completed for each group prior to enrolling and treating subjects at the next higher dose level. Enrollment and dose escalation continue until a maximum tolerable dose is achieved or the study is terminated.

Cannulation is via the left subclavian artery until tip of catheter is at the proper hepatic artery junction. Cannulation is guided by ultrasonography as described (Lin et al., Clinical effects of intra-arterial infusion chemotherapy with cisplatin, mitomycin C, leucovor and 5-Fluorouracil for unresectable advanced hepatocellular carcinoma, 2004, *J. Chin. Med. Assoc.* 67:602-10).

Primary Outcome Measures

Safety: Systemic and locoregional adverse events are graded according to CTCAS and coded according to MedRA. The adverse events data for all subjects at a single dose range will be evaluated prior to dose escalation. The final safety assessment incorporates data from all dose ranges.

Secondary Outcome Measures

Lesion distribution and retention of LV-FDPS following locoregional administration and subsequent biopsy or necropsy to obtain tissues.

Objective response rate (ORR) in target and measurable non-local lesions (if present) by physical analysis, medical imaging or biopsy during 3 months after treatment.

Levels of LV-FDPS in blood stream during 10 minutes, 30 minutes, 1 hour and 1 day after local injection.

Changes in markers of hepatic function including ALP, ALT, ASAT, total bilirubin and GGT during 3 months after treatment.

Disease free survival beyond historical control (no LV-FDPS) patients in ad hoc analysis.

Inclusion Criteria

Greater than 18 years and including both males and females.

Diagnosis confirmed by histology or cytology or based on currently accepted clinical standards of hepatocellular carcinoma of parenchyma cell origin that is not amenable, at the time of screening, to resection, transplant or other potentially curative therapies.

Treating physician determines that the lesion is amenable to locoregional targeted delivery.

Target lesion must represent measurable disease with a unidimensional longest diameter of ≥ 1.0 cm by computed tomography; the maximum longest diameter is ≤ 5.0 cm.

Karnofsky performance score 60-80% of ECOG values. Life expectancy ≥ 12 weeks.

Hematopoietic function: WBC $\geq 2,500/\text{mm}^3$; ANC $\geq 1000/\text{mm}^3$; Hemoglobin ≥ 8 g/dL; Platelet count $\geq 50,000/\text{mm}^3$; Coagulation INR ≤ 1.3 .

AST and ALT < 5 times ULN; ALPS < 5 time ULN. Bilirubin ≤ 1.5 times ULV; Creatine ≤ 1.5 times ULN and eGFR ≥ 50 .

Thyroid function: Total T3 or free T3, total T4 or free T4 and THCAE Grade 2 abnormality.

Renal, cardiovascular and respiratory function adequate in the opinion of the attending physician.

Immunological function: Circulating Vgamma9Vdelta2+ T cells $\geq 30/\text{mm}^3$; no immunodeficiency disease.

Negative for HIV by serology and viral RNA test.

Written informed consent.

Exclusion Criteria

Intolerant to or unwilling to continue bisphosphonate adjunct therapy.

Objective clinical response after bisphosphonate therapy.

Target lesion contiguous with, encompasses or infiltrates blood vessel.

Primary HCC amenable to resection, transplantation or other potentially curative therapies.

Hepatic surgery or chemoembolization within the past 4 months.

Hepatic radiation or whole body radiation therapy within past 4 months.

Chemotherapy excluding bisphosphonate, within 4 weeks or any use of nitrosourea, mitomycin C or cisplatin.

Investigational agents within 4 weeks or < 5 drug half-lives.

Impaired wound healing due to diabetes.

Significant psychiatric illness, alcohol dependence or illicit drug use.

Unwilling to comply with study protocols and reporting requirements.

Presence of clinically significant cardiovascular, cerebrovascular (stroke), immunological (except hepatitis B or C virus infection, viral hepatitis or cirrhosis), endocrine or central nervous system disorders; current encephalopathy; variceal bleeding requiring hospitalization or transfusion within past 4 months.

History of HIV or acquired immune deficiency syndrome. Current or prior treatment with antiretroviral medications.

Pregnant, lactating or refusal to adopt barrier or chemical contraceptive use throughout trial and follow-up interval.

Example 15—Treatment of a Subject with Chronic Viral Disease(s) of the Liver

LV-FDPS is a Genetic Medicine Delivered by a Lentivirus Vector Via Local Administration to Liver for the Treatment of Hepatitis B Virus, Hepatitis C Virus, HIV or Other Viral Infection of the Liver

A Phase I clinical trial will test safety and feasibility of delivering LV-FDPS to virally infected liver using ultrasound guided cannulation. It is rationally predicted that this study will result in the successful treatment of infections of the liver. The study is an open label, 4x3 dose escalation (4 dose ranges, up to 3 subjects per dose) to identify the maximum tolerable dose of LV-FDPS in patients 18 years or older with chronic viral disease of the liver that is resistant to chemotherapy.

LV-FDPS is a genetic therapy designed to reduce expression in tumor cells of the enzyme farnesyl diphosphate synthase. Experimental studies show that tumor cells modified by LV-FDPS induce human gamma delta T cells, including a capacity for cellular cytotoxicity against virally-infected cells.

Subjects with confirmed viral infection of the liver including hepatitis B virus, hepatitis C virus, HIV or other viruses are enrolled into the next available LV-FDPS dosing category. A maximum of 3 subjects are recruited for each dosage group. The LV-FDPS dose is a number of transducing units of LV-FDPS as described in the product release criteria, delivered via intrahepatic cannulation in a single bolus with volume not to exceed 25 mL. The minimum dose is 1×10^9 transducing units and escalation is 10-fold to a next dose of 1×10^{10} transducing units, the next dose is 1×10^{11} transducing units, and a maximum dose of 1×10^{12} transducing units based on reported experience with recombinant adenovirus therapy for HCC (Sangro, et al., A phase I clinic trial of thymidine kinase-based gene therapy in advanced hepatocellular carcinoma, 2010, *Cancer Gene Ther.* 17:837-43). Subjects are enrolled, treated and evaluated for 3 months. All safety evaluations are completed for each group prior to enrolling and treating subjects at the next higher dose level. Enrollment and dose escalation continue until a maximum tolerable dose is achieved or the study is terminated.

Cannulation is via the left subclavian artery until tip of catheter is at the proper hepatic artery junction. Cannulation is guided by ultrasonography as described (Lin et al., Clinical effects of intra-arterial infusion chemotherapy with cisplatin, mitomycin C, leucovor and 5-Fluorouracil for unresectable advanced hepatocellular carcinoma, 2004, *J. Chin. Med. Assoc.* 67:602-10).

Primary Outcome Measures

Safety: Systemic and locoregional adverse events are graded according to CTCAS and coded according to MedRA. The adverse events data for all subjects at a single dose range will be evaluated prior to dose escalation. The final safety assessment incorporates data from all dose ranges.

Secondary Outcome Measures

Lesion distribution and retention of LV-FDPS following locoregional administration and subsequent biopsy or necropsy to obtain tissues.

Objective response rate (ORR) measured as a Sustained Viral Response (SVR) within the organ or systemically during 3 months after treatment.

Levels of LV-FDPS in blood stream during 10 minutes, 30 minutes, 1 hour and 1 day after local injection.

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Changes in markers of hepatic function including ALP, ALT, ASAT, total bilirubin and GGT during 3 months after treatment.
Disease free survival beyond historical control (no LV-FDPS) patients in ad hoc analysis.

Inclusion Criteria

Greater than 18 years and including both males and females.
Diagnosis confirmed by histology or cytology or based on currently accepted clinical standards of chronic viral infection of the liver that is not amenable, at the time of screening, to resection, transplant or other potentially curative therapies.
Treating physician determines that the lesion is amenable to locoregional targeted delivery.
Karnofsky performance score 60-80% of ECOG values.
Life expectancy ≥ 12 weeks.
Hematopoietic function: WBC $\geq 2,500/\text{mm}^3$; ANC $\geq 1000/\text{mm}^3$; Hemoglobin ≥ 8 g/dL; Platelet count $\geq 50,000/\text{mm}^3$; Coagulation INR ≤ 1.3 .
AST and ALT < 5 times ULN; ALPS < 5 time ULN. Bilirubin ≤ 1.5 times ULV; Creatine ≤ 1.5 times ULN and eGFR ≥ 50 .
Thyroid function: Total T3 or free T3, total T4 or free T4 and THCAE Grade 2 abnormality.
Renal, cardiovascular and respiratory function adequate in the opinion of the attending physician.
Immunological function: Circulating Vgamma9Vdelta2+ T cells $\geq 30/\text{mm}^3$; no immunodeficiency disease.
Negative for HIV by serology and viral RNA test.
Written informed consent.

Exclusion Criteria

Chronic viral disease amenable to resection, transplantation or other potentially curative therapies.
Hepatic surgery or chemoembolization within the past 4 months.
Hepatic radiation or whole body radiation therapy within past 4 months.
Investigational agents within 4 weeks or < 5 drug half-lives.
Current (within past 4 weeks) or ongoing receipt of bisphosphonate therapy.
Impaired wound healing due to diabetes.
Significant psychiatric illness, alcohol dependence or illicit drug use.
Unwilling to comply with study protocols and reporting requirements.
Presence of clinically significant cardiovascular, cerebrovascular (stroke), immunological (except virus infection, viral hepatitis or cirrhosis), endocrine or central nervous system disorders; current encephalopathy; variceal bleeding requiring hospitalization or transfusion within past 4 months.
Pregnant, lactating or refusal to adopt barrier or chemical contraceptive use throughout trial and follow-up interval.

LV-FDPS is a Genetic Medicine Delivered by a Lentivirus Vector Via Local Administration to Liver for the Treatment of Hepatitis B Virus, Hepatitis C Virus, HIV or Other Viral Infection of the Liver—Concomitant Adjunct Bisphosphonate Therapy

A Phase I clinical trial will test safety and feasibility of delivering LV-FDPS to virally infected liver using ultrasound guided cannulation. It is rationally predicted that this study will result in the successful treatment of infections of the liver. The study is an open label, 4x3 dose escalation (4 dose ranges, up to 3 subjects per dose) to identify the

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maximum tolerable dose of LV-FDPS in patients 18 years or older with chronic viral disease of the liver that is resistant to chemotherapy.

LV-FDPS is a genetic therapy designed to reduce expression in tumor cells of the enzyme farnesyl diphosphate synthase. Experimental studies show that tumor cells modified by LV-FDPS induce human gamma delta T cells, including a capacity for cellular cytotoxicity against virally-infected cells. Prior experimental studies also showed the potential for positive interactions of LV-FDPS and specific bisphosphonate drugs that may be prescribed during infectious disease. For this study, subjects will receive dose escalating amounts of LV-FDPS with continuous standard of care dosing with Aredia® (pamidronate), Zometa® (zoledronic acid) or Actonel® (risedronate) according to physician advice and subject preference.

Subjects with confirmed viral infection of the liver including hepatitis B virus, hepatitis C virus, HIV or other viruses will initiate bisphosphonate therapy for 45 days before re-screening to meet enrollment criteria for LV-FDPS treatment of infectious disease. Eligible subjects are enrolled into the next available LV-FDPS dosing category. A maximum of 3 subjects are recruited for each dosage group. The LV-FDPS dose is a number of transducing units of LV-FDPS as described in the product release criteria, delivered via intra-hepatic cannulation in a single bolus with volume not to exceed 25 mL. The minimum dose is 1×10^9 transducing units and escalation is 10-fold to a next dose of 1×10^{10} transducing units, the next dose is 1×10^{11} transducing units, and a maximum dose of 1×10^{12} transducing units based on reported experience with recombinant adenovirus therapy for HCC (Sangro, et al., A phase I clinic trial of thymidine kinase-based gene therapy in advanced hepatocellular carcinoma, 2010, *Cancer Gene Ther.* 17:837-43). Subjects are enrolled, treated and evaluated for 3 months. All safety evaluations are completed for each group prior to enrolling and treating subjects at the next higher dose level. Enrollment and dose escalation continue until a maximum tolerable dose is achieved or the study is terminated.

Cannulation is via the left subclavian artery until tip of catheter is at the proper hepatic artery junction. Cannulation is guided by ultrasonography as described (Lin et al., Clinical effects of intra-arterial infusion chemotherapy with cisplatin, mitomycin C, leucovor and 5-Fluorouracil for unresectable advanced hepatocellular carcinoma, 2004, *J. Chin. Med. Assoc.* 67:602-10).

Primary Outcome Measures

Safety: Systemic and locoregional adverse events are graded according to CTCAS and coded according to MedRA. The adverse events data for all subjects at a single dose range will be evaluated prior to dose escalation. The final safety assessment incorporates data from all dose ranges.

Secondary Outcome Measures

Lesion distribution and retention of LV-FDPS following locoregional administration and subsequent biopsy or necropsy to obtain tissues.
Objective response rate (ORR) measured as a Sustained Viral Response (SVR) within the organ or systemically during 3 months after treatment.
Levels of LV-FDPS in blood stream during 10 minutes, 30 minutes, 1 hour and 1 day after local injection.
Changes in markers of hepatic function including ALP, ALT, ASAT, total bilirubin and GGT during 3 months after treatment.
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Inclusion Criteria

Greater than 18 years and including both males and females.

Diagnosis confirmed by histology or cytology or based on currently accepted clinical standards of chronic viral infection of the liver that is not amenable, at the time of screening, to resection, transplant or other potentially curative therapies.

Treating physician determines that the lesion is amenable to locoregional targeted delivery.

Karnofsky performance score 60-80% of ECOG values.

Life expectancy ≥ 12 weeks.

Hematopoietic function: WBC $\geq 2,500/\text{mm}^3$; ANC $\geq 1000/\text{mm}^3$; Hemoglobin ≥ 8 g/dL; Platelet count $\geq 50,000/\text{mm}^3$; Coagulation INR ≤ 1.3 .

AST and ALT < 5 times ULN; ALPS < 5 time ULN. Bilirubin ≤ 1.5 times ULN; Creatine ≤ 1.5 times ULN and eGFR ≥ 50 .

Thyroid function: Total T3 or free T3, total T4 or free T4 and THC $\leq$ CTCAE Grade 2 abnormality.

Renal, cardiovascular and respiratory function adequate in the opinion of the attending physician.

Immunological function: Circulating Vgamma9Vdelta2+ T cells $\geq 30/\text{mm}^3$; no immunodeficiency disease.

Negative for HIV by serology and viral RNA test.

Written informed consent.

Exclusion Criteria

Chronic viral disease amenable to resection, transplantation or other potentially curative therapies.

Hepatic surgery or chemoembolization within the past 4 months.

Hepatic radiation or whole body radiation therapy within past 4 months.

Investigational agents within 4 weeks or < 5 drug half-lives.

Impaired wound healing due to diabetes.

Significant psychiatric illness, alcohol dependence or illicit drug use.

Unwilling to comply with study protocols and reporting requirements.

Presence of clinically significant cardiovascular, cerebrovascular (stroke), immunological (except virus infection, viral hepatitis or cirrhosis), endocrine or central nervous system disorders; current encephalopathy; variceal bleeding requiring hospitalization or transfusion within past 4 months.

Pregnant, lactating or refusal to adopt barrier or chemical contraceptive use throughout trial and follow-up interval

Sequences

The following sequences are referred to herein:

SEQ ID NO:	Description	Sequence
1	FDPS shRNA sequence #1	GTCTGGAGTACAATGCCATTCTCGAGAATGGCATTGTA CTCCAGGACTTTTT
2	FDPS shRNA sequence #2	GCAGGATTTTCGTTACGACTTCTCGAGAAGTGCTGAACG AAATCCTGCTTTTT
3	FDPS shRNA sequence #3	GCCATGTACATGGCAGGAATTCTCGAGAATTCCTGCCAT GTACATGGCTTTTT
4	FDPS shRNA sequence #4	GCAGAAGGAGGCTGAGAAAGTCTCGAGACTTTCTCAGCC TCCTTCTGCTTTTT
5	miR30 FDPS sequence #1	AAGGTATATTGCTGTTGACAGTGAGCGACACTTTCTCAG CCTCCTTCTGCGTGAAGCCACAGATGGCAGAAGGAGGCT GAGAAAGTGCTGCCTACTGCCTCGGACTTCAAGGGGCT
6	miR30 FDPS sequence #2	AAGGTATATTGCTGTTGACAGTGAGCGACACTTTCTCAG CCTCCTTCTGCGTGAAGCCACAGATGGCAGAAGGAGGCT GAAAGTGCTGCCTACTGCCTCGGACTTCAAGGGGCT
7	miR30 FDPS sequence #3	TGCTGTGACAGTGAGCGACTTTCTCAGCCTCCTTCTGCG TGAAGCCACAGATGGCAGAAGGAGGCTGAGAAAGTTGC CTACTGCCTCGGA
8	miR155 FDPS sequence #1	CCTGGAGGCTTGCTGAAGGCTGTATGCTGACTTTCTCAG CCTCCTTCTGCTTTTGGCCACTGACTGAGCAGAAGGGCT GAGAAAGTCAGGACACAAGGCCTGTTACTAGCACTCA
9	miR21 FDPS sequence #1	CATCTCCATGGCTGTACCACCTTGTGCGGACTTTCTCAGC CTCCTTCTGCCTGTTGAATCTCATGGCAGAAGGAGGCGA GAAAGTCTGACATTTTGGTATCTTTCATCTGACCA
10	miR185 FDPS sequence #1	GGCCTGGCTCGAGCAGGGGGCGAGGGGATACTTTCTCAG CCTCCTTCTGCTGGTCCCTCCCGCAGAAGGAGGCTGA GAAAGTCCCTCCCTCCCAATGACCGCGTCTTCGTGC
11	Rous Sarcoma virus (RSV) promoter	GTAGTCTTATGCAATACTCTTGTAGTCTTGCAACATGGTA ACGATGAGTTAGCAACATGCCTTACAAGGAGAGAAAAA GCACCGTGATGCCGATTGGTGAAGTAAGGTGGTACGA TCGTGCCTTATTAGGAAGGCAACAGACGGGTCTGACATG GATTGGACGAACCACTGAATTGCCGCATTGCAGAGATAT TGTATTTAAGTGCCCTAGCTCGATACAATAAACG

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SEQ ID NO:	Description	Sequence
12	5' Long terminal repeat (LTR)	GGTCTCTCTGGTTAGACCAGATCTGAGCCTGGGAGCTCT CTGGCTAACTAGGGAACCCACTGCTTAAGCCTCAATAAA GCTTGCCCTTGAGTGCTTCAAGTAGTGTGTGCCGTCTGTT GTGTGACTCTGGTAACTAGAGATCCCTCAGACCCTTTTA GTCAGTGTGAAAATCTCTAGCA
13	Psi Packaging signal	TACGCCAAAAATTTGACTAGCGGAGGCTAGAAGGAGA GAG
14	Rev response element (RRE)	AGGAGCTTTGTTCTTGGGTTCTTGGGAGCAGCAGGAAG CACTATGGGCGCAGCCTCAATGACGCTGACGGTACAGGC CAGACAATTATTTGCTGGTATAGTGCAGCAGCAGAACAA TTTGCTGAGGGCTATTGAGGCGCAACAGCATCTGTTGCA ACTCACAGTCTGGGGCATCAAGCAGCTCCAGGCAAGAAT CCTGGCTGTGAAAAGATACCTAAGGATCAACAGCTCC
15	Central polypurine tract (cPPT)	TTTTAAAAGAAAAGGGGGGATTGGGGGGTACAGTGCAG GGGAAAGAATAGTAGACATAATAGCAACAGACATACAA ACTAAAGAATTACAAAACAAATTACAAAATTCAAAATT TTA
16	Polymerase III shRNA promoters; H1 promoter	GAACGCTGACGTCATCAACCCGCTCCAAGGAATCGCGGG CCCAGTGTCACTAGGCGGGAACCCAGCGCGCTGCGC CCTGGCAGGAAGATGGCTGTGAGGGAAGGGGAGTGGC GCCCTGCAATATTTGCATGTCGCTATGTGTTCTGGGAAAT CACCATAAACGTGAAATGTCTTTGGATTGGGAATCTTA TAAGTTCTGTATGAGACCATT
17	Long WPRE sequence	AATCAACCTCTGATTACAAAATTTGTGAAAGATTGACTG GTATTCCTTAACATATGTTGCTCCTTTTACGCTATGTGGATA CGCTGCTTTAATGCCTTTGTATCATGCTATGCTTCCCGT ATGGCTTTCATTTTCTCCTCCTGTATAAATCCTGGTTGC TGTCTCTTATGAGGAGTTGTGGCCCGTTGTCAAGCAAC GTGGCGTGGTGTGCACTGTGTTTGTGACGCAACCCCA CTGGTTGGGGCATTGCCACCACCTGTGAGCTCCTTTCCGG GACTTTCGCTTTCCTCCCTCCTATTGCCACGGCGAACTC ATCGCCGCTGCCTTGCCCGCTGTGACAGGGGCTCGG CTGTTGGGCACTGACAATTCGTGGTGTGTCGGGGAAA TCATCGTCTTCTCTGGCTGCTCGCTGTGTTGCCACCT GGATTCTGCGCGGACGTCTTCTGCTACGTCCTTCGGC CCTCAATCCAGCGGACCTTCTTCCCGCGGCTGCTGCC GGCTCTGCGGCTCTTCCGCTCTTCGCTTCGCCCTCAG ACGAGTCGGATCTCCCTTTGGGCGGCTCCCCGCT
18	3' delta LTR	TGGAAGGGCTAATTAACCTCCCAACGAAGATAAGATCTGC TTTTTGCTTGTACTGGGTCTCTGTTAGACCAGATCTG AGCCTGGGAGCTCTCTGGCTAACTAGGGAACCCACTGCT TAAGCCTCAATAAAGCTTGCTTGAAGTGTCTCAAGTAGT GTGTGCCGCTCTGTTGTGACTCTGGTAACTAGAGATC CCTCAGACCTTTTAGTCAGTGTGAAAATCTCTAGCAG TAGTAGTTCATGTCA
19	Helper/Rev; Chicken beta actin (CAG) promoter; Transcription	GCTATTACCATGGGTCGAGGTGAGCCCCACGTTCTGCTT CACTCTCCCATCTCCCCCCCCCTCCCAACCCCAATTTT TATTTATTTATTTTAAATTATTTGTGACGATGGGGG CGGGGGGGGGGGGGCGCGCCAGGCGGGGGGGGGC GGGCGAGGGGCGGGCGGGCGAGGCGGAGAGGTGC GGCGGCAGCCAATCAGAGCGGCGGCTCCGAAAGTTTCC TTTTATGGGAGGGCGGGCGGGCGGGCGGCTATAAAAA GCGAAGCGCGCGGGGGC
20	Helper/Rev; HIV Gag; Viral capsid	ATGGGTGCGAGAGCGTCAGTATTAAGCGGGGAGAATT AGATCGATGGGAAAAAATTCGGTTAAGGCCAGGGGGAA AGAAAAATATAAATTAAACATATAGTATGGGCAAGC AGGGAGCTAGAACGATTCGAGTTAATCTGGCCTGTTA GAAACATCAGAAGCTGTAGACAAATACTGGGACAGCT ACAACCATCCCTTCAGACAGGATCAGAAGAACTTAGATC ATTATATAATACAGTAGCAACCTCTATTGTGTGCATCA AAGGATAGAGATAAAAGACCAAGGAAGCTTTAGACA AGATAGAGGAAGAGCAAAACAAAGTAAGAAAAAGC ACAGCAAGCAGCAGCTGACACAGGACACAGCAATCAGG TCAGCCAAAATTACCTATAGTGCAGAACATCCAGGGGC AAATGGTACATCAGGCCATATCACCTAGAACTTTAAATG CATGGGTAAAGTAGTAGAAGAGAAGGCTTTACGCCCA GAAGTGATACCAATGTTTTCAGCATTATCAGAAGGAGCC ACCCCAAGATTTAAACACCATGCTAAACACAGTGGG

SEQ ID NO:	Description	Sequence
		GGACATCAAGCAGCCATGCAAAATGTTAAAAGAGACCAT CAATGAGGAAGCTGCAGAATGGGATAGAGTGCATCCAG TGCATGCAGGGCCTATTGCACCGGCCAGATGAGAGAA CCAAGGGGAAGTGACATAGCAGGAACCTACTAGTACCCCT CAGGAACAAATAGGATGGATGACACATAATCCACCTATC CCAGTAGGAGAAATCTATAAAAGATGGATAATCCTGGG ATTAATAAAATAGTAAGAATGTATAGCCCTACCAGCAT TCTGGACATAAGACAAGGACCAAGGAACCCCTTAGAG ACTATGTAGACCGATTCTATAAACTCTAAGAGCCGAGC AAGCTTCACAAGAGGTAAAAAATTGGATGACAGAACC TTGTTGGTCCAAAATGCGAACCCAGATTGTAAGACTATT TTAAGACATTGGGACCAGGAGCGACACTAGAAGAAAT GATGACAGCATGTGAGGAGTGGGGGACCCGGCCATA AAGCAAGAGTTTGGCTGAAGCAATGAGCCAAGTAACA AATCCAGCTACCAATAATGATACAGAAAGGCAATTTAGG AACCAAGAAAGACTGTTAAGTGTTCATTTGTGGCAA GAAGGGCACATAGCCAAAAATTCAGGGCCCTTAGGAA AAAGGGCTGTTGGAAATGTGGAAGGAAGGACACCAAA TGAAAGATTGTACTGAGAGACAGGCTAATTTTAGGGA AGATCTGGCCTTCCCACAAGGAAGGCCAGGGAATTTTC TTCAGAGCAGACCAGAGCCAACAGCCACCAGAAAGAG AGCTTCAGGTTTGGGGAAGAGACAACAACCTCCCTCTCAG AAGCAGGAGCCGATAGACAAGGAAGTATCCTTTAGCT TCCCTCAGATCACTCTTGGCAGCGACCCCTCGTCACAA AA
21	Helper/Rev; HIV Pol; Protease and reverse transcriptase	ATGAATTTGCCAGGAAGATGGAACCAAAATGATAGG GGGAATTGGAGGTTTTATCAAAGTAGGACAGTATGATCA GATACTCATAGAAATCTGCGGACATAAAGCTATAGGTAC AGTATTAGTAGGACCTACACCTGTCAACATAATTGGAAG AAATCTGTTGACTCAGATTGGCTGCCTTTAAATTTTCCC ATTAGTCTTATGAGACTGTACCAGTAAATTAAGGCCA GGAATGGATGGCCCAAAAGTTAAACAATGGCCATTGAC AGAAGAAAAATAAAAGCATTAGTAGAAATTTGTACAG AAATGGAAGGAAGGAAAAATTTCAAAATTTGGGCCT GAAAAATCCATACAATACTCCAGTATTTGCCATAAAGAAA AAAGACAGTACTAAATGGAGAAAAATTAGTAGATTTCAG AGAACTTAATAAGGAACTCAAGATTTCTGGGAAGTTCA ATTAGGAATACCAATCCTGCAGGGTTAAACAGAAAA AATCAGTAACAGTACTGGATGTGGCGCATGCATTTTTT CAGTTCCCTTAGATAAAGACTTCAGGAAGTATAGTCAT TTACCATACCTAGTATAAACAATGAGACACCAGGGATTA GATATCAGTACAATGTGCTTCCACAGGGATGGAAGGAT CACCAGCAATATTCAGTGTAGCATGACAAAAATCTTAG AGCCTTTTAGAAAAACAAATCCAGACATAGTCATCTATC AATACATGGATGATTGTATGTAGGATCTGACTTAGAAA TAGGGCAGCATAGAACAAAAATAGAGGAAGTGAACAA CATCTGTTGAGGTGGGATTACACACCAGACAAAAAA CATCAGAAAGAACCTCCATTCTTTGGATGGGTTATGAA CTCCATCCTGATAAATGGACAGTACAGCCTATAGTGTG CCAGAAAAGGACAGCTGGACTGTCAATGACATACAGAA ATTAGTGGGAAATTTGAATTGGGCAAGTCAGATTATGC AGGGATTAAAGTAAGGCAATTATGTAACTTCTTAGGGG AACCAGAGCTAACAGAAAGTAGTACCACTAACAGAA AAGCAGAGCTAGAACTGGCAGAAAAACAGGGAGATTCTA AAAGAACCGGTACATGGAGTGTATTATGACCCATCAAAA GACTTAATAGCAGAAATACAGAAAGCAGGGCAAGGCCA ATGGACATATCAAATTTATCAAGAGCCATTAAAAATCT GAAAAACAGGAAAAATGCAAGAAATGAAGGGTGCCACA CTAATGATGTGAAACAATTAAACAGAGGCAGTACAAAA ATAGCCACAGAAAGCATAGTAATATGGGGAAGACTCC TAAATTTAAATTACCCATACAAAAGGAACATGGGAAG CATGGTGGACAGAGTATTGGCAAGCCACCTGGATTCTG AGTGGGAGTTTGTCAATACCCCTCCCTTAGTGAAGTTAT GGTACAGTTAGAGAAAGAACCCATAATAGGAGCAGAA ACTTCTATGTAGATGGGGCAGCCAAAGGGAACTAAA TTAGGAAAAGCAGGATATGTAAGTACAGAGGAAGACA AAAAGTTGTCCCCCTAACGGACACAACAATCAGAAGA CTGAGTTACAAGCAATTCATCTAGCTTTCAGGATTTCG GATTAGAAGTAAACATAGTGACAGACTCACAATATGCAT TGGGAATCATTCAGGCACAACAGATAAGAGTGAATCA GAGTTAGTCAGTCAAATAATAGAGCAGTTAATAAAAA GGAAAAAGTCTACCTGGCATGGGTACCAGCACACAAAG GAATTGGAGGAAATGAACAAGTAGTGGTTGGTCAGT GCTGGAATCAGGAAAGTACTA

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SEQ ID NO:	Description	Sequence
22	Helper Rev; HIV Integrase; Integration of viral RNA	TTTTATAGATGGAATAGATAAGGCCCAAGAAGAACATGA GAAATATCACAGTAATTGGAGAGCAATGGCTAGTGATT TAACCTACCACCTGTAGTAGCAAAAGAAATAGTAGCCAG CTGTGATAAATGTCAGCTAAAAGGGGAAGCCATGCATG GACAAGTAGACTGTAGCCCAGGAATATGGCAGCTAGATT GTACACATTTAGAAGGAAAAGTTATCTTGGTAGCAGTTC ATGTAGCCAGTGGATATATAGAAGCAGAAGTAATTCCAG CAGAGACAGGGCAAGAAACAGCATACTTCCTCTTAAAT TAGCAGGAAGATGGCCAGTAAAAACAGTACATACAGAC AATGGCAGCAATTTACCAGTACTACAGTTAAGGCCGCC TGTGGTGGGCGGGGATCAAGCAGGAATTTGGCATTCCC TACAATCCCCAAAGTCAAGGAGTAATAGAATCTATGAAT AAAGAATTAAAGAAAATTATAGGACAGGTAAGAGATCA GGCTGAACATCTTAAGACAGCAGTACAAATGGCAGTATT CATCCCAATTTTAAAGAAAAGGGGGGATTGGGGGGT ACAGTGCAGGGGAAGAATAGTAGACATAATAGCAACA GACATACAACTAAAGAATTACAAAAACAAATTACAAA AATTCAAAATTTTCGGGTTTATTACAGGGACAGCAGAGA TCCAGTTTGGAAAGGACCAGCAAGCTCCTTGGAAGG TGAAGGGGAGTAGTAATACAAGATAATAGTGACATAA AAGTAGTGCCAAAGAAGAAAAGCAAGATCATCAGGGAT TATGGAACAAGATGGCAGGTGATGTTGTGTGGCAAGT AGACAGGATGAGGATTAA
23	Helper/Rev; HIV RRE; Binds Rev element	AGGAGCTTTGTTCTTGGGTTCTTGGGAGCAGCAGGAAG CACTATGGGCGCAGCGTCAATGACGCTGACGGTACAGGC CAGACAATTATGTCTGGTATAGTGACGACGAGCAACAA TTTGCTGAGGGCTATTGAGGCGCAACAGCATCTGTTGCA ACTCACAGTCTGGGCGCATCAAGCAGCTCCAGGCAAGAAT CCTGGCTGTGGAAAGATACCTAAAGGATCAACAGCTCCT
24	Helper/Rev; HIV Rev; Nuclear export and stabilize viral mRNA	ATGGCAGGAAGAAGCGGAGACAGCGACGAAGAACTCCT CAAGGCAGTCAGACTCATCAAGTTTCTCTATCAAAGCAA CCCACCTCCAATCCCGAGGGGACCCGACAGGCCCGAA GGAATAGAAGAAGAAGGTGGAGAGAGACAGAGACA GATCCATTTCGATTAGTGAACGGATCCTTAGCATTATCT GGGACGATCTGCGGAGCCTGTGCCTCTTCAGCTACCACC GCTTGAGAGACTTACTCTTGATTGTAACGAGGATTGTGG AACTTCTGGGACGACGGGGTGGGAAGCCCTCAAATATT GGTGGAATCTCCTACAATATTGGAGTCAGGAGCTAAAGA ATAG
25	Envelope; CMV promoter; Transcription	ACATTGATTATTGACTAGTTATTAATAGTAATCAATTACG GGGTCATTAGTTCATAGCCCATATATGGAGTCCCGGTT ACATAACTTACGGTAAATGGCCCGCTGGCTGACCGCCC AACGACCCCGCCCATTTGACGTCAATAATGACGTATGTT CCCATAGTAACGCCAATAGGGACTTTCATTGACGTCAA TGGGTGGAGTATTACGGTAACTGCCCACTTGGCAGTA CATCAAGTGATCATATGCCAAGTACGCCCCCTATTGAC GTCAATGACGGTAAATGGCCCGCTGGCATTATGCCCG TACATGACCTTATGGGACTTTCCTACTTGGCAGTACATCT ACGTATTAGTCATCGCTATTACATGGTGATGCGGTTTGT GCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCAG GGGATTTCCAAGTCTCCACCCCATTTGACGTCAATGGGAG TTTGTTTGGCACCAAAATCAACGGGACTTCCAAAATG TCGTAACAACCTCCGCCCATTTGACGCAATGGGCGGTAG GCGTGACGGTGGGAGGTCTATATAAGC
26	Envelope; VSV- G; Glycoprotein envelope-cell entry	ATGAAGTGCCTTTTGTACTTAGCCTTTTATTTCATTGGGG TGAATTGCAAGTTCACCATAGTTTTCACACACACCAAAA AAGGAAACTGGAAAAATGTTCTTCTAATTACCATTTAT GCCCGTCAAGCTCAGATTTAAATTGGCATAATGACTTAA TAGGCACAGCCTTACAAGTCAAATGCCCAAGAGTCACA AGGCTATTCAAGCAGACGGTTGGATGTGTATGCTTCCA AATGGGTCACTACTTGTGATTTCCGCTGGTATGGACCGA AGTATATAACACATTCCATCCGATCCTTCACTCCATCTGT AGAACATGCAAGGAAAGCATTGAACAAACGAAACAAG GAACCTGGCTGAATCCAGGCTTCCCTCCTCAAAGTTGTG GATATGCAACTGTGACGGATGCCGAAGCAGTGATTGTCC AGGTGACTCCTCACCATGTGCTGGTTGATGAATACACAG GAGAATGGGTTGATTACAGTTTATCAACGGAATGCA GCAATTACATATGCCCCACTGTCCATAACTCTACAACCT GGCATTCTGACTATAAGGTCAAAGGGCTATGTGATTCTA ACCTCATTTCCATGGACATCACCTTCTTCTCAGAGGACG GAGAGCTATCATCCTGGGAAAGGAGGGCACAGGGTTC AGAAGTAACTACTTTGCTTATGAACCTGGAGGCCAAGGCC

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SEQ ID NO:	Description	Sequence
		TGCAAAATGCAATACTGCAAGCATTGGGGAGTCAGACTC CCATCAGGTGTCTGGTTCGAGATGGCTGATAAGGATCTC TTGTCTGCAGCCAGATTCCCTGAATGCCCAGAAGGGTCA AGTATCTCTGCTCCATCTCAGACCTCAGTGGATGTAAGT CTAATTCAGGACGTTGAGAGGATCTTGGATTATTCCTCT GCCAAGAAACCTGGAGCAAATCAGAGCGGGTCTTCCA ATCTCTCCAGTGGATCTCAGCTATCTTGCTCCTAAAAACC CAGGAACCGGTCTGCTTTCACCATAATCAATGGTACCC TAAATACTTTGAGACCAGATACATCAGAGTCGATATTG CTGCTCCAATCCTCTCAAGAATGGTCGGAATGATCAGTG GAACTACCACAGAAAGGGAACGTGGGATGACTGGGCA CCATATGAAGACGTGGAATGGACCAATGGAGTTCTG AGGACCACTTCAGGATATAAGTTTCTTTATACATGATT GGACATGGTATGTTGGACTCCGATCTTCATCTTAGCTCA AAGGCTCAGGTGTTGCAACATCTCATTCAAGACGCT GCTTCGCAACTTCTGATGATGAGAGTTATTTTGGTG ATACTGGGCTATCCAAAAATCCAATCGAGCTTGTAGAAG GTTGGTTCAGTAGTTGGAAGAGCTCTATTGCCTCTTTTT CTTTATCATAGGGTAATCATTGGACTATTCTGGTTCTC CGAGTTGGTATCCATCTTTGCATTAAATTAAAGCACACC AAGAAAAGACAGATTTATACAGACATAGAGATGA
27	Helper/Rev; CMV early (CAG) enhancer; Enhance Transcription	TAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCAT AGCCCATATATGGAGTTCGCGTTACATAACTTACGGTA AATGGCCCGCTGGCTGACCGCCCAACGACCCCGCCCA TTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCA ATAGGGACTTTCCATTGACGTCAATGGGTGGACTATTTA CGGTAAACTGCCACTTGGCAGTACATCAAGTGTATCAT ATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAA TGGCCCCCTGGCATTATGCCAGTACATGACCTTATGG GACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATC
28	Helper/Rev; Chicken beta actin intron; Enhance gene expression	GGAGTCGCTGCGTTGCCTTCGCCCCGTGCCCGCTCCGC GCCGCTCGCGCCGCCCGCCCGCTCTGACTGACCGCG TTAATCCACAGGTGAGCGGGCGGACGGCCCTTCTCCT CCGGGCTGTAATTAGCGCTTGGTTTAATGACGGCTCGTTT CTTTTCTGTGGCTGCGTGAAAGCCTTAAAGGGCTCCGGG AGGGCCCTTTGTGCGGGGGGAGCGGCTCGGGGGTGC GTGCGTGTGTGTGCGTGGGGAGCGCCGCTGCGGCC GCGCTGCCCGCGGCTGTGAGCGCTGCGGGCGCGCGC GGGGCTTTGTGCGCTCCGCGTGTGCGGAGGGGAGCGCG GCCGGGGCGGTGCCCGCGGTGCGGGGGGCTGCGAG GGGAACAAAGGCTGCGTGCGGGGTGTGTGCTGGGGG GTGAGCAGGGGTGTGGGCGCGGCTCGGGCTGTAAAC CCCCCCTGCACCCCTCCCGAGTTGCTGAGCACGGC CCGGCTTCGGGTGCGGGCTCCGTGCGGGGCTGGCGCG GGGCTCGCCGTGCCGGCGGGGGTGGCGGCAGGTGGG GGTGCCGGCGGGCGGGGCGCCCTCGGGCCGGGAGG GCTCGGGGAGGGGCGCGCGGCCCGGAGCGCGCGG GCTGTCGAGGCGCGGCGAGCCGAGCCATTGCCTTTTAT GGTAATCGTGCGAGAGGGCGCAGGACTTCCTTTGTCCC AAATCTGCGGAGCCGAAATCTGGGAGGCGCCCGCA CCCCCTCTAGCGGCGCGGGCGAAGCGGTGCGGCGCG GCAGGAAGGAAATGGCGGGGAGGGCTTCGTGCGTCG CCGCGCGCGCTCCCTTCTCCATCTCCAGCCTCGGGGT GCCGCAGGGGACGGCTGCCTTCGGGGGGACGGGGCA GGGCGGGGTTCGGCTTCTGGCGTGTGACCGCGG
29	Helper/Rev; Rabbit beta globin poly A; RNA stability	AGATCTTTTCCCTCTGCCAAAAATTATGGGGACATCAT GAAGCCCTTGAGCATCTGACTTCTGGCTAATAAAGGAA ATTTATTTTATTGCAATAGTGTGTTGGAATTTTGTGT CTCTCACTCGGAAGGACATATGGGAGGGCAAATCATTTA AAACATCAGAATGAGTATTTGGTTTAGAGTTGGCAACA TATGCCATATGCTGGCTGCCATGAACAAAGTGGCTATA AAGAGGTCATCAGTATATGAAACAGCCCCCTGCTGTCCA TTCCTTATTCATAGAAAAGCCTTGACTTGAGGTTAGATT TTTTTATATTTGTTTTGTGTTATTTTTCTTTAACATCC CTAAAATTTTCTTACATGTTTACTAGCCAGATTTTCC TCCTCTCTGACTACTCCAGTCATAGCTGCCCTCTCT CTTATGAAGATC
30	Envelope; Beta globin intron; Enhance gene expression	GTGAGTTTGGGGACCTTGATTGTTCTTTCTTTTCGCTA TTGTAAATTCATGTTATATGGAGGGGCAAGTTTCA GGGTGTGTGTTAGAATGGGAAGATGTCCTTGTATCACC ATGGACCCTCATGATAATTTGTTTCTTCACTTTCTACT CTGTGACAACCATGTCCTCTTATTTCTTTTCATTTT

SEQ ID NO:	Description	Sequence
		CTGTAACCTTTTCGTTAACTTTAGCTTGCATTGTAAACG AATTTTAAATTCACCTTTTGTATTGTGTCAGATTGTAAG TACTTTCTCTAATCACTTTTTTTCAAGGCAATCAGGGTA TATTATATTGTACTTCAGCACAGTTTTAGAGAACAATTGT TATAATTAAATGATAAGGTAGAATATTTCTGCATATAAA TTCTGGCTGGCGTGGAAATATTCTTATGGTAGAAACAA CTACACCCTGGTCATCATCCTGCCTTTCTCTTATGGTTA CAATGATATACACTGTTTGAGATGAGGATAAAATACTCT GAGTCCAAACCGGGCCCTCTGCTAACCATGTTTCATGCC TTCTTCTCTTTCTACAG
31	Envelope; Rabbit beta globin poly A; RNA stability	AGATCTTTTCCCTCTGCCAAAAATTATGGGGACATCAT GAAGCCCCTTGAGCATCTGACTTCTGGCTAATAAAGGAA ATTTATTTTCATTGCAATAGTGTGTGGAATTTTTGTGT CTCTCACTCGGAAGGACATATGGGAGGGCAAAATCATTTA AAACATCAGAATGAGTATTTGGTTTAGAGTTGGCAACA TATGCCCATATGCTGGCTGCCATGAACAAAGGTTGGCTA TAAAGAGGTCACTAGTATATGAACAGCCCCCTGCTGTC CATTCTTATTCCATAGAAAAGCCTTGACTTGAGGTTAG ATTTTTTTTATATTTGTTTTGTGTTATTTTTTCTTTAACA TCCCTAAATTTTCTTACATGTTTACTAGCCAGATTTT TCCTCCTCTCTGACTACTCCAGTCATAGCTGTCCCTCT TCTCTTATGGAGATC
32	Primer	TAAGCAGAATTCATGAATTTGCCAGGAAGAT
33	Primer	CCATACAATGAATGGACACTAGGCGGCCGCAGAAAT
34	Gag, Pol, Integrase fragment	GAATTCATGAATTTGCCAGGAAGATGGAAACCAAAAT GATAGGGGGAATTTGGAGGTTTATCAAAGTAAGACAGT ATGATCAGATACTCATAGAAATCTGCGGACATAAAGCTA TAGGTACAGTATTAGTAGGACCTACACCTGTCAACATAA TTGGAAGAAATCTGTTGACTCAGATTGGCTGCACTTTAA ATTTTCCATTAGTCTCTATTGAGACTGTACCAGTAAAT AAAGCCAGGAATGGATGGCCCAAAGTTAAACAATGGC CATTGACAGAAGAAAAAATAAAGCATAGTAGAAATTT TGACAGAAATGGAAAAGGAAGGAAAAATTTCAAAAT TGGGCTGAAAAATCCATACAATACTCCAGTATTTGCCAT AAAGAAAAAGACAGTACTAAATGGAGAAATTAGTAG ATTTAGAGAACTTAATAAGAGAACTCAAGATTTCTGGG AAGTTCAATTAGGAATACCACATCCTGCAGGGTTAAAC AGAAAAAATCAGTAACAGTACTGGATGTGGCGCATGCA TATTTTTCAGTTCCCTTAGATAAAGACTTCAGGAAGTATA CTGCATTTACCATACCTAGTATAAACAATGAGACACCAG GGATTAGATATCAGTACAATGTGCTTCCACAGGGATGGA AAGGATCACCAGCAATATTCCAGTGTAGCATGACAAA ATCTTAGAGCCTTTAGAAAAAATAATCCAGACATAGTC ATCTATCAATACATGGATGATTGTATGTAGGATCTGAC TTAGAAATAGGGCAGCATAGAACAAAAATAGAGGAACT GAGACAACATCTGTTGAGGTGGGGATTTACCACACCAGA CAAAAAACATCAGAAAGAACCTCCATTCTCTTGGATGGG TTATGAACTCCATCTGATAAATGGACAGTACAGCCTAT AGTGCTGCCAGAAAAGGACAGCTGGACTGTCAATGACA TACAGAAATTAGTGGGAAATTTGAATTTGGGCAAGTCAG ATTTATGCAGGGATTAAAGTAAGGCAATTATGTAACTT CTTAGGGGAACCAAAGCACTAACAGAAGTAGTACCCT AACAGAGAAGCAGAGCTAGAACTGGCAGAAAAACAGGG AGATTCTAAAGAACCGGTACATGGAGTGTATTATGACC CATCAAAAGACTTAATAGCAGAAATACAGAAGCAGGGG CAAGGCCAATGGACATATCAAATTTATCAAGAGCCATTT AAAAATCTGAAACAGGAAAGTATGCAAGAATGAAGGG TGCCCACTAATGATGTGAAACAATTAACAGAGGCAGT ACAAAAAATAGCCACAGAAAGCATAGTAATATGGGGAA AGACTCTAAATTTAAATTACCATACAAAGGAAACAT GGGAAGCATGGTGGACAGAGTATTGGCAAGCCCTGG ATTCCTGAGTGGGAGTTTGTCAATACCCCTCCCTTAGTGA AGTTATGGTACCAGTTAGAGAAAGAACCCATAATAGGA GCAGAACTTTCTATGTAGATGGGGCAGCCAATAGGGA AACTAAATTAGGAAAAGCAGGATATGTAACGTACAGAG GAAGACAAAAAGTTGTCCCCCTAACGACACACAAAT CAGAAGACTGAGTTACAAGCAATTCATCTAGCTTTGCAG GATTCGGGATTAGAAGTAAACATAGTGACAGACTCACA ATATGCATTGGGAATCATTCAAGCACACCAGATAAGAG TGAATCAGAGTTAGTCAGTCAAATAATAGAGCAGTTAAT AAAAAGGAAAAAGTCTACCTGGCATGGGTACCAGCAC ACAAAGGAATTTGGAGGAATGAACAAGTAGATAAATTG

SEQ ID NO:	Description	Sequence
		GTCAGTGCTGGAATCAGGAAAGTACTATTTTATAGATGGA ATAGATAAGGCCCAAGAAGACATGAGAAATATCACAG TAATTGGAGAGCAATGGCTAGTGATTTTAACTTACCACC TGTAGTAGCAAAGAAATAGTAGCCAGCTGTGATAAAT GTCAGCTAAAAGGGGAAGCCATGCATGGACAAGTAGAC TGTAGCCAGGAATATGGCAGCTAGATTGTACACATTTA GAAGGAAAAGTTATCTTGGTAGCAGTTTCATGTAGCCAGT GGATATATAGAAGCAGAAGTAATTCAGCAGAGACAGG GCAAGAAACAGCATACTTCTCTTAAATTTAGCAGGAAG ATGGCCAGTAAAAACAGTACATACAGACAATGGCAGCA ATTTCCAGTACTACAGTTAAGGCCGCTGTGTGGTGGG CGGGGATCAAGCAGGAATTTGGCATTCCCTACAATCCCC AAAGTCAAGGAGTAATAGAACTATGAATAAAGAATTA AAGAAAATTATAGGACAGGTAAGAGATCAGGCTGAACA TCTTAAGACAGCAGTACAAATGGCAGTATTATCCACAA TTTTAAAGAAAAGGGGGGATTGGGGGTACAGTGACAG GGGAAAGAATAGTAGACATAATAGCAACAGACATACAA ACTAAAGAATTACAAAAACAAATTACAAAAATTCAAAA TTTTCGGGTTTATTACAGGACAGCAGAGATCCAGTTTG GAAAGGACCAGCAAAGCTCCTCTGAAAGGTGAAGGGG CAGTAGTAATACAAGATAATAGTGACATAAAGTAGTG CCAAGAAGAAAAGCAAAGATCATCAGGGATTATGAAA ACAGATGGCAGGTGATGATTGTGTGGCAAGTAGACAGG ATGAGGATTAA
35	DNA Fragment containing Rev, RRE and rabbit beta globin poly A	TCTAGAATGGCAGGAAGAAGCGGAGACAGCGACGAAGA GCTCATCAGAACAGTCAGACTCATCAAGCTTCTCTATCA AAGCAACCCACCTCCCAATCCCGAGGGGACCCGACAGG CCCGAAGGAATAGAAGAAGAAGGTGGAGAGAGAGACA GAGACAGATCCATTTCGATTAGTGAACGGATCCTTGGCAC TTATCTGGGACGATCTGCGGAGCCTGTGCCCTCTCAGCT ACCACCGCTTGAGAGACTTACTCTTGATTGTAACGAGGA TTGTGGAACCTCTGGGACGCAAGGGGTGGGAAGCCCTCA AATATTGGTGGAACTCTCTACAATATTGGAGTCAGGAGC TAAAGAATAGAGGAGCTTTGTTCTTGGGTTCTTGGGAG CAGCAGGAAGCACTATGGGCGCAGCGTCAATGACGCTG ACGGTACAGGCCAGACAATTATTGTCTGGTATAGTGACG CAGCAGAAACAATTTGCTGAGGGCTATTGAGGCGCAACA GCATCTGTTGCAACTCACAGTCTGGGGCATCAAGCAGCT CCAGGCAAGAATCCTGGCTGTGGAAGATACCTAAAGG ATCAACAGCTCCTAGATCTTTTCCCTCTGCCAAAAATTA TGGGGACATCATGAAGCCCTTGAGCATCTGACTTCTGG CTAATAAAGGAAATTTATTTTCATTGCAATAGTGTGTG GAATTTTTTGTGCTCTCACTCGGAAGGACATATGGGAG GGCAAAATCATTTAAACATCAGAATGAGTATTTGGTTTA GAGTTTGGCAACATATGCCATATGCTGGCTGCCATGAAC AAAGGTGGCTATAAAGAGGTATCAGTATATGAAACAG CCCCTGCTGTCCATTCTTATTCATAGAAAAGCCTTGA CTTGAGGTTAGATTTTTTTATATTTTGTGTTTATTT TTTTCTTTAACATCCCTAAAATTTTCTTACATGTTTACT AGCCAGATTTTCTCTCTCTCTGACTACTCCAGTCATA GCTGTCCCTCTTCTTATGAAGATCCCTCGACCTGCAGC CCAAGCTTGGCGTAATCATGGTCATAGCTGTTTCTGTGT GAAATTGTTATCCGCTCACAAATCCACACAACATACGAG CCGGAAGCATAAAGTGTAAGCCTGGGGTGCTTAATGA GTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCC GCTTTCAGTCGGGAAACCTGTCGTGCCAGCGGATCCGC ATCTCAATTAGTCAGCAACCATAGTCCCGCCCCTAATC CGCCCATCCCGCCCCTAATCCGCCCAGTTCCGCCCATT TCCGCCCCATGGCTGACTAATTTTTTTATTTATGCAGAG GCCGAGGCCGCTCGGCCTGAGCTATTCCAGAAGTAG TGAGGAGGCTTTTTTGAGGCGCTAGGCTTTTGCAAAAAG CTAACCTTGTTTATGTCAGCTTATAATGGTTACAAATAAA GCAATAGCATCACAAATTTCAAAATAAAGCATTTTTTT CACTGCATTCTAGTTGTGGTTTGTCCAACTCATCAATGT ATCTTATCAGCGCGCCCGCCGG
36	DNA fragment containing the CAG enhancer/promoter/ intron sequence	ACGCGTTAGTTATTAATAGTAATCAATTACGGGTCATT AGTTCATAGCCATATATGGAGTTCGCGTTACATAACT TACGGTAAATGGCCCGCTGGCTGACCGCCCAACGACCC CCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGT AACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGA CTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGT GTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGA CGGTAAATGGCCCGCTGGCATATGCCAGTACATGAC CTTATGGGACTTTCTACTTGGCAGTACATCTACGTATTA

SEQ ID NO:	Description	Sequence
		GTCATCGCTATTACCATGGGTCGAGGTGAGCCCCACGTT CTGCTTCACTCTCCCCATCTCCCCCCCCCTCCCCACCCCCA ATTTTGTATTATTATTTTAAATTATTTTGTGCAGCGAT GGGGGCGGGGGGGGGGGGGCGCGCCAGCGGGG GGGGCGGGGCGAGGGCGGGGCGGGGCGAGGCGGAGA GGTGCGGCGGCAGCAATCAGAGCGGCGCTCCGAAA GTTTCCTTTTATGGCGAGGCGGCGGCGGCGGCGCCTA TAAAAAGCGAAGCGCGCGGGCGGGGAGTCGCTGCGT TGCCTTCGCCCCGTGCCCCGCTCCGCGCGCCTCGCGCC GCCCGCCCCGCTCTGACTGACCGGTACTCCACAGG TGAGCGGGCGGACGGCCCTTCTCCTCCGGGCTGTAATT AGCGCTTGGTTAATGACGCTCGTTTCTTTCTGTGGCT GCGTGAAAGCCTTAAAGGGCTCCGGGAGGGCCCTTTGTG CGGGGGGAGCGGCTCGGGGGTGCGTGCGTGTGTGTG TCGTGGGGAGCGCCGCTGCGGCGCGCTGCCGCGG GCTGTGAGCGTGCGGCGCGGCGCGGGCTTTGTGCGC TCCGCGTGTCGCGAGGGGAGCGCGCGGGGGCGGTG CCCC CGCGTGCGGGGGGGCTGCGAGGGGAACAAAGGCT GCGTGCGGGGTGTGTGCGTGCGGGGGTGAGCAGGGGT GTGGGCGCGCGGTGCGGGCTGTAACCCCCCTGCACCC CCCTCCCCGAGTTGCTGAGCACGGCCCGGCTTCGGGTGC GGGGCTCCGTGCGGGCGTGCGCGGGGCTCGCGTGCC GGGCGGGGGTGGCGGCAAGTGGGGTGCCGGGCGGGG CGGGGCGCGCTCGGGCCGGGAGGGCTCGGGGAGGGG CGCGGCGCGCCCGAGCGCGCGGCTGTGAGGCGCG GCGAGCGCAGCCATTGCCTTTTATGGTAATCGTGCGAG AGGGCGCAGGACTTCCTTTGTCCAAATCGCGGAGC CGAAATCTGGGAGGCGCGCGCACCCCTTAGCGGGC GCGGGCGAAGCGTGCGGCGCGGCGAGGAAGGAATGG GCGGGGAGGGCTTCGTGCGTCGCGCGCGCGCTCCCC TTCTCCATCTCCAGCTCGGGCTGCGCGAGGGGACGG CTGCCTTCGGGGGGACGGGGAGGGCGGGTTTCGCTT CTGGCGTGACCGGCGGAATTC
37	DNA fragment containing VSV-G	GAATTCATGAAGTGCCTTTTGTACTTAGCCTTTTATTCA TTGGGGTGAATTGCAAGTTCACCATAGTTTTCACACA ACCAAAAGGAACTGGAAAATGTTCTTCTAATTACC ATTATTGCCCGTCAAGCTCAGATTAAATTGGCATAATG ACTTAATAGGCACAGCCTTACAGTCAAAATGCCAAGA GTCAAGGCTATTCAAGCAGACGGTTGGATGTGTATG CTTCCAAATGGGTCACTACTTGTGATTTCCGCTGGTATGG ACCGAAGTATATAACACATCCATCCGATCCTTCACTCC ATCTGTAGAACAATGCAAGGAAGCATTGAACAAACGA AACAAAGGAATTGGCTGAATCCAGGCTTCCCTCCTCAA GTTGTGGATATGCAACTGTGACGGATGCCGAAGCAGTGA TTGTCCAGGTGACTCCTCACCATGTGCTGGTTGATGAAT ACACAGGAGAATGGGTTGATTACAGTTCATCAACGGAA AATGCAGCAATTACATATGCCCCACTGTCCATAACTCTA CAACCTGGCATTCGACTATAAGTCAAAGGGCTATGTG ATTCTAACCTCATTTCCATGGACATCACCTTCTTCTCAGA GGACGGAGAGCTATCATCCCTGGGAAAGGAGGGCACAG GGTTCAAGTAATACTTTGCTTATGAACTGGAGGCA AGGCCTGCAAAATGCAATCTGCAAGCATTGGGGAGTC AGACTCCCATCAGGTGTCTGGTTCGAGATGGCTGATAAG GATCTCTTTGCTGCGCCAGATTCCCTGAATGCCAGAA GGGTCAAGTATCTCTGCTCCATCTCAGACCTCAGTGGAT GTAAGTCTAATTCAGGACGTTGAGAGGATCTTGGATTAT TCCCTCTGCCAAGAACTGGAGCAAAATCAGAGCGGT CTTCCAATCTCTCAGTGGATCTCAGTATCTTGCTCCTA AAAACCCAGGAACCGGTCCTGCTTACCATAATCAATG GTACCCATAAAATACTTTGAGACCAGATACATCAGAGTCG ATATTGCTGCTCCATCTCTCAAGATGGTCGGAATGA TCAGTGGAATACACAGAAAGGAACTGTGGGATGAC TGGGACCATATGAAGACGTGGAAATTGGACCCAATGG AGTTCTGAGGACAGTTCAGGATATAAGTTTCCTTTATA CATGATTGGACATGGTATGTTGGACTCCGATCTTCATCTT AGCTCAAAGGCTCAGGTGTTCAACATCCTCACATTCAA GACGCTGCTTCGCAACTTCTGATGATGAGAGTTTATTTT TTGGTGATCTGGGCTATCCAAAAATCCAATCGAGCTTG TAGAAGGTTGGTTCAGTAGTTGGAAGCTCTATTGCCT CTTTTTCTTTATCATAGGGTTAATCATTGGACTATTCTT GGTTCTCCGAGTTGGTATCCATCTTGCATTAAATTAAAG CACACCAAGAAAAGACAGATTTATACAGACATAGAGAT GAGAATTC

SEQ ID NO:	Description	Sequence
38	RSV promoter and HIV Rev	CAATTGCGATGTACGGGCCAGATATACGCGTATCTGAGG GGACTAGGGTGTGTTTAGGCGAAAAGCGGGCTTCGGTT GTACGCGGTAGGAGTCCCCTCAGGATATAGTAGTTTCG CTTTTGATAGGGAGGGGAAATGTAGTCTTATGCAATA CACTTGTAGTCTTGCAACATGGTAACGATGAGTTAGCAA CATGCCTTACAAGGAGAGAAAAAGCACCGTCATGCCG ATTGGTGAAGTAAGTGGTACGATCGTGCTTATTAGG AAGGCAACAGACAGGTCTGACATGGATTGGACGAACCA CTGAATTCGCGATTGCAGAGATAATTGATTTAAGTGCC TAGCTCGATACAATAAACGCCATTGACCATTACCCACA TTGGTGTGCACCTCCAAGCTCGAGCTCGTTTAGTGAACC GTGAGATCGCCTGGAGACGCCATCCACGCTGTTTGACC TCCATAGAAGACACCGGGACCGATCCAGCCTCCCTCGA AGCTAGCGATTAGGCATCTCCTATGGCAGGAAGAAGCG GAGACAGCGACGAAGAACTCCTCAAGGCAGTCAGACTC ATCAAGTTTCTCTATCAAAGCAACCCACCTCCCAATCCC GAGGGGACCCGACAGGCCCGAAGGAATAGAAGAAGAA GGTGGAGAGAGAGACAGAGACAGATCCATTGATTAGT GAACGGATCCTTAGCACTTATCTGGGACGATCTGCGGAG CCTGTGCCTCTTCAGCTACCACCGCTTGAGAGACTTACTC TTGATTGTAACGAGGATTGTGGAATCTCTGGGACGCAGG GGGTGGGAAGCCCTCAAATATTGGTGAATCTCTACAA TATTGGAGTCAGGAGCTAAAGAATAGTCTAGA
39	Elongation Factor- 1 alpha (EF1- alpha) promoter	CCGGTGCCTAGAGAAGGTGGCGCGGGGTAAACTGGGAA AGTGATGTCGTGACTGGCTCCGCCTTTTCCCGAGGGTG GGGGAGAACCGTATATAAGTGCAGTAGTCGCCGTGAAC GTTCTTTTTCGCAACGGGTTTGCCGCCAGAACACAGGTA AGTGCCGTGTGTGGTTCCCGCGGGCTTGGCCTCTTTACG GGTTATGGCCCTTGCGTGCTTGAATTACTTCCACGCCCC TGGCTGCAGTACGTGATTCTTGATCCCGAGCTTCGGGTT GGAAGTGGGTGGGAGAGTTCGAGGCCTTGCGCTTAAGG AGCCCCCTTCGCCTCGTGCTTGAGTTGAGGCTTGGCCTGG GCGCTGGGCGCCGCGCTGCGAATCTGGTGGCACCTTCG CGCTGTCTCGCTGCTTTTCGATAAGTCTCTAGCCATTTAA AATTTTGTGATGACCTGCTGCGACGCTTTTTCCTGGCAAG ATAGTCTTGTAATGCGGGCCAAGATCTGCACACTGGTA TTTCGGTTTTTGGGGCGCGGGCGGCGACGGGGCCCGTG CGTCCAGCGCACATGTTGCGCGAGGCGGGGCTGCGAG CGCGGCCACCGAGAATCGGACGGGGTAGTCTCAAGCT GGCGGCGCTGCTCTGGTGCTTGGCCTCGCGCCGCGGTGT ATCGCCCCGCTTGGGCGGAAGGCTGGCCCGGTGCGCA CCAGTTGCGTGAGCGGAAAGATGGCCGCTTCCCGGCCCT GCTGCAGGGAGCTCAAAATGGAGGACGCGCGCTCGGG AGAGCGGGCGGGTGAGTCACCCACACAAAGGAAAAGGG CCTTTCGCTCCTCAGCCGTGCTTCATGTGACTCCACGGA GTACCGGGCGCGTCCAGGCACCTCGATTAGTTCTCGAG CTTTTGAGTACGTCGCTTTAGGTTGGGGGAGGGGTT TTATGCGATGGAGTTTCCCACTGAGTGGGTGGAGAC TGAAGTTAGGCCAGCTTGGCACTTGATGTAATTCTCCTTG GAATTTGCCCTTTTGTAGTTTGGATCTTGTTTATTCTCA AGCCTCAGACAGTGGTTCAAAGTTTTTTCTTCCATTCTCA GGTGTCTGTGA
40	Promoter; PGK	GGGGTTGGGTTGCGCCTTTTCCAAGCGAGCCCTGGGTT TGCGCAGGGACGCGGCTGCTCTGGGCTGGTTCCGGGAA ACGCAGCGGCGCCGACCTGGGTCTCGCACATTCTTCAC GTCCGTTTCGAGCGTCAACCGGATCTTCGCGCTACCCCT GTGGGCCCCCGGCGACGCTTCTGCTCCGCCCTAAGT CGGGAAGGTTCCCTGCGGTTGCGGCGTGCCGGACGTGA CAAACGGAAGCGCACGTCTCACTAGTACCCTCGCAGAC GGACAGCGCCAGGAGCAATGGCAGCGCGCCGACCGCG ATGGGCTGTGGCCAATAGCGGCTGCTCAGCAGGCGCGCG CGAGAGCAGCGGCCGGAAGGGCGGTGCGGAGGCGCG GGTGTGGGCGGTAGTGTGGGCCCTGTTCTGCCCCGCGC GGTGTCCGCAATTCTGCAAGCCTCCGAGCGCACGTCGG CAGTCGGCTCCCTCGTTGACCGAATACCCGACCTCTCTCC CCAG
41	Promoter; UbC	GCGCCGGGTTTTTGGCGCCTCCCGCGGGCGCCCCCTCCT CACGGCGAGCGCTGCCACGTGAGACGAAGGCGCAGGA GCGTTCTGTATCCTTCGCCCCGACGCTCAGGACAGCGG CCCGCTGCTCATAAGACTCGGCCTTAGAACCCAGTATC AGCAGAAGGACATTTTAGGACGGGACTTGGGTGACTCTA GGGCACTGGTTTTCTTCCAGAGAGCGGAACAGGCGAGG AAAAGTAGTCCCTTCTCGCGATTCTGCGGAGGATCTC

SEQ ID NO:	Description	Sequence
		CGTGGGGCGGTGAACGCCGATGATTATATAAGGACGCG CCGGGTGTGGCAGCTAGTTCCGTCGCAGCCGGGATTT GGGTCGCGGTTCTTGTTTGTGGATCGCTGTGATCGTCACT TGGTGAGTTGCGGGCTGCTGGGCTGGCCGGGGCTTTCTGT GGCCGCCGGGCCGCTCGGTGGGACGGAAGCGTGTGGAG AGACGCCCAAGGGCTGTAGTCTGGGTCGCGAGCAAGG TTGCCCTGAACCTGGGGGTTGGGGGAGCGCACAAAATG GCGGCTGTTCCCGAGTCTTGAATGGAAGACGCTTGTAAAG GCGGGCTGTGAGGTCGTTGAAACAAGGTGGGGGCGATG GTGGCGGCAAGAACCAGGTCTTGAGGCTTCGCTAA TGCGGGAAAGCTCTTATTCGGGTGAGATGGGCTGGGGCA CCATCTGGGACCTGACGTGAAGTTTGTCACTGACTGG AGAACTCGGGTTTGTCTGTTGCGGGGGCGGAGTT ATGCGGTGCCGTTGGGCGAGTGACCCGTACCTTTGGGAG CGCGCGCTCGTCTGTCTGACGTACCCGTTCTGTTGG CTTATAATGCAGGGTGGGGCCACCTGCCGTTAGGTGTGC GGTAGGCTTTTCTCCGTCGAGGACGAGGGTTCGGGCC TAGGGTAGGCTCTCTGAATCGACAGGCGCCGACCTCT GGTGAGGGAGGGATAAGTGAGGCGTCAGTTCTTTGGT CGGTTTTATGTACCTATCTTCTTAAGTAGCTGAAGCTCCG GTTTTGAATATGCGCTCGGGGTTGGCGAGTGTGTTTGT GAAGTTTTTATAGGCACCTTTGAAATGTAATCATTTGGGT CAATATGTAATTTTCAGTGTAGACTAGTAAA
42	Poly A; SV40	GTTTATTGCAGCTTATAATGGTTACAAATAAGCAATAG CATCACAAATTTACAAATAAGCATTTTTTCACTGCAT TCTAGTTGTGGTTTGTCCAACTCATCAATGTATCTTATC A
43	Poly A; bGH	GACTGTGCCTTCTAGTTGCCAGCCATCTGTTGTTTGGCCC TCCCCGTGCCTTCTTGACCTTGAAGGTGCCACTCCCA CTGTCTTTTCTAATAAAATGAGGAAATTCATCGCATT GTCTGAGTAGGTGTCATTCTATTCTGGGGGTTGGGGTGG GGCAGGACAGCAAGGGGAGGATTGGGAAGACAATAGC AGGCATGCTGGGATGCGGTGGGCTCTATGG
44	Envelope; RD114	ATGAAACTCCCAACAGGAATGGTCATTTTATGTAGCCTA ATAATAGTTCGGGCAGGGTTTGACGACCCCGCAAGGCT ATCGCATTAGTACAAAACAACATGGTAAACCATGCGA ATGCAGCGGAGGGCAGGTATCCGAGGCCCAACCGAACT CCATCCAACAGGTAACCTTGCCAGGCAAGACGGCCTACT TAATGACCAACCAAAATGGAATGCAAGATCACTCCA AAAAACTCACCCCTAGCGGGGAGAACTCCAGAACTG CCCCTGTAACACTTTCCAGGACTCGATGCACAGTTCTTGT TATACTGAATACCGGCAATGACGGGCGAATAATAAGAC ATACTACAGGCCACCTTGCTTAAATACGCTCTGGGAG CCTCAACGAGGTACAGATATTACAAAACCCCAATCAGCT CCTACAGTCCCTTGTAGGGGCTCTATAATCAGCCCGT TTGCTGGAGTGCCACAGCCCCATCCATATCTCCGATGG TGGAGGACCCCTCGATACTAAGAGAGTGTGGACAGTCCA AAAAAGGCTAGAACAAATCATAAGGCTATGCATCCTGA ACTTCAATACCACCCTTAGCCCTGCCCAAAGTCAGAGA TGACCTTAGCCTTGATGCACGGACTTTTGATATCCTGAAT ACCACTTTTAGGTTACTCCAGATGTCCAATTTTAGCCTTG CCCAAGATTGTTGGCTCTGTTTAAACTAGGTACCCCTA CCCCTCTTGCATACCCACTCCCTCTTTAACCTACTCCCT AGCAGACTCCCTAGCGAATGCCCTCTGTGATATATACC TCCCCTCTTGGTTCAACCGATGAGTTCTCCAACCTCGTCC TGTTTATCTTCCCCTTTCATTAAAGATACGGAACAAATAG ACTTAGGTGACGTACCTTTACTAACTGCACCTCTGTAGC CAATGTGAGTAGTCTTTATGTGCCCTAAACGGGTGAGT CTTCTCTGTGGAATAACATGGCATAACCTATTTACCC CAAAACCTGGACAGGACTTTGCGTCCAAGCCTCCCTCCTC CCCGACATTGACATCATCCCGGGGATGAGCCAGTCCCC ATTCTGCCATTGATCATTATATACATAGACCTAAACGA GCTGTACAGTTTATCCCTTTACTAGCTGGACTGGGAATC ACCGCAGCATTACACCCGGAGCTACAGGCCTAGGTGTC TCCGTACCCAGTATACAAAATATCCCATCAGTTAATA TCTGATGTCCAAGTCTTATCCGGTACCATAACAAGATTAC AAGACCAGGTAGACTCGTTAGCTGAAGTAGTTCTCCAAA ATAGGAGGGGACTGGACCTACTAACGGCAGAACAAAGGA GGAATTTGTTTAGCCTTACAAGAAAAATGCTGTTTTTATG CTAACAAGTCAGGAATTGTGAGAAACAAAATAAGAACC CTACAAGAAGAAATACAAAACGAGGAAAGCCTGGC ATCCAACCTCTCTGGACCGGGCTGCAGGGCTTCTTCC GTACCTCCTACCTCTCTGGGACCCCTACTCACCTCCTA

SEQ ID NO:	Description	Sequence
		CTCATACTAACCATTGGGCCATGCGTTTTCAATCGATTGG TCCAATTTGTTAAAGACAGGATCTCAGTGGTCCAGGCTC TGGTTTTGACTCAGCAATATCACCAGCTAAAACCCATAG AGTACGAGCCATGA
45	Envelope; GALV	ATGCTTCTCACCTCAAGCCCGCACCCCTTCGGCACCAG ATGAGTCTCGGAGCTGGAAAAGATGATCATCCTCTTA AGCTGCGTATTTCGGAGACGGCAAAACGAGTCTGCAGAA TAAGAACCCCCACCAGCCTGTGACCCCTCAGCTGGCAGGT ACTGTCCAAACTGGGACGTTGTCTGGACAAAAGGC AGTCCAGCCCCCTTTGGACTTGGTGGCCCTCTTTACACCT GATGTATGTGCCCTGGCGGCCGCTCTTGAGTCTCGGGAT ATCCCGGATCCGATGTATCGTCTCTTAAAGAGTTAGA CCTCCTGATTCAGACTTACTGCCGCTTATAAGCAAATC ACCTGGGGAGCCATAGGGTGACGTACCTCGGGCTAGG ACCAGGATGGCAAATTCCCCCTTCTACGTGTGTCCCGA GCTGGCCGAACCCATTGAGAAGCTAGGAGGTGTGGGGG GCTAGAATCCCTATAGTGAAGAATGGAGTTGTGAGAC CACGGGTACCGTTTATTGGCAACCAAGTCTCATGGGA CCTCATAACTGTAAATGGGACCAAAATGTGAAATGGG AGCAAAAATTTCAAAGTGTGAACAAACCGGCTGGTGT AACCCCTCAAGATAGACTTCACAGAAAAGGAAAAC CTCCAGAGATTGGATAACGGAAAAACCTGGGAATTAA GGTTCTATGTATATGGACACCCAGGCATACAGTTGACTA TCCGCTTAGAGGTCACTAACATGCCGTTGTGGCAGTGG GCCCAGACCCCTGTCTTGGGAACAGGGACCTCCTAGCA AGCCCTCACTCTCCCTCTCTCCCAACGGAAGCGCCGC CCACCCCTCTACCCCGCGCGCTAGTGAGCAACCCCTG CGGTGCATGGAGAACTGTTACCTTAACTCTCCGCTC CCACCACTGGCGACCGACTCTTGGCCTTGTGAGGGGG CCTTCTAACCTTGAATGCTACCAACCCAGGGGCCACTA AGTCTTGCTGGCTCTGTTTGGGCATGAGCCCCCTTATTA TGAAGGGATAGCCTCTTCAGGAGAGGTCGCTTATACCTC CAACCATACCGATGCCACTGGGGGCCCAAGGAAGC TTACCCCTCACTGAGGTCTCCGACTCGGGTCATGCATAG GGAAGGTGCTCTTACCCATCAACATCTTTGCAACCA CCTTACCATCAATTCTCTTAAACCATCAGTATCTGCT CCCCTCAAACCATAGCTGGTGGGCTGCAGCACTGGCCT CACCCCTGCTCTCCACCTCAGTTTTAATCAGTCTAAA GACTTCTGTGTCCAGGTCCAGCTGATCCCCGCATCTATT ACCATTCTGAAGAACTTGTACAAAGCCTATGACAAAT CACCCCCCAGGTTTAAAGAGAGCCTGCCTCACTTACCC TAGCTGTCTTCTGGGGTTAGGGATTGCGGCAGGTATAG GTACTGGCTCAACCGCCCTAATTAAAGGGCCCATAGACC TCCAGCAAGGCCTAACAGCCTCCAAATCGCCATTGACG CTGACCTCCGGGCCCTTCAGGACTCAATCAGCAAGCTAG AGGACTCACTGACTTCCCTATCTGAGGTAGTACTCCAA ATAGGAGAGGCCTTGACTTACTATTCTTAAAGAAGGAG GCCTCTGCGCGGCCCTAAAGAAGAGTGTGTTTTATG TAGACCACTCAGGTGCAGTACGAGACTCCATGAAAAA CTTAAAGAAAGACTAGATAAAGACAGTTAGAGCGCCA GAAAAACCAAACTGGTATGAAGGGTGGTTCAATAACT CCCCTTGGTTTACTACCTACTATCAACCATCGCTGGGCC CCTATTGCTCCTCTTTTGTACTCACTCTTGGGCCCTGC ATCATCAATAAATTAATCCAATTCATCAATGATAGGATA AGTGCAGTCAAAATTTTAGTCCCTAGACAGAAATATCAG ACCTAGATAACGAGGAAAACCTTTAA
46	Envelope; FUG	ATGGTTCGCGAGGTTCTTTTGTGTACTCCTTCTGGGTT TTTCGTTGTGTTTCGGGAAGTTCCCATTTACACGATACC AGACGAACTGGTCCCTGGAGCCCTATTGACATACACCA TCTCAGCTGTCCAAATAACCTGGTTGTGGAGGATGAAGG ATGTACCAACCTGTCCGAGTCTCCTACATGGAACCTCAA AGTGGGATACATCTCAGCCATCAAAGTGAACGGGTTTAC TTGCACAGGTGTTGTGACAGAGGCAGAGACCTACACCAA CTTTGTTGGTTATGTCAACACCATTCAGAGAAAGCA TTTCCGCCCCACCCAGACGATGTAGAGCCGCGTATAA CTGGAAGATGGCCGCTGACCCAGATATGAAGATCCCT ACACAATCCATACCCGACTACCACTGGCTTCGAACGT AAGAACCACCAAGAGTCCCTCATATCATATCCCCAAG TGTGACAGATTGGACCCATATGACAAATCCCTTCACTC AAGGGTCTTCCCTGGCGAAAGTGTCTCAGGAATAACGGT GTCTCTACCTACTGCTCAACTAACCATGATTACACCAT TGGATGCCCCGAGAATCCGAGACCAAGGACACCTTGTGAC ATTTTACCAATAGCAGAGGGAAGAGGCATCCAACGG GAACAAGACTTGGCGCTTTGTGGATGAAAGAGCCTGTA

SEQ ID NO:	Description	Sequence
		TAAGTCTCTAAAAGGAGCATGCAGGCTCAAGTTATGTGG AGTTCTTGGACTTAGACTTATGGATGGAACATGGGTGCG GATGCAAACATCAGATGAGACCAATGGTGCCCTCCAG ATCAGTTGGTGAATTGACGACTTTCGCTCAGACGAGA TCGAGCATCTCGTTGTGGAGGAGTTAGTTAAGAAAAGAG AGGAATGTCTGGATGCATTAGAGTCCATCATGACCACCA AGTCAGTAAGTTTCAGACGTCTCAGTCACCTGAGAAAAC TTGTCCCAGGGTTTGGAAAAGCATATACCATATTCAACA AAACCTTGATGGAGGCTGATGCTCACTACAAGTCAGTCC GGACCTGGAATGAGATCATCCCTCAAAGGGTGTGTTGA AAGTTGGAGGAAGGTGCCATCCTCATGTGAACGGGGTGT TTTTCAATGGTATAATATTAGGGCTGACGACCATGTCTT AATCCCAGAGATGCAATCATCCCTCCTCCAGCAACATAT GGAGTTGTTGGAACTTTCAGTTATCCCCCTGATGCACCCC CTGGCAGACCTTCTACAGTTTCAAAGAAGGTGATGAG GCTGAGGATTTGTGTAAGTTCACCTCCCGATGTGTAC AAACAGATCTCAGGGGTGACCTGGGTCTCCCGAAGTGG GGAAGTATGTATTGATGACTGCAGGGGCCATGATTGGC CTGGTGTGATATTTCCCTAATGACATGGTGCAGAGTTG GTATCCATCTTGCATTAAATTAAGCACACCAAGAAAA GACAGATTATACAGACATAGAGATGAACCGACTTGGA AAGTAA
47	Envelope; LCMV	ATGGGTGAGATTGTGACAATGTTTGAGGCTCTGCCTCAC ATCATCGATGAGGTGATCAACATTGTCAATTATTGTGCTTA TCGTGATCACGGGTATCAAGGCTGTCTACAATTTTGCCA CCTGTGGGATATTGCGATTGATCAGTTTCTTACTTCTGGC TGGCAGGTCTGTGGCATGTACGGTCTTAAGGGACCCGA CATTTACAAAGGAGTTTACCAATTTAAGTCAGTGGAGTT TGATATGTCACATCTGAACCTGACCATGCCCAACGCATG TTCAGCCAAACAACCTCCACCATTACATCAGTATGGGGAC TTCTGGACTAGAATTGACCTTCACCAATGATTCCATCATC AGTCACAACCTTTTGAATCTGACCTCTGCCTTCAACAAA AAGACCTTTGACCACACACTCATGAGTATAGTTTCGAGC CTACACCTCAGTATCAGAGGGAACCTCAACTATAAGGCA GTATCCTGCGACTTCAACAATGGCATAACCATCCAATAC AACTTGACATTTCTCAGATCGACAAAGTGCTCAGAGCCAG TGTAAGAACCTTCAGAGGTAGAGTCTTAGATATGTTTAGA ACTGCCTTCGGGGGGAATACATGAGGAGTGGCTGGGG CTGGACAGGCTCAGATGGCAAGACCCTGGTGTAGCCA GACGAGTTACCAATACCTGATTATACAAAATAGAACCTG GGAAAACCATGTCACATATGCAGGTCTTTTGGGATGTC CAGGATTCTCTTTCCCAAGAGAAGACTAAGTTCTTCAC TAGGAGACTAGCGGGCACATTACCTGGACTTTGTGAGA CTCTTCAGGGGTGGAGAATCCAGGTGGTTATTGCTGAC CAAATGGATGATTCTTGCTGCAGAGCTTAAGTGTTCGG GAACACAGCAGTTGCGAAATGCAATGTAAATCATGATGC CGAATTCTGTGACATGCTGCGACTAATTGACTACAACAA GGCTGCTTTGAGTAAGTTCAAAGAGGACGTAGAATCTGC CTTGCATTTATTCAAAACAACAGTGAATCTTTGATTTC GATCAACTACTGATGAGGAACCACTTGAGAGATCTGATG GGGGTGCCATATTGCAATTACTCAAAGTTTGGTACCTA GAACATGCAAGACCGGCGAAACTAGTGTCCCAAGTG CTGGCTTGTCAACATGGTTCTTACTTAAATGAGACCCA CTTCAAGTATCAAATCGAACAGGAAGCCGATAACATGAT TACAGAGATGTTGAGGAAGGATTACATAAAGAGGCAGG GGAGTACCCCTTAGCATTGATGGACCTTCTGATGTTTT CACATCTGCATATCTAGTCAGCATCTTCTGACCTTGTG AAAATACCAACACACAGGCACATAAAGGTGGCTCATG TCCAAAGCCACACCGATTAAACCAAGGAATTTGTAG TTGTGGTGCATTTAAGGTGCCTGGTGTAAAACCGTCTG GAAAAGACGCTGA
48	Envelope; FPV	ATGAACACTCAAATCCTGGTTTTTCGCCCTTGTGGCAGTC ATCCCCACAAATGCAGACAAAATTTGTCTTGGACATCAT GCTGTATCAAATGGCACCAGTAACACACTCACTGAG AGAGGAGTAGAAGTTGTCAATGCAACGGAAACAGTGGA GCGGACAAACATCCCCAAAATTTGCTCAAAGGGGAAAA GAACCACTGATCTTGGCCAATGCGGACTGTTAGGGACCA TTACCGGACCACCTCAATGCGACCAATTTCTAGAATTTT AGCTGATCTAATAATCGAGAGACGAGAAGGAAATGATG TTTGTTACCCGGGAAGTTTGTAAATGAAGAGGCATTGC GACAAATCCTCAGAGGATCAGGTGGGATTGACAAAGAA ACAATGGGATTACATATAGTGAATAAGGACCAACGG AACAACAGTGCATGTAGAAGATCAGGTCTTCAATCTA TGCAGAAATGGAGTGGCTCTGTCAAATACAGACAATGC

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SEQ ID NO:	Description	Sequence
		TGCTTTCCACAAATGACAAATCATACAAAACACAAG GAGAGAATCAGCTCTGATAGTCTGGGGAATCCACCATT AGGATCAACCACCGAAGACAGACCAACTATATGGGAGTG GAAATAAACTGATAACAGTCGGGAGTTCCAAATATCATC AATCTTTTGTGCCGAGTCCAGGAACACGACCGCAGATAA ATGGCCAGTCCGGACGGATTGATTTTCATTGGTTGATCTT GGATCCCAATGATACAGTTACTTTTAGTTTCAATGGGGC TTTCATAGCTCCAAATCGTGCCAGCTTCTTGAGGGGAAA GTCCATGGGGATCCAGAGCGATGTGCAGGTTGATGCCAA TTGCGAAGGGGAATGCTACCAAGTGGAGGGACTATAA CAAGCAGATTGCCTTTTCAAACATCAATAGCAGAGCAG TTGGCAAATGCCAAGATATGTAACAGGAAAGTTTAT TATTGGCAACTGGGATGAAGAAGCTTCCGAACCTTCCA AAAAAGGAAAAAAGAGGCCTGTTTGGCGCTATAGCA GGGTTTATTGAAATGGTTGGGAAGGTCGGTCGACGGG TGTACGGTTTCAGGCATCAGAATGCACAAGGAGAAGG AACTGCAGCAGACTACAAAGCACCAATCGGCAATTG ATCAGATAACCGGAAAGTTAAATAGACTCATTGAGAAA ACCAACCAGCAATTGAGCTAATAGATAATGAATTCCT GAGGTGGAAAAGCAGATTGGCAATTTAATTAAGTGGACC AAAGACTCCATCAGAGAAGTATGGTCTTACAATGCTGAA CTTCTGTGGCAATGGAACACAGCACACTATTGATTG GCTGATTAGAGATGAACAAGCTGTATGAGCGAGTGAG GAAACAATTAGGGAAAAATGCTGAAGAGGATGGCACTG GTTGCTTTGAAATTTTTCATAATGTGACGATGATTGTAT GGCTAGTATAAGGAACAATACTTATGATCACAGCAAATA CAGAGAAGAAGCGATGCAAAATAGAAACAAATGAC CAGTCAAATTGAGTAGTGGCTACAAGATGTGATACTTT GGTTTAGCTTCGGGCATCATGCTTTTGTCTCTTGCCAT TGCAATGGGCCTTGTTCATATGTGTGAAGAACGGAAA CATGCGGTGCACATTTTGTATATAA
49	Envelope; RRV	AGTGTAACAGAGCACTTTAATGTGTATAAGGCTACTAGA CCATACCTAGCACATTGCGCCGATTGCGGGACGGGTAC TTCTGCTATAGCCAGTTGCTATCGAGGAGATCCGAGAT GAGGCGTCTGATGGCATGCTTAAGATCCAAGTCTCCGCC CAAATAGGTCTGGACAAGGCAGGCACCCACGCCACAC GAAGCTCCGATATATGGCTGGTCATGATGTTTCAGGAATC TAAGAGAGATTCTTTGAGGGTGTAACGTCGCGAGCGTG CTCCATACATGGGACGATGGGACACTTCATCGTCGCACA CTGTCCACCAGGCGACTACCTCAAGGTTTCGTTTCGAGGA CGCAGATTCCGACGTAAGGCATGTAAGGTCCAATACAA GCACAATCCATTGCCGGTGGGTAGAGAGAAGTTTCGTGGT TAGACCACACTTTGGCGTAGAGCTGCCATGCACCTCATA CCAGCTGACAACGGCTCCACCGACGAGGAGATTGACAT GCATACCCGCGAGATATACCGGATCGCACCTGTATC ACAGACGGCGGCAACGTCAAAATAACAGCAGGCGGCA GGACTATCAGGTACAACGTGACCTGCGGGCGTGACAACG TAGGCACTACCACTACTGACAAGACCATCAACACATGCA AGATTGACCAATGCCATGCTGCCGTACCAGCCATGACA AATGGCAATTACCTCTCCATTTGTTCCAGGGCTGATCA GACAGCTAGGAAGGCAAGGTACACGTTCCGTTCCCTCT GACTAACGTCACTGCCGAGTGCCGTGGCTCGAGCGCC GGATGCCACCTATGGTAAGGAGGAGTGACCTGAGATT ACACCCAGATCATCCGACGCTTCTCTATAGGAGTTT AGGAGCCGAACCGCACCCGTACGAGGAATGGGTTGACA AGTTCTCTGAGCGCATCATCCAGTGACGGAAGAAGGGA TTGAGTACCAGTGGGGCAACAACCGCCGGTCTGCCTGT GGGCGCAACTGACGACCGAGGGCAACCCCATGGCTGG CCACATGAAATCATTCAGTACTATTATGGACTATACCC GCCGCCACTATTGCCGAGTATCCGGGGCAGTCTGATG GCCCTCCTAACTCTGGCGGCCACATGCTGCATGCTGGCC ACCGCGAGGAGAAAGTGCCTAACCCGTACGCCCTGAC GCCAGAGCGGTGGTACCGTTGACACTGGGGCTGCTTTG CTGCGACCGAGGGCGAATGCA
50	Envelope; MLV 10A1	ATGGAAGGTCCAGCGTTCTCAAAACCCCTTAAAGATAAG ATTAACCCGTGGAAGTCCCTAATGGTATGCGGGGTCTAT TTAAGAGTAGGGATGGCAGAGAGCCCCATCAGGTCTTT AATGTAACCTGGAGAGTCAACACCTGATGACTGGGCGT ACGCCAATGCCACCTCCCTTTTAGGAAGTGTACAAGAT GCCTTCCAAGATTATATTTTGATCTATGTGATCTGGTCTG GAGAAGAGTGGGACCTTCAGACCAGGAACCATATGTC GGGTATGGTGAATAACCCCGAGGGAGAAAGCGGAC CCGACTTTTGACTTTTACGTGTGCCCTGGGCATACCGTA AAATCGGGTGTGGGGGCCAAGAGAGGGCTACTGTGG

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SEQ ID NO:	Description	Sequence
		TGAATGGGGTTGTGAAACCAACCGGACAGGCTTACTGGAA GCCCACATCATCATGGGACCTAATCTCCCTTAAGCGCGG TAACACCCCCCTGGGACACGGGATGCTCCAAAATGGCTTG TGGCCCTGCTACGACCTCTCCAAAGTATCCAATTCCCTTC CAAGGGGCTACTCGAGGGGGCAGATGCAACCCCTCTAGTC CTAGAATTCACTGATGCAGGAAAAAGGCTAATTGGGA CGGGCCCAAATCGTGGGGACTGAGACTGTACCGGACAG GAACAGATCCTATTACCATGTTCTCCCTGACCCGCCAGG TCCTCAATATAGGGCCCCGCATCCCCATTGGGCCTAATC CCGTGATCACTGGTCAACTACCCCCCTCCGACCCGTGC AGATCAGGCTCCCAGGCCTCCTCAGCCTCCTCCTACAG GCGCAGCCTCTATAGTCCCTGAGACTGCCCCACCTTCTC AACAACTGGGACGGGAGACAGGCTGCTAAACCTGGTA GAAGGAGCCTATCAGGCGCTTAACCTCACCATCCCGAC AAGACCAAGAATGTTGGCTGTGCTTAGTGTCGGGACCT CCTTATTACGAAGGAGTAGCGGTCTGGGCACTTATACC AATCATTCTACCGCCCCGGCCAGCTGTACGGCCACTTCC CAACATAAGCTTACCCTATCTGAAGTGACAGGACAGGGC CTATGCATGGGAGCACTACCTAAACTCACCAGGCCTTA TGTAACACCACCCAAAGTGCCGGCTCAGGATCCTACTAC CTGTCAGACCCCGCTGGAACAATGTGGGCTTGTAGCACT GGATTGACTCCCTGCTTGTCCACCACGATGCTCAATCTA ACCACAGACTATTGTGTATTAGTTGAGCTCTGGCCAGA ATAATTTACCACTCCCCGATTATATGTATGGTCAGCTTG AACAGCGTACCAAATATAAGAGGGAGCCAGTATCGTTG ACCCTGGCCCTTCTGCTAGGAGGATTAACCATGGGAGGG ATTGCAGCTGGAAAGGGACGGGACCACTGCCCTAATC AAAACCCAGCAGTTTGAGCAGCTTCACGCCGCTATCCAG ACAGACCTCAACGAAGTCGAAAAATCAATTACCAACCTA GAAAAGTCACTGACCTCGTTGTCTGAAGTAGTCTACAG AACCGAAGAGGCCTAGATTGCTCTTCTAAAGAGGGA GGTCTCTGCGCAGCCCTAAAAGAAGAATGTTGTTTTAT GCAGACCACACGGGACTAGTGAGAGACAGCATGGCCAA ACTAAGGGAAGGCTTAATCAGAGACAAAACCTATTG AGTCAGGCCAAGGTTGGTTCGAAGGGCAGTTTAATAGAT CCCCTGGTTTACACCTTAATCTCCACCATCATGGGACC TCTAATAGTACTCTTACTGATCTTACTCTTGGACCCGTGC ATTCTCAATCGATTGGTCCAATTTGTTAAAGACAGGATC TCAGTGGTCCAGGCTCTGGTTTGTACTCAACAATATCAC CAGCTAAACCTATAGAGTACGAGCCATGA
51	Envelope; Ebola	ATGGGTGTTACAGGAATATTGCAGTTACCTCGTGATCGA TTCAAGAGGACATCATTCTTCTTTGGGTAATTATCCTTT TCCAAAGAACATTTTCCATCCCACTTGGAGTCATCCACA ATAGCATTACAGGTTAGTGATGTCGACAAACTGGTTT GCCGTGACAACTGTCTCCACAAATCAATTGAGATCAG TTGGACTGAATCTCGAAGGGAATGGAGTGGCAACTGAC GTGCCATCTGCAACTAAAGATGGGGCTCAGGTCCGGT GTCCACCAAGGTGGTCAATTATGAAGCTGGTGAATGG GCTGAAAACCTGCTACAATCTTGAATCAAAAAACCTGAC GGGAGTGAGTGTCTACAGCAGCGCCAGACGGGATTCTG GGGCTTCCCCGGTGCCGGTATGTGCACAAAGTATCAGG AACGGGACCGTGTGCCGAGACTTTGCCTTCCACAAAGA GGGTGCTTTCTCTGTATGACCGACTTGCTTCCACAGTT ATCTACCGAGGAACGACTTTCGCTGAAGGTGTCGTTGCA TTTCTGATACTGCCCCAAGCTAAGAAGGACTTCTTCAGC TCACACCCCTTGAGAGAGCCGGTCAATGCAACGGAGGA CCGTCTAGTGGCTACTATTCTACCACAATTAGATATCA AGCTACCGGTTTTTGAACCAATGAGACAGAGTATTTGTT CGAGGTTGACAATTTGACCTACGTCCAACTGAATCAAG ATTCACACCACAGTTTCTGCTCCAGCTGAATGAGACAAT ATATACAAGTGGGAAAAGGAGCAATACCACGGGAAAC TAATTTGGAAGGTCAACCCCGAAATTGATACAACAATCG GGGAGTGGGCTTCTGGGAACTAAAAAACCTCACTA GAAAAATTCGAGTGAAGAGTTGTCTTTCACAGCTGTAT CAAACAGAGCCAAAAACATCAGTGGTCAGAGTCCGGCG CGAACTTCTTCCGACCCAGGGACCAACACAACAACAGGAA GACCACAAAATCATGGCTTCAGAAAAATCCTCTGCAATG GTTCAAGTGACAGTCAAGGAAGGGAAGCTGCAGTGTC GCATCTGACAACCTTGCCCAATCTCCACGAGTCTCTCA ACCCCCCACAACCAACCAAGGTCCGGACAACAGCACCC ACAATACACCCGTGTATAAACTTGACATCTCTGAGGCAA CTCAAGTTGAACAACATCACCGCAGAACAGACAACGAC AGCACAGCCTCCGACACTCCCCCGCCACGACCGCAGCC GGACCCCTAAAGCAGAGAACCAACACAGAGCAAGGG TACCGACCTCTGGACCCCGCCACCACAACAGTCCCCA

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SEQ ID NO:	Description	Sequence
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52	Polymerase III shRNA promoters; U6 promoter	TTTCCCATGATTCCTTCATATTTGCATATACGATACAAGG CTGTTAGAGAGATAAATTGGAATTAATTTGACTGTAAACA CAAAGATATTAGTACAAAATACGTGACGTAGAAAGTAA TAATTTCTTGGGTAGTTTGCAGTTTAAAAATTATGTTTTA AAATGGACTATCATATGCTTACCGTAACCTGAAAGTATT TCGATTTCTTGGCTTTATATATCTTGTGGAAGGACGAA AC
53	Polymerase III shRNA promoters; 7SK promoter	CTGCAGTATTTAGCATGCCCCACCATCTGCAAGGCATT CTGGATAGTGTCAAAACAGCCGGAATCAAGTCCGTTTA TCTCAAACCTTAGCATTTTGGGAATAAATGATTTTGCTA TGCTGGTTAAATTAGATTTTAGTTAAATTTCTGCTGAAG CTCTAGTACGATAAGCAACTTGACCTAAGTGTAAGTTG AGATTTCTTCAGGTTTATATAGCTTGTGCGCCGCTGGC TACCTC
54	FDPS target sequence #1	GTCTGGAGTACAATGCCATT
55	FDPS target sequence #2	GCAGGATTTCTGTTAGCACTT
56	FDPS target sequence #3	GCCATGTACATGGCAGGAATT
57	FDPS target sequence #4	GCAGAAGGAGGCTGAGAAAGT
58	Non-targeting sequence	GCCGCTTTGTAGGATAGAGCTCGAGCTCTATCCTACAAA GCGGCTTTTT
59	Forward primer	AGGAATTGATGGCGAGAAGG
60	Reverse primer	CCCAAAGAGGTCAAGGTAATCA
61	Forward primer	AGCGCGCTACAGCTTCA
62	Reverse primer	GGCGACGTAGCACAGCTTCT
63	Left Inverted Terminal Repeat (Left ITR)	CCTGCAGGCAGCTGCGCGCTCGCTCGCTCACTGAGGCCG CCCGGGCGTCGGGCGACCTTTGGTCGCCCAGGCTCAGTG AGCGAGCGAGCGCGCAGAGAGGAGTGCCCAACTCCAT CACTAGGGGTTCTT
64	Right Inverted Terminal Repeat (Right ITR)	GAGCGGCCGAGGAACCCCTAGTGATGGAGTTGGCCACT CCCTCTCTGCGCGCTCGCTCGCTCACTGAGGCCGGGCGA CCAAAGGTCGCCCAGCGCCGGGCTTTGCCCGGGCGGCC TCAGTGAGCGAGCGAGCGCAGCTGCCTGCAGG
65	RRE/rabbit poly A beta globin	TCTAGAAGGAGCTTTGTTCTTGGGTTCTTGGGAGCAGC AGGAAGCACTATGGGCGCAGCGTCAATGACGCTGACGG TACAGGCCAGACAATTATTGTTCTGGTATAGTGACAGC AGAACAATTTGCTGAGGGCTATTGAGGCGCAACAGCATC TGTTGCAACTCACAGTCTGGGCGATCAAGCAGCTCCAGG CAAGAATCCTGGCTGTGGAAGATACCTAAAGGATCAA CAGCTCCTAGATCTTTTCCCTCTGCCAAAATTATGGGG ACATCATGAAGCCCTTGAGCATCTGACTTCTGGCTAAT

SEQ ID NO:	Description	Sequence
		AAAGGAAATTTATTTTCATTGCAATAGTGTGTTGGAATTT TTTGTGTCTCTCACTCGGAAGGACATATGGGAGGGCAAA TCATTTAAACATCAGAATGAGTATTTGGTTTAGAGTTT GGCAACATATGCCATATGCTGGCTGCCATGAACAAAGGT GGCTATAAAGAGGTCATCAGTATATGAAACAGCCCCCTG CTGTCCATTCTTATTCATAGAAAAGCCTTGACTTGAGG TTAGATTTTTTTTATATTTTGTGTTTGTGTTATTTTTCTTT AACATCCCTAAAAATTTCTTACATGTTTACTAGCCAGA TTTTTCCTCTCTCTGACTACTCCAGTCATAGCTGTCC CTCTCTCTTATGAAGATCCCTCGACCTGCAGCCCAAGCT TGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTG TTATCCGCTCACAATCCACACACATACGAGCCGGAAG CATAAAGTGTAAGCCTGGGGTGCCTAATGAGTGAGCTA ACTCACATTAATTGCGTTGCGCTCACTGCCCCGCTTTCCAG TCGGGAAACCTGTGTCGACGCGGATCCGCATCTCAATT AGTCAGCAACCATAGTCCCGCCCTAACTCCGCCCATCC CGCCCCTAACCTCCGCCAGTTCCGCCCATTTCTCCGCCCA TGGGTGACTAATTTTTTTTATTTATGAGAGGCGGAGGCC GCCTCGCCTCTGAGCTATTCCAGAAGTAGTGAGGAGGC TTTTTTGGAGGCCAGGCTTTTGCAAAAAGCTAACTGTT TATTGCAGCTTATAATGGTTACAAATAAGCAATAGCAT CACAAATTTACAAATAAAGCATTTTTTCACTGCATTCT AGTTGTGGTTTGTCCAAACTCATCAATGTATCTTATCACC CGGG

While certain of the preferred embodiments of the present invention have been described and specifically exemplified above, it is not intended that the invention be limited to such

embodiments. Various modifications may be made thereto without departing from the scope and spirit of the present invention.

SEQUENCE LISTING

Sequence total quantity: 65

SEQ ID NO: 1 moltype = DNA length = 53
FEATURE Location/Qualifiers
misc_feature 1..53
 note = FDPS shRNA sequence #1
source 1..53
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1
gtcctggagt acaatgccat tctcgagaat ggcatgtac tccaggactt ttt 53

SEQ ID NO: 2 moltype = DNA length = 53
FEATURE Location/Qualifiers
misc_feature 1..53
 note = FDPS shRNA sequence #2
source 1..53
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 2
gcaggatttc gttcagcact tctcgagaag tgctgaacga aatcctgctt ttt 53

SEQ ID NO: 3 moltype = DNA length = 53
FEATURE Location/Qualifiers
misc_feature 1..53
 note = FDPS shRNA sequence #3
source 1..53
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 3
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SEQ ID NO: 4 moltype = DNA length = 53
FEATURE Location/Qualifiers
misc_feature 1..53
 note = FDPS shRNA sequence #4
source 1..53
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 4
gcagaaggag gctgagaaag tctcgagact ttctcagcct ccttctgctt ttt 53

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SEQ ID NO: 5          moltype = DNA  length = 116
FEATURE              Location/Qualifiers
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source                1..116
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 5
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agatggcaga aggaggctga gaaagtgtcg cctactgcct cggacttcaa ggggct 116

SEQ ID NO: 6          moltype = DNA  length = 114
FEATURE              Location/Qualifiers
misc_feature          1..114
                      note = miR30 FDPS sequence #2
source                1..114
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 6
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agatggcaga agggtgaga aagtgtgtcc tactgcctcg gacttcaagg ggct 114

SEQ ID NO: 7          moltype = DNA  length = 91
FEATURE              Location/Qualifiers
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                      note = miR30 FDPS sequence #3
source                1..91
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 7
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ggaggctgag aaagttgctt actgctcgg a 91

SEQ ID NO: 8          moltype = DNA  length = 115
FEATURE              Location/Qualifiers
misc_feature          1..115
                      note = miR155 FDPS sequence #1
source                1..115
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 8
cctggaggct tgctgaaggc tgtatgctga ctttctcagc ctccttctgc ttttgccac 60
tgactgagca gaagggtgga gaaagtcagg acacaaggcc tgttactagc actca 115

SEQ ID NO: 9          moltype = DNA  length = 114
FEATURE              Location/Qualifiers
misc_feature          1..114
                      note = miR21 FDPS sequence #1
source                1..114
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 9
catctccatg gctgtaccac cttgtcggga ctttctcagc ctccttctgc ctgttgaatc 60
tcatggcaga aggaggcgag aaagtctgac attttggtat ctttcatctg acca 114

SEQ ID NO: 10         moltype = DNA  length = 114
FEATURE              Location/Qualifiers
misc_feature          1..114
                      note = miR185 FDPS sequence #1
source                1..114
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 10
gggcctggct cgagcagggg gcgagggata ctttctcagc ctccttctgc tggccccctc 60
ccgcgagaag gaggtgaga aagtccttcc ctcccaatga ccgcgtcttc gtcg 114

SEQ ID NO: 11         moltype = DNA  length = 228
FEATURE              Location/Qualifiers
misc_feature          1..228
                      note = Rous Sarcoma virus (RSV) promoter
source                1..228
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 11
gtagtcttat gcaatactct tgtagtcttg caacatggta acgatgagtt agcaacatgc 60
cttacaagga gagaaaaagc accgtgcatg ccgattggtg gaagtaaggt ggtacgatcg 120
tgccctatta ggaaggcaac agacgggtct gacatggatt ggacgaacca ctgaattgcc 180
gcattgcaga gatattgtat ttaagtgcct agctcgatac aataaacg 228

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SEQ ID NO: 12 moltype = DNA length = 180
 FEATURE Location/Qualifiers
 misc_feature 1..180
 note = 5' Long terminal repeat (LTR)
 source 1..180
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 12
 ggtctctctg gttagaccag atctgagcct gggagctctc tggctaacta gggaaccac 60
 tgcttaagcc tcaataaagc ttgccttgag tgcttcaagt agtgtgtgcc cgtctgttgt 120
 gtgactctgg taactagaga tccttcagac ccttttagtc agtgtggaaa atctctagca 180

 SEQ ID NO: 13 moltype = DNA length = 41
 FEATURE Location/Qualifiers
 misc_feature 1..41
 note = Psi Packaging signal
 source 1..41
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 13
 tacgccaaaa attttgacta gcggaggcta gaaggagaga g 41

 SEQ ID NO: 14 moltype = DNA length = 233
 FEATURE Location/Qualifiers
 misc_feature 1..233
 note = Rev response element (RRE)
 source 1..233
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 14
 aggagctttg ttccttgggt tcttgggagc agcaggaagc actatgggcg cagcctcaat 60
 gacgtgacg gtacaggcca gacaattatt gtctgggata gtgcagcagc agaacaattt 120
 gctgagggct attgaggcgc aacagcatct gttgcaactc acagtctggg gcatacagca 180
 gctccaggca agaatcctgg ctgtggaaag atacctaaag gatcaacagc tcc 233

 SEQ ID NO: 15 moltype = DNA length = 118
 FEATURE Location/Qualifiers
 misc_feature 1..118
 note = Central polypurine tract (cPPT)
 source 1..118
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 15
 ttttaaaaga aaagggggga ttggggggta cagtgccagg gaaagaatag tagacataat 60
 agcaacagac atacaacta aagaattaca aaaacaaatt acaaaattca aaatttta 118

 SEQ ID NO: 16 moltype = DNA length = 217
 FEATURE Location/Qualifiers
 misc_feature 1..217
 note = Polymerase III shRNA promoters; H1 promoter
 source 1..217
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 16
 gaacgctgac gtcatacaacc cgctccaagg aatcgcgggc ccagtgtcac taggcgggaa 60
 caccagcgcg gcgtgcgccc tggcaggaag atggctgtga gggacagggg agtggcgccc 120
 tgcaatattt gcatgtcgct atgtgttctg ggaaatcacc ataaacgtga aatgtctttg 180
 gatttgggaa tcttataagt tctgtatgag accactt 217

 SEQ ID NO: 17 moltype = DNA length = 590
 FEATURE Location/Qualifiers
 misc_feature 1..590
 note = Long WPRE sequence
 source 1..590
 mol_type = other DNA
 organism = synthetic construct

 SEQUENCE: 17
 aatcaacctc tgattacaaa atttgtgaaa gattgactgg tattcttaac tatgttgctc 60
 cttttacgct atgtggatac gctgctttaa tgccctttga tcatgctatt gcttcccgtg 120
 tggttttcat tttctctccc ttgtataaat cctgggttgc gtctctttat gaggagtgtg 180
 ggcccgttgt caggcaacgt ggcgtggtgt gcactgtgtt tgcagacgca acccccactg 240
 gttggggcat tgccaccacc tgtcagctcc tttccgggac tttcgtcttc cccctcccta 300
 ttgccacggc ggaactcatc gccgcctgcc ttgcccgctg ctggacaggg gctcggctgt 360
 tgggcaactg caattccgtg gtgttgtcgg ggaaatcacc gtcctttcct tggctgctcg 420
 cctgtgttgc cacctggatt ctgcgcggga cgtccttctg ctacgtccct tcggccctca 480
 atccagcgga ccttccctcc cgcggcctgc tgccggctct gcggcctctt ccgctcttcc 540
 gccttcgccc tcagacgagt cggatctccc tttgggcgcg ctccccgcct 590

 SEQ ID NO: 18 moltype = DNA length = 250

FEATURE	Location/Qualifiers					
misc_feature	1..250					
	note = 3' delta LTR					
source	1..250					
	mol_type = other DNA					
	organism = synthetic construct					
SEQUENCE: 18						
tggaagggtc	aattcaactcc	caacgaagat	aagatctgct	ttttgcttgt	actgggtctc	60
tctgtgttaga	ccagattctga	gctctggagc	tctctggcta	acttagtgaac	ccactgctta	120
agcctcaata	aagcttgctc	tgagtgtctc	aagtagatgtg	tgcctgtctg	ttgtgtgact	180
ctggtaacta	gagatccctc	agaccctttt	agtcagtgtg	gaaaatctct	agcagtagta	240
gttcattgtca						250
SEQ ID NO: 19				moltype = DNA length = 290		
FEATURE	Location/Qualifiers					
misc_feature	1..290			note = Helper/Rev; Chicken beta actin (CAG) promoter;		
				Transcription		
source	1..290			mol_type = other DNA		
				organism = synthetic construct		
SEQUENCE: 19						
gctattaccac	tgggtgcgag	tgagccccac	gtttctgttc	actctcccac	tctccccccc	60
ctccccaccc	tcttatttgt	atttatttat	tttttaatta	ttttgtgcag	cgatgggggc	120
ggggggggggg	ggggcgcgcg	ccaggcgggg	cgggcggggg	cgaggggcgg	ggcggggcga	180
ggcgagagag	tgccgcgcca	gccaatcaga	gcgcgcgcct	cggaaagtgt	ccttttatgg	240
cgaggcgggc	gcgcggcgcg	ccctataaaa	agcgaagcgc	gcgcggggcg		290
SEQ ID NO: 20				moltype = DNA length = 1503		
FEATURE	Location/Qualifiers					
misc_feature	1..1503			note = Helper/Rev; HIV Gag; Viral capsid		
				1..1503		
source	1..1503			mol_type = other DNA		
				organism = synthetic construct		
SEQUENCE: 20						
atgggtgcga	gagcgtcagt	attaagcggg	ggagaattag	atcgatggga	aaaaattcgg	60
ttaaggccag	ggggaaagaa	aaaaataaaa	ttaaaacata	tagtatgggc	aagcaggggag	120
ctagaacgat	tcgcagttta	tctctgcctt	ttagaacaat	cagaaggctg	tagacaaaata	180
ctgggacagc	tacaaccatc	ccttcagaca	ggatcagaag	aacttagatc	attatataat	240
acagtagcaa	ccctctattg	tgtgcatcaa	aggatagaga	taaaagacac	caaggaagct	300
ttagacaaga	tagaggaaga	gcaaaacaaa	agtaagaaaa	aagcacagca	agcagcagct	360
gacacaggac	acagcaatca	ggtcagccaa	aattacccta	tagtgagaaa	catccagggg	420
caaatggtag	atcaggccat	atcacctaga	actttaaatg	catgggtaaa	agtagtagaa	480
gagaagcgct	tcagcccaga	agtgataccc	atgttttcag	cattatcaga	aggagccacc	540
ccacaagatt	taaacaccat	gctaaacaca	gtggggggac	ataaagcagc	catgcgaatg	600
ttaaaagaga	ccatcaatga	ggaagctgca	gaattgggata	gagtgcattc	agtgcatgca	660
gggcctattg	caccaggcca	gatgagagaa	ccaaggggaa	gtgacatagc	aggaactact	720
agtacccttc	aggaacaaat	aggatgggat	acacataatc	cacctatccc	agtaggagaa	780
attctataaaa	gatggataat	cctggggata	aataaaaatg	taagaatgta	tagccctacc	840
agcattctcg	aagaaatgat	agacccaaag	gaacctttta	gagactatgt	agaccgattc	900
tataaaactc	taagagccga	gcaagcttca	caagaggtaa	aaaattggat	gacagaaaac	960
ttgttgcttc	aaaatgcgaa	cccagattgt	aagactattt	taaaagcatt	gggaccagga	1020
gcgacatcag	aagaaatgat	gacagcatgt	caggvgatgg	ggggaccggc	ccataaagca	1080
agagtttttg	ctgaagcaat	gagccaagta	acaaatccag	ctaccataat	gatacagaaa	1140
ggcaattttt	ggaaccaaa	aaagactgtt	aagtgtttca	attgtggcaa	agaagggcac	1200
atcttaaatg	attgcagggc	ccctaggaaa	aagggtgttc	ggaaattgtg	aaaggaagga	1260
caccaaatga	aagattgtac	tgagagacag	gctaattttt	taggggaagt	ctggccttcc	1320
cacaggggga	ggctcaggga	ttttcttcag	agcagaccag	agccaacagc	cccaccagaa	1380
gagagcttca	gttttgggga	agagacaaca	actccctctc	agaagcagga	cccgatagac	1440
aaggaactgt	atccctttag	ttccctcaga	tcactctttg	gcagcgaccc	ctcgtcacaa	1500
taa						1503
SEQ ID NO: 21				moltype = DNA length = 1872		
FEATURE	Location/Qualifiers					
misc_feature	1..1872			note = Helper/Rev; HIV Pol; Protease and reverse		
				transcriptase		
source	1..1872			mol_type = other DNA		
				organism = synthetic construct		
SEQUENCE: 21						
atgaatttgc	caggaagatg	gaaacaaaaa	atgatagggg	gaattggagg	ttttatcaaa	60
gtaggacagt	atgatcagat	actcatagaa	atctgcggac	ataaagctat	aggtacagta	120
ttagtaggac	ctacacctgt	caacataaat	ggaagaaatc	tgttgactca	gatttggtcg	180
acttttaaat	ttccattag	tcctatttag	actgtaccag	taaaattaaa	gccaggaatg	240
gatggcccaa	aagttaaaca	atggccattg	acagaagaaa	aaataaaagc	attagtagaa	300
atttgtacag	aaatggaaaa	ggaagggaaa	atttcaaaaa	tgggcctgta	aaatccatac	360
aataactcac	tttgcgcat	aaqaaaaaaa	qacqactact	attggaqaaa	attagtagat	420

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ttcagagaac ttaataagag aactcaagat ttctgggaag ttcaattagg aataccacat 480
cctgcagggt taaaacagaa aaaatcagta acagtactgg atgtgggcga tgcataatctt 540
tcagtccctc tagataaaga cttcaggaag tatactgcat ttaccatacc tagtataaac 600
aatgagacac cagggattag atatcagtac aatgtgtctc cacagggatg gaaaggatca 660
ccagcaatat tccagtgtag catgacaaaa atcttagagc cttttagaaa aaaaaatcca 720
gacatagtc tctatcaata catggatgat ttgtatgtag gatctgactt agaaataggg 780
cagcatagaa caaaaataga ggaactgaga caacatctgt tgaggtgggg atttaccaca 840
ccagacaaaa aacatcagaa agaacctcca ttcctttgga tgggttatga actccatcct 900
gataaatgga cagtacagcc tatagtgtct ccagaaaagg acagctggac tgtcaatgac 960
atacagaat tagtgggaaa attgaattgg gcaagtcaga tttatgcagg gattaaagta 1020
aggcaattat gtaaacctct taggggaacc aaagcactaa cagaagtagt accactaaca 1080
gaagaagcag agctagaact ggcagaaaac agggagattc taaagaacc ggtacatgga 1140
gtgtattatg aacctcaaa agacttaata gcagaaatc agaagcagg gcaaggccaa 1200
tggacatatc aaatttatca agagccattt aaaaatctga aaacaggaaa atatgcaaga 1260
atgaagggtg cccacactaa tgatgtgaaa caattaacag aggcagtaca aaaaatagcc 1320
acagaagca tagtaatatg gggaaagact cctaaattta aattacccat acaaaaggaa 1380
acatgggaag catgggttag agagtattgg caagccacct ggattcctga gtgggagttt 1440
gtcaataccc ctcccttagt gaagttagt taccagttag agaaagaacc cataatagga 1500
gcagaactct tctatgtaga tggggcagcc aatagggaaa ctaaattagg aaaagcagga 1560
tatgtaactg acatgagaag acaaaaagtt gtcccctaa cggacacaac aaatcagaag 1620
actgagttac aagcaattca tctagctttg caggattcgg gattagaagt aaacatagtg 1680
acagactcac aatatgcatt gggaaatcatt caagcacac cagataagag tgaatcagag 1740
ttagtcagtc aaataataga gcagttaata aaaaaggaaa aagtcctacc ggcatgggta 1800
ccagcacaca aaggaaattg aggaaatgaa caagtagatg ggttggtcag tgctggaatc 1860
aggaaagtac ta 1872

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SEQ ID NO: 22      moltype = DNA length = 867
FEATURE
misc_feature      1..867
                  note = Helper Rev; HIV Integrase; Integration of viral RNA
source            1..867
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 22
tttttagatg gaatagataa ggcccaagaa gaacatgaga aatatcacag taattggaga 60
gcaatgggcta gtgattttta cctaccacct gtagtagcaa aagaatagt agccagctgt 120
gataaatgtc agctaaaagg ggaagccatg catggacaag tagactgtag cccagggaata 180
tggcagctag attgtacaca tttagaagga aaagtattct tggtagcagt tcatgtagcc 240
agtggatata tagaagcaga agtaattcca gcagagacag ggcaagaaac agcatacttc 300
ctcttaaaat tagcaggaaag atggccagta aaaaacagta atacagacaa tggcagcaat 360
ttcaccagta ctacagttta ggccgcctgt ttgtgggcgg ggatcaagca ggaatttggc 420
attccctaca atccccaag tcaaggagta atagaatcta tgaataaaga attaaagaaa 480
attataggac aggtaaagaga tcaggctgaa catcttaaga cagcagtaca aatggcagta 540
ttcatccaca attttaaaag aaaagggggg attggggggg acagtgcagg ggaagaata 600
gtagacataa tagcaacaga catacaaact aaagaattac aaaaacaaat tacaaaaatt 660
caaaatttct ggggtttatta cagggacagc agagatccag tttggaaaagg accagcaaaag 720
ctcctctgga aaggtgaagg ggcagtagta atacaagata atagtgcacat aaaagtagtg 780
ccaagaagaa aagcaagat catcagggat tatggaaaac agatggcagg tgatgattgt 840
gtggcaagta gacaggatga ggattaa 867

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SEQ ID NO: 23      moltype = DNA length = 234
FEATURE
misc_feature      1..234
                  note = Helper/Rev; HIV RRE; Binds Rev element
source            1..234
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 23
aggagctttg ttccttgggt tcttgggagc agcaggaagc actatgggag cagcgtcaat 60
gacgctgacg gtacaggcca gacaattatt gtctgggata gtgcagcagc agaacaattt 120
gctgagggct attgaggcgc aacagcatct gttgcaactc acagtctggg gcatcaagca 180
gctccaggca agaactctgg ctgtggaaaag atacctaaag gatcaacagc tcct 234

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SEQ ID NO: 24      moltype = DNA length = 351
FEATURE
misc_feature      1..351
                  note = Helper/Rev; HIV Rev; Nuclear export and stabilize
                  viral mRNA
source            1..351
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 24
atggcaggaa gaagcggaga cagcgacgaa gaactcctca aggcagtcag actcatcaag 60
tttctctatc aaagcaaccc acctcccaat cccgagggga cccgacaggc ccgaaggaaat 120
agaagaagaa ggtggagaga gagacagaga cagatccatt cgattagtga acggatcctt 180
agcacttata tgggacgacg tcgggagcct gtgcctcttc agctaccacc gcttgagaga 240
cttactcttg attgtaacga ggattgtgga acttctggga cgcagggggg gggaaagccct 300
caaatattgg tggaatctcc tacaatattg gagtcaggag ctaaagaata g 351

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SEQ ID NO: 25      moltype = DNA length = 577
FEATURE           Location/Qualifiers
misc_feature       1..577
                   note = Envelope; CMV promoter; Transcription
source            1..577
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 25
acattgatta ttgactagtt attaatagta atcaattacg gggtcattag ttcataagccc 60
atatatggag ttccgcggtta cataacttac ggtaaatggc ccgcctggct gaccgcccac 120
cgacccccgc ccattgacgt caataatgac gtatgttccc atagtaacgc caatagggac 180
tttccattga cgtcaatggg tggagtattt acggtaaacg gcccaacttg cagtacatca 240
agtgtatcat atgccaagta cgcgccctat tgacgtcaat gacggtaaat ggcccgcctg 300
gcattatgcc cagtacatga ccttatggga ctttcctact tggcagtaca tctacgtatt 360
agtcacgctg attaccatgg tgatgcggtt ttggcagtag atcaatgggc gtggatagcg 420
gtttgactca cgggggatttc caagtctcca ccccatggac gtcaatggga gtttgttttg 480
gcaccaaaat caacgggact ttccaaaatg tcgtaaacac tccgcccacat tgacgcaaat 540
gggcggtagg cgtgtacggt gggaggtcta tataagc 577

SEQ ID NO: 26      moltype = DNA length = 1519
FEATURE           Location/Qualifiers
misc_feature       1..1519
                   note = Envelope; VSV-G; Glycoprotein envelope-cell entry
source            1..1519
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 26
atgaagtgcc ttttgcactt agccttttta ttcattgggg tgaattgcaa gttcaccata 60
gtttttccac acaacccaaa aggaaactgg aaaaatgttc cttctaatta ccattattgc 120
cgttcaagct cagattttaa ttggcataat gacttaatat gcacagcctt acaagtcaaa 180
atgcccaaga gtcacaaggc tattcaagca gacggttgga tgtgtcatgc ttccaaatgg 240
gtcaactact gtgatttccg ctggatgga cgaagtata taacacattc catccgatcc 300
tcaactccat ctgtagaaca atgcaaggaa agcattgaac aaacgaaaca aggaacttgg 360
ctgaatccag gcttccctcc tcaaaagtgt ggatagcaa ctgtgacgga tgcgaagca 420
gtgattgtcc aggtgactcc tcaccatgtg ctgggtgatg aatacacagg agaatgggtt 480
gattcacagt tcatcaacgg aaaatgcagc aattacatat gcccactgt ccataactct 540
acaaactggc attctgacta taagggtcaaa gggctatgtg attctaacct catttccatg 600
gacatcacct tcttctcaga ggacggagag ctatcatccc tgggaaagga gggcacaggg 660
ttcagaagta actactttgc ttatgaaact ggaggcaagg cctgcaaaaat gcaatactgc 720
aagcattggg gagtcagact cccatcaggt gtctgtgtcg agatggctga taaggatctc 780
tttgcgtcag ccagattccc tgaatgccca gaagggtcaa gtatctctgc tccatctcag 840
acctcagtg atgtaagtct aattcaggac gttgagagga tcttgagta ttccctctgc 900
caagaaacct ggagcctaaa cagagcgggt cttccaatct ctccagtga tctcagctat 960
cttgcctcta aaaacccagg aaccggctct gctttcacca taatcaatgg taccctaaaa 1020
tactttgaga ccagatacat cagagtcgat attgtgtctc caatcctctc aagaatgggtc 1080
ggaatgatca gtggaactca cacagaaagg gaactgtggg atgactgggc accatatgaa 1140
gacgtggaaa ttggacccaa tggagttctg aggaccagtt caggatataa gtttctttaa 1200
tacatgattg gacatggtat gttggactcc gatcttctc ttagctcaaa ggctcagggtg 1260
ttcgaacatc ctcacattca agacgtgctg tcgcaacttc ctgatgatga gagtttattt 1320
tttggtgata ctgggctatc caaaaatcca atcgagcttg tagaagggtg gttcagtagt 1380
tggaaaagct ctattgcctc tttttcttt atcatagggg taatcattgg actattcttg 1440
gttctccgag ttggtatcca tctttgcatt aaattaaagc acaccaagaa aagacagatt 1500
tatacagaca tagagatga 1519

SEQ ID NO: 27      moltype = DNA length = 352
FEATURE           Location/Qualifiers
misc_feature       1..352
                   note = Helper/Rev; CMV early (CAG) enhancer; Enhance
                   Transcription
source            1..352
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 27
tagttattaa tagtaatcaa ttacgggggc attagtcat agcccatata tggagttccg 60
cgttacataa cttacggtaa atggcccgcc tggctgaccg cccaacgacc ccgcccatt 120
gacgtcaata atgacgtatg ttcccatagt aacgccaata gggactttcc attgacgtca 180
atgggtggac tatttacggg aaactgccca ctgggcagta catcaagtgt atcatatgcc 240
aagtacgccc cctattgacg tcaatgacgg taaatggccc gcctggcatt atgccagta 300
catgacctta tgggactttc ctacttgcca gtacatctac gtattagtca tc 352

SEQ ID NO: 28      moltype = DNA length = 960
FEATURE           Location/Qualifiers
misc_feature       1..960
                   note = Helper/Rev; Chicken beta actin intron; Enhance gene
                   expression
source            1..960
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 28

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ggagtcgctg cgttgccctt gcccgctgcc ccgctccgcg ccgctccgcg ccgccccccc 60
cggctctgac tgaccgcggt actccacacg gtgagcgggc gggacggccc ttctctcccg 120
ggctgtaatt agcgtttggt ttaatgacgg ctgctttctt ttctgtggct gcgtgaaagc 180
cttaaagggc tccgggaggg ccctttgtgc gggggggagc ggctcggggg gtgcgtgcgt 240
gtgtgtgtgc gtggggagcg ccgctgctgc ccgctgctgc ccgctgctgc tgagcgtgc 300
gggcgcggcg cggggctttg tgcgtccgcg gtgtgcccga ggggagcgcg gccggggggc 360
gtgccccgcg gtgcgggggg gctgcgaggg gaacaaaggc tgcgtgcggg gtgtgtgcgt 420
ggggggggtg gcaggggggt tgggcgcggc ggtcgggctg taaccccccc ctgcaccccc 480
ctccccgagt tgetgagcac ggcgggctt cgggtgcggg gctccgtgcg gggcgtggcg 540
cggggctcgc cgtgccgggc ggggggtggc ggcaggtggg ggtgccgggc ggggcggggc 600
cgctccgggc cggggagggg tcgggggagg ggcgcggcgg ccccgagcgc ccggcggctg 660
tcgagggcgcg gcgagcgcca gccattgcct tttatggtaa tcgtgcgaga gggcgaggg 720
acttcctttg tcccaaatct ggcggagcgc aaatctggga ggcgcgcgcg caccctctct 780
agcggggcgcg gcgaaagcgc tcggcgcgcg gcaggaagga aatggcgggg gagggccttc 840
gtgcgtcgcc gcgcgcgctg ccccttctcc atctccagcc tcggggctgc cgcaggggga 900
cggctgcctt cggggggggc ggggcagggc ggggttcggc ttctggcggtg tgaccggcgg 960

SEQ ID NO: 29      moltype = DNA length = 448
FEATURE           Location/Qualifiers
misc_feature       1..448
                   note = Helper/Rev; Rabbit beta globin poly A; RNA stability
source            1..448
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 29
agatcttttt ccctctgcc aaaaattatg ggacatcatg aagccccctg agcatctgac 60
ttctggctaa taaaggaaat ttattttcat tgcaatagtg tggtggaatt tttgtgtct 120
ctcaactcga aggacatag ggagggcaaa tcatttaaaa catcagaatg agtatttgg 180
ttagagtttg gcaacatag ccacatgctg gctgccatga acaaaggttg ctataaagag 240
gtcatcagta tatgaaacag cccctgctg tccattcctt attccataga aaagccttga 300
cttgaggtta gatctttttt atattttgtt ttgtgttatt tttttcttta acatccctaa 360
aatcttctct acatgtttta ctagccagat ttttctctct ctctgacta ctcccagtc 420
tagctgtccc tctctcttta tgaagatc 448

SEQ ID NO: 30      moltype = DNA length = 573
FEATURE           Location/Qualifiers
misc_feature       1..573
                   note = Envelope; Beta globin intron; Enhance gene expression
source            1..573
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 30
gtgagtttgg ggacccttga ttgttctttc ttttctgcta ttgtaaaatt catgttatat 60
ggagggggca aagttttcag ggtgtgtgtt agaattggga gatgtccctt gtatcaccat 120
ggaccctcat gataattttg ttctttcac ttctactctt gttgacaacc attgtctcct 180
cttattttct tttcattttc tgtaaacttt tcgttaaaact ttagottgca tttgtaacga 240
atttttaaat tcacttttgt ttatttgtca gattgtaagt actttctcta atcaactttt 300
ttcaaggca atcagggtat attatattgt acttcagcac agtttttagag aacaatttgt 360
ataattaaat gataaggtag aatatttctg catataaatt ctggctggcg tggaaatatt 420
cttatttgta gaacaacta caccctggtc atcactctgc ctttctcttt atgggttaca 480
tgatatacac tgtttgagat gaggataaaa tactctgagt ccaaaccggg cccctctgct 540
aacatgttc atgccttctt ctctttctca cag 573

SEQ ID NO: 31      moltype = DNA length = 450
FEATURE           Location/Qualifiers
misc_feature       1..450
                   note = Envelope; Rabbit beta globin poly A; RNA stability
source            1..450
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 31
agatcttttt ccctctgcc aaaaattatg ggacatcatg aagccccctg agcatctgac 60
ttctggctaa taaaggaaat ttattttcat tgcaatagtg tggtggaatt tttgtgtct 120
ctcaactcga aggacatag ggagggcaaa tcatttaaaa catcagaatg agtatttgg 180
ttagagtttg gcaacatag ccacatgctg ggctgccatg aacaaaggtt ggctataaag 240
aggctcatcg tatatgaac agccccctgc tgtccattcc ttattccata gaaaagcctt 300
gacttgaggt tagatttttt ttatattttg ttttgtgtta tttttttctt taacatccct 360
aaaaatttcc ttacatgttt tactagccag atttttctct ctctctgac tactccagtc 420
catagctgtc cctcttctct tatggagatc 450

SEQ ID NO: 32      moltype = DNA length = 31
FEATURE           Location/Qualifiers
misc_feature       1..31
                   note = Primer
source            1..31
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 32
taagcagaat tcatagaatt gccaggaaga t 31

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SEQ ID NO: 33          moltype = DNA length = 36
FEATURE               Location/Qualifiers
misc_feature          1..36
                      note = Primer
source                1..36
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 33
ccatacaatg aatggacact aggcggccgc acgaat          36

SEQ ID NO: 34          moltype = DNA length = 2745
FEATURE               Location/Qualifiers
misc_feature          1..2745
                      note = Gag, Pol, Integrase fragment
source                1..2745
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 34
gaattcatga atttgccagg aagatggaaa ccaaaaatga tagggggaat tggaggtttt 60
atcaaagtaa gacagtatga tcagatactc atagaaatct gcggacataa agctataggt 120
acagtattag taggacctac acctgtcaac ataattggaa gaaatctgtt gactcagatt 180
ggctgcactt taaatttttc cattagtcct attgagactg taccagtaaa attaaagcca 240
ggaatggatg gcccaaaagt taacaatgg ccattgacag aagaaaaaat aaaagcatta 300
gtagaaattt gtacagaatg ggaaggaa ggaaaattt caaaaattgg gcctgaaaat 360
ccatacaata ctccagtatt tgcataaag aaaaaagaca gtactaaatg gagaaaatta 420
gtagatttca gagaacttaa taagagaact caagatttct gggaagtcca attaggaata 480
ccacatcccg cagggtttaa acagaaaaaa tcagtaacag tactggatgt gggcgatgca 540
tatttttcag ttcccttaga taagacttc aggaagtata ctgcatttac catacctagt 600
ataaacaatg agacaccagg gattagatat cagtacaatg tgcttcacac gggatggaaa 660
ggatcaccag caatattcca gtgtagcatg acaaaaatct tagagccttt tagaaaacaa 720
aatccagaca tagtcatcta tcaatacatg gatgatttgt atgtaggatc tgacttagaa 780
atagggcagc atagaacaaa aatagaggaa ctgagacaac atctgttgag gtggggattt 840
accacaccag acaaaaaaca tcagaaagaa cctccattcc tttggatggg ttatgaactc 900
catcctgata aatggacagt acagcctata gtgctgccag aaaaggacag ctggactgtc 960
aatgacatag agaaattagt gggaaaattg aattgggcaa gtcagattta tcaggaggatt 1020
aaagtaaggc aattatgtaa acttcttagg ggaaccaaag cactaacaga agtagacca 1080
ctaacagaag aagcagagct agaactggca gaaaacaggg agatttctaa agaaccggta 1140
catggagtgt attatgaccc atcaaaagac ttaatagcag aaatacagaa gcaggggcaa 1200
ggccaatgga cttatcaaat ttatcaagag ccatttaaaa atctgaaaac aggaagtat 1260
gcaagaatga aggggtgccc cactaatgat gtgaaacaat taacagaggc agtacaaaaa 1320
atagccacag aaagcatagt aatatgggga aagactccta aatttfaat acccatacaa 1380
aaggaaacat ggggaagcatg gtggacagag tattggcaag ccacctggat tcctgagtgg 1440
gagtttgtca atacccttc cttagtgaag ttatgtgacc agttagagaa agaaccata 1500
ataggagcag aaactttcta tgtagatggg gcagccaata gggaaactaa attagaaaaa 1560
gcaggatagc taactgacag aggaagacaa aaagtgtgct ccctaacgga cacaacaaat 1620
cagaagactg agttacaagc aattcatcta gctttgcagg attcgggatt agaagtaaac 1680
atagtgcagc actcacaata tgcatgggga atcattcaag cacaaccaga taagagtga 1740
tcagagttag tcagtcaaat aatagagcag ttaataaaaa aggaaaaagt ctacctggca 1800
tgggtaccag cacacaaagg aattggagga aatgaacaag tagataaatt ggtcagtgtc 1860
ggaatcagga aagtactatt tttagatgga atagataagg cccaagaaga acatgagaaa 1920
tatcacagta attggagcag aatggctagt gattttaacc taccacctgt agtagcaaaa 1980
gaaatagtag ccagctgtga taaatgtcag ctaaaagggg aagccatgca tggacaagta 2040
gactgtagcc caggaatatg gcagctagat tgtacacatt tagaaggaaa agttatcttg 2100
gtagcagttc atgtagccag ttgatatata gaagcagaag taattccagc agagacaggg 2160
caagaaacag catacttctc tttaaaatta gcaggaagat ggccagtaaa aacagtacat 2220
acagacaatg gcagcaattt caccagtact acagttaagg ccgcctgttg gtggcgggg 2280
atcaagcagg aatttggcat tcctacaat ccccaaagtc aaggagtaat agaattctatg 2340
aataaagaat taaagaaaat tataggacag gtaagagatc aggtgaaca tcttaagaca 2400
gcagtacaaa tggcagtatt catccacaat tttaaaagaa aaggggggat tgggggggtac 2460
agtgccaggg aaagaatagt agacataata gcaacagaca tacaactaa agaattacaa 2520
aaacaaatta caaaaattca aaattttcgg gtttattaca gggacagcag agatccagtt 2580
tggaaaggac cagcaagact cctctggaaa ggtgaagggg cagtagtaat acaagataat 2640
agtgacataa aagtagtgcc aagaagaaaa gcaaagatca tcagggatta tggaaaacag 2700
atggcagggtg atgattgtgt ggcaagtaga caggatgagg attaa          2745

SEQ ID NO: 35          moltype = DNA length = 1586
FEATURE               Location/Qualifiers
misc_feature          1..1586
                      note = DNA Fragment containing Rev, RRE and rabbit beta
                      globin poly A
source                1..1586
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 35
tctagaatgg caggaagaag cggagacagc gacgaagagc tcatcagaac agtcagactc 60
atcaagcttc tctatcaaa caaccacct cccaatcccg aggggacccg acaggcccga 120
aggaatagaa gaagaagggt gagagagaga cagagacaga tccattcgat tagtgaacgg 180
atccttgcca cttatctggg acgatctgag gagcctgtgc ctcttcagct accaccgctt 240

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gagagactta ctcttgattg taacgaggat tgtggaactt ctgggacgca ggggggtggga 300
agccctcaaa tattgggtgga atctcctaca atattggagt caggagctaa agaatagagg 360
agctttgttc cttgggttct tgggagcagc aggaagcact atgggacgag cgtcaatgac 420
gctgacggta caggccagac aattattgtc tggatatgtg cagcagcaga acaatttgc 480
gagggctatt gaggcgcaac agcatctgtt gcaactcaca gtctggggca tcaagcagct 540
ccaggcaaga atcctggctg tggaaagata cctaaaggat caacagctcc tagatctttt 600
tccctctgcc aaaaattatg gggacatcat gaagcccctt gagcatctga cttctggcta 660
ataaaggaaa tttattttca ttgcaatagt gtgttggaat tttttgtgc tctcactcgg 720
aaggacatat gggagggcaa atcatttaaa acatcagaat gagtatttgg tttagagttt 780
ggcaacatat gccatatgct ggtgcccattg aacaaagggtg gctataaaga ggatcatcagt 840
atatgaaaca gcccctgtct gtccattcct tattccatag aaaagccttg acttgagggt 900
agattttttt tatattttgt tttgtgttat tttttctttt aacatcccta aaattttcct 960
tacatgtttt actagccaga tttttcctcc tctcctgact actccagtc atagctgtcc 1020
ctcttctctt atgaagatcc ctgcacctgc agcccaagct tggcgtaatc atggtcatag 1080
ctgtttcctg tgtgaaattg ttatccgctc acaattccac acaacatagc agccggaagc 1140
ataaagtgtg aagcctgggg tgccaatga gtgagctaac tcacattaat tgcgttgccg 1200
tcactgcccg ctttccagtc gggaaacctg tctgtccagc ggatccgcat ctcaattagt 1260
cagcaacatt agtcccgccc ctaactccgc ccatcccgcc cctaactccg cccagttccg 1320
cccatttccc gcccatggc tgactaattt tttttattta tgcagaggcc gaggcgcgct 1380
cggcctctga gctattccag aagtagtgag gaggtttttt tggaggccta ggcttttgca 1440
aaaagctaac ttgtttattg cagcttataa tgggtacaaa taaagcaata gcatcacaaa 1500
tttcacaaat aaagcatttt ttccactgca ttctagttgt gggttgccca aactcatcaa 1560
tgtatcttat cagcgccgcg cccggg

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SEQ ID NO: 36      moltype = DNA length = 1614
FEATURE            Location/Qualifiers
misc_feature       1..1614
                    note = DNA fragment containing the CAG
                    enhancer/promoter/intron sequence
source             1..1614
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 36
acgcgttagt tattaatagt aatcaattac ggggtcatta gttcatagcc catatatgga 60
gttcgcgctt acataactta cggtaaatgg cccgcctggc tgaccgcccac acgacccccg 120
ccatttgacg tcaataatga cgtatgttcc catagtaaac ccaataggga ctttccattg 180
acgtcaatgg gtggactatt tacggtaaac tgcccacttg gcagtacatc aagtgtatca 240
tatgccaaat acgcccccta ttgacgtcaa tgacggtaaa tggccccgct ggcatattgc 300
cagttacatg accttatggg actttcctac ttggcagtag atctacgtat tagtcatcgc 360
tattaccatg ggtcgagggt agccccacgt tctgcttcac tctcccatc tccccccct 420
ccccaccccc aattttgtat ttatttattt tttaattatt ttgtgcagcg atggggggcg 480
gggggggggg ggcgcgcgcc aggcggggcg gggcgggggc agggggcggg cggggagagg 540
cggagagggt ggcgggcagc caatcagagc ggcgcgctcc gaaagtctc tttatggcg 600
agggggcggc ggcgggcggc ctataaaaaa cgaagcgccg ggcggggcgg agtcgctgcg 660
tgccttcgc cccgtgcgcc gctccgcgcc gcctcgccgc gcccgcccg gctctgactg 720
accgcgttac tcccacaggt gacggggcg gacggccctt ctctccggg ctgtaattag 780
cgcttggttt aatgacggct cgtttctttt ctgtggctgc gtgaaagcct taaagggctc 840
cgggagggcc ctttgtgcgg gggggagcgg ctcggggggt gcgtgcgtgt gtgtgtgcgt 900
ggggagcgcc gcgtgcggcc cgcgctgccc ggcggctgtg agcgtgcgg gcgcggcgcg 960
gggctttgtg gcctcccgct gtgcgcgagg ggagcgcgcc cggggggcgt gccccgcggt 1020
gcgggggggg tgccagggga acaaaaggct cgtgcgggggt gtgtgcgtgt ggggggtgagc 1080
aggggggtgt ggcgcggcgg tcgggctgta accccccct gcacccccct ccccgagttg 1140
ctgagcacgg cccggcttcg ggtgcggggc tccgtgcggg gcgtggcgcg gggctcgccg 1200
tgccggggcg ggggtgcccg caggtggggg tgccggggcg ggcggggcgg cctcgggccg 1260
gggagggctc gggggagggg cgcggcgccc ccggagcgcc ggcggctgtc gaggcgcggc 1320
gagccgcgag catcgctttt tatggtaaac gtgcgagagg gcgcagggac ttctttgtc 1380
ccaaatctgg cggagccgaa atctgggagg cgcgcgcgca cccctctag cgggcgcggg 1440
cgaagcgggt cggcgccggc aggaaggaaa tgggcgggga gggccttcgt gcgtcgccgc 1500
gcgcgcgctc cctctccat ctccagcctc gggggtgcgg caggggggag gctgccttcg 1560
ggggggacgg ggcaggcgcg ggttcggctt ctggcggtgtg accggcggga attc 1614

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SEQ ID NO: 37      moltype = DNA length = 1531
FEATURE            Location/Qualifiers
misc_feature       1..1531
                    note = DNA fragment containing VSV-G
source             1..1531
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 37
gaattcatga agtgcccttt gtacttagcc tttttattca ttggggtgaa ttgcaagttc 60
accatagttt ttccacacaa ccaaaaagga aactggaaaa atgttctctt taattaccat 120
tattgcccggt caagctcaga tttaaattgg cataatgact taataggcac agccttacia 180
gtcaaaatgc ccaagagtca caaggctatt caagcagacg gttggatgtg tcatgcttcc 240
aaatgggtca ctacttgtga tttccgtgg tatggaccga agtatataac acattccatc 300
cgatccttca ctccatctgt agaacaatgc aaggaaaagca ttgaacaaac gaaacaagga 360
acttggtctg atccaggctt cctcctccta agttgttgat atgcaactgt gacggatgcc 420
gaagcagtgat ttgtccaggt gactcctcac catgtgctgt ttgatgaata cacaggagaa 480
tgggttgatt cacagttcat caacggaaaa tgcagcaatt acatatgcc cactgtccat 540
aactctacaa cctggcattc tgactataag gtcaaaagggc tatgtgattc taacctcatt 600

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tccatggaca tcaccttctt ctcagaggac ggagagctat catccctggg aaaggagggc 660
acagggttca gaagtaacta ctttgcttat gaaactggag gcaaggcctg caaaatgcaa 720
tactgcaagc attggggagt cagactccca tcaggtgtct ggctcgagat ggctgataag 780
gatctctttg ctgcagccag attccctgaa tgcccagaag ggtcaagtat ctctgctcca 840
tctcagacac cagtggtatg aagtctaatt caggacgttg agaggatctt ggattattcc 900
ctctgccaa gaaacctggag caaaatcaga gcgggtcttc caatctctcc agtggatctc 960
agctatcttg ctctaaaaa cccaggaacc ggtcctgctt tcaccataat caatggtacc 1020
ctaaaaacta ttgagaccag atacatcaga gtcgatattg ctgctccaat cctctcaaga 1080
atggctcgga tgatcagtg aactaccaca gaaagggaac tgtgggatga ctgggcacca 1140
tatgaagacg tggaaattgg acccaattga gttctgagga ccagttcagg atataagttt 1200
cctttataca tgattggaca tggtatgttg gactcogac ttcattctag ctcaaaggct 1260
caggtgtttg aacatctcca catccaagac gctgcttcgc aactctctga tgatgagagt 1320
ttattttttg gtgatactgg gctatccaaa aatccaatcg agctttaga aggttggttc 1380
agtagttgga aagatctctat tgctctttt ttctttatca tagggtaaat cattggacta 1440
ttcttggttc tccgagttg tatccatctt tgcattaaat taaagcacac caagaaaaga 1500
cagatttata cagacataga gatgagaatt c 1531

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SEQ ID NO: 38      moltype = DNA length = 884
FEATURE           Location/Qualifiers
misc_feature       1..884
                   note = RSV promoter and HIV Rev
source            1..884
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 38
caattgcat gtacgggcca gatatacgcg tatctgaggg gactagggtg tgtttaggcg 60
aaaagcgggg cttcggttgt acgcggttag gactccctc aggatatagt agtttcgctt 120
ttgcataggg aggggggaaat gtagtcttat gcaatacact ttagtctttg caacatggta 180
acgatgagtt agcaacatgc cttacaagga gagaaaaagc accgtgcatg ccgatgggtg 240
gaagtaaggt ggtacgatcg tgcttatta ggaaggcaac agacaggtct gacatggatt 300
ggacgaacca ctgaattccg cattgcagag ataattgtat ttaagtgcct agctcgatac 360
aataaaacgc attgacctt tcaccacatt ggtgtgcacc tccaagctcg agctcgttta 420
gtgaaccgct agatcgctcg gagacgcat ccaagctgtt ttgacctcca tagaagacac 480
cgggaccgat ccagcctccc ctgaaagcta gcgattaggg atctcctatg gcaggaagaa 540
gcggagacag cgacgaagaa ctctcaagg cagtcagact catcaagttt ctctatcaaa 600
gcaacccacc tcccaatccc gaggggaccc gacaggcccg aaggaaataga agaagaaggt 660
ggagagagag acagagacag atccattcga ttagtgaacg gatccttagc acttatctgg 720
gacgatctgc ggaagcctgtg cctcttcagc taccaccgct tgagagactt actcttgatt 780
gtaacgagga ttgtggaact tctgggacgc aggggggtgg aagccctcaa atattggtg 840
aatctcttac aatattggag tcaggagcta aagaatagtc taga 884

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SEQ ID NO: 39      moltype = DNA length = 1104
FEATURE           Location/Qualifiers
misc_feature       1..1104
                   note = Elongation Factor-1 alpha (EF1-alpha) promoter
source            1..1104
                   mol_type = other DNA
                   organism = synthetic construct

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```

SEQUENCE: 39
ccggtgccta gagaagggtg cgcggggtaa actgggaaag tgatgtcgtg tactggctcc 60
gcctttttcc cgagggtggg ggagaaccgt atataagtgc agtagtcgcc gtgaacgttc 120
tttttcgcaa cgggtttgccc gccagaacac aggtaaagtgc cgtgtgtggt tcccgcgggc 180
ctggcctctt tacgggttat ggccttgcg tgccctgaat tacttccacg cccctggctg 240
cagtagctga ttcttgatcc cgagcttcgg gttggaagtg ggtgggagag ttcgaggcct 300
tgcgcttaag gagccctccc gccctgtgct tgagttgagg cctggcctg gcgctggggc 360
cgccgcgtgc gaattctggtg gcaccttcgc gcctgtctcg ctgcttccga taagtctcta 420
gccatttaaa atttttgatg acctgctgcg acgctttttt tctggcaaga tagtcttgta 480
aatgcgggcc aagatctgca cactggtatt tcggtttttg gggccgcggg cggcgacggg 540
gcccgtgcgt cccagcgcac atgttcggcg agggcgggcc tcgagcgcg gccaccgaga 600
atcgagcggg ggtagtctca agctggcccg cctgctctgg tgccctggcct cgcgcgcggc 660
tgtatcgccc cgccctggcg ggcaaggctg gcccggtcgg caccagttgc gtgagcggaa 720
agatggccgc ttcccggccc tgcctcaggg agctcaaaat ggaggacgcg gcgctcgga 780
gagcgggcgg gtgagtcacc cacacaaagg aaaaggccct ttccgtcctc agccgtcgct 840
tcagtgtact ccacggagta ccgggcgcgg tccaggcacc tcgattagtt ctcgagcttt 900
tggagtacgt cgtcttttag ttggggggag ggggtttatg cgatggagtt tccccacact 960
gagtggtgtg agactgaagt tagggcagct tggcacttga tgtaattctc cttggaattt 1020
gccctttttg agtttgatc ttggttcatt ctcaagcctc agacagtggt tcaaagtttt 1080
tttcttccat ttcagggtgc gtga 1104

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SEQ ID NO: 40      moltype = DNA length = 511
FEATURE           Location/Qualifiers
misc_feature       1..511
                   note = Promoter; PGK
source            1..511
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 40
gggggtgggg ttgcgccttt tccaaggcag ccctgggttt gcgcagggac gcggtgctc 60
tgggcgtggt tccgggaaac gcagcggcgc gcacctggg tctcgacat tcttcacgtc 120

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cgttcgcagc gtcacccgga tcttcgcgcg tacccttggt gggcccccg cgaacgttcc 180
tgctccgccc ctaagtcggg aaggttcctt gcggttcgcg gcgtgccgga cgtgacaaac 240
ggaagccgca cgtctcacta gtaccctcgc agacggagac gcgccaggag caatggcagc 300
gcgcgcagcg cgatgggctg tggccaatag cggctgctca gcagggcgcg ccgagagcag 360
cggccgggaa gggcggtg gggaggcggt gtgtggggcg gtagtggtgg cctgtttcct 420
gcccgcgcgg tgttcgcgat tctgcaagcc tccggagcgc acgtcggcag tcggctccct 480
cgttgaccga atcacccgac tctctcccca g                               511

SEQ ID NO: 41          moltype = DNA length = 1162
FEATURE               Location/Qualifiers
misc_feature          1..1162
                        note = Promoter; UbC
source                1..1162
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 41
gcgcggggtt ttggcgctc cgcggggcgc cccctctctc acggcgagcg ctgccacgtc 60
agacgaaggg cgcaggagcg ttctgatcc ttccgcccgc acgctcagga cagcggcccg 120
ctgctcataa gactcggcct tagaacccca gtatcagcag aaggacattt taggacggga 180
cttgggtgac tctagggcac tggttttctt tccagagagc ggaacaggcg aggaaaagta 240
gtccctttct ggcgattctg cggagggatc tccgtggggc ggtgaacgcc gatgattata 300
taaggacgcg ccgggtgtgg cacagctagt tccgtcgcag ccgggatttg ggtcgcggtt 360
cttgttttgt gatcgtgtg atcgtcactt ggtgagttgc gggctgtgtg gctggccggg 420
gctttcgttg ccgcggggcc gctcgtggg acggaagcgt gtggagagac cgccaagggc 480
tgtagtctgg gtcgcgcagc aaggttgccc tgaactgggg gttgggggga gcgcacaaaa 540
tggcggtcgt tcccgagtct tgaatggaag acgcttgtaa ggcgggctgt gaggtcgttg 600
aaacaaggtg gggggcatgg tggcgggcaa gaacccaagg tcttgaggcc ttcgctaatt 660
cgggaaagct cttattcggg tgagatgggc tggggcacca tctggggacc ctgacgtgaa 720
gtttgtcact gactggagaa ctcggtttg tctgtctggt gcggggcgcg cagttatgcg 780
gtgcccgttg gcagtgacac cgtacctttg ggagcgcgcg cctcgtcgtg tctgtacgtc 840
accggttctg ttgcttata atgcagggtg gggccacctg ccggtaggtg tgcggtaggc 900
ttttctccgt cgcaggacgc aggggtccgg cctagggtag gctctcctga atcgacaggc 960
gccggacctc tggtgagggg agggataagt gaggcgctcag tttctttggt cggttttatg 1020
tacctatctt cttaaagtagc tgaagctccg gttttgaact atgcgctcgg ggttggcgag 1080
tgtgttttgt gaagtttttt aggcaccttt tgaaatgtaa tcatttgggt caatatgtaa 1140
ttttcagtgt tagactagta aa                               1162

SEQ ID NO: 42          moltype = DNA length = 120
FEATURE               Location/Qualifiers
misc_feature          1..120
                        note = Poly A; SV40
source                1..120
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 42
gtttattgca gcttataatg gttacaaata aagcaatagc atcacaaatt tcacaaataa 60
agcatttttt tcaactgcatt ctagtgtgtg tttgtccaaa ctcatcaatg tatcttatca 120

SEQ ID NO: 43          moltype = DNA length = 227
FEATURE               Location/Qualifiers
misc_feature          1..227
                        note = Poly A; bGH
source                1..227
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 43
gactgtgcct tctagttgcc agccatctgt tgtttgcccc tccccgtgc cttccttgac 60
cctggaaggt gccactccca ctgtcctttc ctaataaaat gaggaatttg catcgacattg 120
tctgagtagg gtgcattcta ttctgggggg tggggtgggg caggacagca agggggaggga 180
ttgggaagac aatagcaggc atgctgggga tgcggtgggc tctatgg                               227

SEQ ID NO: 44          moltype = DNA length = 1695
FEATURE               Location/Qualifiers
misc_feature          1..1695
                        note = Envelope; RD114
source                1..1695
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 44
atgaaactcc caacaggaat ggtcatttta tgtagcctaa taatagttcg ggcagggttt 60
gacgaccccc gcaaggctat cgcattagta caaaaacaac atggtaaacc atcgcaatgc 120
agcggagggg aggtatccga gggcccaccg aactccatcc aacaggtaac ttgccagggc 180
aagacggcct acttaatgac caacaaaaaa tggaaatgca gagtcactcc aaaaaatctc 240
accctagcgg gggggagaact ccagaactgc cctgttaaca cttccaggga ctcgatgcac 300
agttcttgtt atactgaata ccggcaatgc agggcggaata ataagacata ctacagggcc 360
accttgctta aaatacggtc tgggagcctc aacgaggtag agatattaca aaaccccaat 420
cagctcctac agtccccttg taggggctct ataaatcagc ccgtttgctg gagtgccaca 480
gcccccatcc atatctccga tgggtggagg cccctcgata ctaagagagt gtggacagtc 540
caaaaaggcg tagaacaat tcataaggct atgcatcctg aacttcaata ccacccctta 600

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gccctgcccc aagtcagaga tgaccttagc cttgatgcac ggacttttga tatectgaat 660
accactttta ggttactcca gatgtccaat tttagocctg cccaagattg ttggctctgt 720
ttaaaactag gtacccctac cctctctgcg ataccacctc cctctttaac ctactcccta 780
gcagactccc tagcgaatgc ctctgtcag attatactc cctctcttgg tcaaccgatg 840
cagttctcca actcgtcctg tttatcttcc cctttcatta acgatacgga acaaatagac 900
ttaggtgcag tcacctttac taactgcacc tctgtagcca atgtcagtag tcctttatgt 960
gccctaaacg ggtcagtctt cctctgtgga aataacatgg catacaccta tttaccccaa 1020
aactggacag gacttttgcg ccaagcctcc ctcctccccg acattgacat catccccggg 1080
gatgagccag tccccattcc tgcattgat cattatata atagacctaa acgagctgta 1140
cagttcatcc ctttactagc tggactggga atcaccgcag cattcaccac cggagctaca 1200
ggcctaggtg tctcgtccac ccagtataca aaattatccc atcagttaat atctgatgtc 1260
caagtcttat ccggtacctt acaagattta caagaccagg tagactcgtt agctgaagta 1320
gttctccaaa ataggagggg actggaccta ctaacggcag aacaaggagg aatttgttta 1380
gccttacaag aaaaaatctg tttttatgct aacaagtcag gaattgtgag aaacaaaata 1440
agaaccctac aagagaattt acaaaaacgc agggaaagcc tggcatccaa cctctctctg 1500
accgggctgc agggctttct tccgtacctc ctacctctcc tgggacccct actcaccctc 1560
ctactcatat taaccattgg gccatgcgtt ttcaatcgat tggccaattt tgttaagac 1620
aggatctcag tggccaggc tctggttttg actcagcaat atcaccagct aaaccacata 1680
gagtcagagc catga                                     1695

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SEQ ID NO: 45      moltype = DNA length = 2013
FEATURE
misc_feature       1..2013
                    note = Envelope; GALV
source             1..2013
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 45
atgctttctca cctcaagccc gcaccacctt cggcaccaga tgagtctctg gagctggaaa 60
agactgatca tcctcttaag ctgcgtattc ggagacggca aaacgagttc gcagaataag 120
aacccccacc agcctgtgac cctcacctgg caggtaactg cccaaactgg ggacgttgtc 180
tgggacaaaa aagcagtcac gccctcttgg acttgggtgc cctctcttac acctgatgta 240
tgtgccctgg cggccggtct tgagtcctgg gatatcccg gatccgatgt atcgtcctct 300
aaaagagtta gacctctcta ttcagactat actgcccgtt ataagcaaat cacctgggga 360
gccataggtg gcagctaccc tgggctagg accaggatgg caaattcccc cttctacgtg 420
tgtccccgag ctggccgaac ccattcagaa gctaggaggt gtggggggct agaatcccta 480
tactgtaaa aatggagtgt tgagaccacg ggtaccgttt attggcaacc caagtctca 540
tgggacctca taactgtaaa atgggaccaa aatgtgaaat gggagcaaaa atttcaaaag 600
tgtgaacaaa ccggtctgtg taacccccctc aagatagact tcacagaaaa aggaaaaactc 660
tcagagattt ggataacgga aaaaacctgg gaattaaggt tctatgtata tggacacca 720
ggcatacagt tgactatccg cttagaggtc actaacatgc cggttgtggc agtggggcca 780
gacctctgct ttgcggaaca gggacctcct agcaagcccc tcaactctcc tctctcccca 840
cggaaaagcg cgcccacccc tctacccccg cgggctagtg agcaaacccc tgcggtgcat 900
ggagaaactg ttaccctaaa ctctccgctt cccaccagtg gcgaccgact ctttggcctt 960
gtgcaggggg ccttccctaac cttgaatgct accaaccagg gggccactaa gtcttgctgg 1020
ctctgtttgg gcatgagccc ccttatttat gaagggatag cctcttcagg agaggctcgt 1080
tatactcca accatacccg atgccactgg ggggccccag gaaagcttac cctcactgag 1140
gtctccggac tcgggtcatg catagggaag gtgcctctta cccatcaaca tctttgcaac 1200
cagaccttac ccataaatc ctctaaaaac catcagatc tgctccctcc aaaccatagc 1260
tggtgggctc gcagcactgg cctcaccccc tgctctcca cctcagtttt taatcagttc 1320
aaagacttct gttgccaggt ccagctgatc ccccgcatct attaccattt tgaagaaacc 1380
ttgttacaag ccttatgaca atcaccccc aggttttaaaa gagagcctgc ctcacttacc 1440
ctagctgtct tcctgggggtt agggattgag gcaggtatag gtactggctc aaccgacctc 1500
attaaagggc ccatagacct ccagcaaggc ctaaccagcc tccaaatcgc cattgacgct 1560
gacctccggg cccttcagga tccaatcagc aagctagagg actcactgac ttccctatct 1620
gaggtagtag tccaaaatag gagaggcctt gacttactat tccttaaaag aggaggcctc 1680
tgccggggcc taaaagaaga gtgctgtttt tatgtagacc actcaggtgc agtacgagac 1740
tccatgaaaa aacttaaaag aagactagat aaaagacagt tagagcgcca gaaaaaccaa 1800
aactggtagt aaggggtggt caataactcc ccttggttta ctaccctact atcaaccatc 1860
gtcggggccc tattgctcct ccttttggtt ctcactcttg ggccctgcat catcaataaa 1920
ttaatccaat tcatcaatga taggataagt gcagtcaaaa ttttagtcct tagacagaaa 1980
tatcagaccc tagataacga ggaaaacctt taa                                     2013

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SEQ ID NO: 46      moltype = DNA length = 1530
FEATURE
misc_feature       1..1530
                    note = Envelope; FUG
source             1..1530
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 46
atggttcctc aggttctttt gtttgtactc cttctgggtt tttcgttgtg tttcgggaag 60
ttccccattt acacgatacc agacgaactt ggtccctgga gccctattga catacaccat 120
ctcagctgtc caaataacct ggttgtggag gatgaaggat gtaccaacct gtccgagttc 180
tcctacatgg aactcaaaag tgggatacat tcagccatca aagtgaacgg gttcactttc 240
acaggtgttg tgacagagcg agagacctac accaactttg ttggttatgt cacaaccaca 300
ttcaagagaa agcattttcc ccccccacca gacgcagtga gagccgcgta taactggaag 360
atggccgggtg accccagata tgaagagtc ctacacaatc catacccga ctaccactgg 420
cttcgaactg taagaaccac caaagagtc ctcattatca tatccccag tgtgacagat 480

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ttggacccat atgacaaatc ccttcaactc agggctcttc ctggcggaat gtgctcagga 540
ataacgggtg cctctaccta ctgctcaact aacctgattt acaccatttg gatgcccgag 600
aatccgagac caaggacacc ttgtgacatt ttaccaata gcagagggaa gagagcatcc 660
aacgggaaca agacttgccg ctttgtggat gaaagaggcc tgtataagtc tctaaaagga 720
gcatgcaggc tcaagttatg tggagttctt ggacttagac ttatggatgg aacatgggtc 780
gcatgcaaaa catcagatga gaccaaattg tgccctccag atcagttggt gaatttgcac 840
gactttcgct cagaagagat cgagcatctc gttgtggagg agttagttaa gaaaagagag 900
gaatgtctgg atgcattaga gtccatcatg accaccaagt cagtaagttt cagacgtctc 960
agtcacctga gaaaacttgt cccagggttt ggaaaagcat ataccatatt caacaaaacc 1020
ttgatggagg ctgatgctca ctacaagtca gtccggacct ggaatgagat catccctca 1080
aaaggggtgt tgaaagttgg aggaagggtg catcctcatg tgaacggggt gtttttcaat 1140
ggtataatat tagggcctga cgaccatgtc ctaatccag agatgcaatc atccctctc 1200
cagcaacata tggagttggt ggaatcttca gttatcccc tgatgcacc cctggcagac 1260
ccttctacag ttttcaaga agtgatgag gctgaggatt ttgttgaag tcacctcccc 1320
gatgtgtaca aacagatctc aggggttgac ctgggtctcc cgaactgggg aaagtatgta 1380
ttgatgactg caggggccat gattggcctg gtgttgatat ttccctaat gacatgggtc 1440
agagttgta tccatctttg cattaaatta aagcacacca agaaaagaca gatttatata 1500
gacatagaga tgaaccgact tggaaagtaa 1530

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SEQ ID NO: 47      moltype = DNA length = 1497
FEATURE            Location/Qualifiers
misc_feature        1..1497
                    note = Envelope; LCMV
source              1..1497
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 47
atgggtcaga ttgtgacaat gtttgaggct ctgcctcaca tcatcgatga ggtgatcaac 60
attgtcatta ttgtgcttat cgtgatcacg ggtatcaagg ctgtctacaa ttttgccacc 120
tgtgggatat tcgcatgatg cagtttcccta cttctggcgtg gcagggtcctg tggcatgtac 180
ggctttaagg gaccgacatc ttacaaagga gtttaccaat ttaagtcagt ggagttgat 240
atgtcacatc tgaacctgac catgcccaac gcattgttcag ccaacaactc ccaccattac 300
atcagttatg ggactttctg actagaattg accttcacca atgattccat catcagtcac 360
aaacttttga atctgacctc tgccttcaac aaaaagacct ttgaccacac actcatgagt 420
atagtttcta gcctacacct cagtatcaga gggaaactcca actataaggc agtatcctgc 480
gacttcaaca atggcataac catccaatac aacttgacat tctcagatcg aaaaagtgtc 540
cagagccagt ttagaacctt cagaggtaga gtccatagata tgtttagaac tgccttcggg 600
gggaaatata tgaggagtgg ctggggctgg acaggctcag atggcaagac cacctgggtg 660
agccagacga gttaccaata cctgattata caaaatagaa cctgggaaaa ccaactgcaca 720
tatgcaggtc cttttgggat gttccaggatt ctcttttccc aagagaagac taagtctctc 780
actaggagac tagcgggcac attcacctgg acctttgtcag actcttcagg ggtggagaat 840
ccagtggttt attgcctgac caaatggatg attctgtctg cagagcttaa gtgttctggg 900
aacacagcag ttgcgaatgc caatgtaaat catgatgccg aattctgtga catgctgcga 960
ctaattgact acaacaaggc tgccttgagt aagttcaaa aggcagtaga atctgctctg 1020
cacttattca aaacaacagt gaattctttg atttcagatc aactactgat gaggaaccac 1080
ttgagagatc tgatgggggt gccatattgc aattactcaa agttttggtt cctagaacat 1140
gcaaaagacc gcgaaactag tgcctccaag tgctggcttg tcaccaatgg ttcttactta 1200
aatgagaccc acttcagtgta tcaaatcgaa caggaagccg ataactgat tacagagatg 1260
ttgaggaagg attacataaa gaggcagggg agtaccccc tagcattgat ggaccttctg 1320
atgttttcca catctgcata cttagtcagc atcttctctg accttgtaa aataccaaca 1380
cacaggcaca taaagggtgg ctcatgtcca aagccacacc gattaaccaa caaagggaat 1440
tgtagttgtg gtgcatttaa ggtgcctggt gtataaaaccg tctggaaaag acgtgta 1497

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SEQ ID NO: 48      moltype = DNA length = 1692
FEATURE            Location/Qualifiers
misc_feature        1..1692
                    note = Envelope; FPV
source              1..1692
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 48
atgaacactc aaatcctggt tttcgccctt gtggcagtc tccccacaaa tgcagacaaa 60
atttgtcttg gacatcatgc tgtatcaaat ggcaccaaag taaacacact cactgagaga 120
ggagtagaag ttgtcaatgc aacggaaaca gtggagcggg caaacatccc caaaatttgc 180
tcaaaagggg aaagaaccac tgatcttggc caatgcggac tgttagggac cattaccgga 240
ccacctcaat gcgaccaatt tctagaattt tcagctgac taataatcga gagacgagaa 300
ggaaatgatg tttgttacc ggggaagttt gttaatgaag aggcattgcg acaaatcctc 360
agaggatcag gtgggattga caaagaaaca atgggattca catatagtgg aataaggacc 420
aacggaacaa ctagtgcatt tagaagatca gggcttctat tcatgcaga aatggagtgg 480
ctcctgtcaa atacagacaa tgctgtcttc ccacaaatga caaaatcata caaaaacaca 540
aggagagaat cagctctgat agtctgggga atccaccatt caggatcaac caccgaacag 600
accaaactat atgggagtgg aaataaactg ataacagtcg ggagttccaa atatcatcaa 660
tcttttctgc cgagtccagg aacacgaccg cagataaatg gccagtcagg acggattgat 720
ttcatgtgtg tgactttgga tcccaatgat acagttactt ttagtttcaa tggggcttct 780
atagctccaa atcgtgcccag cttcttgagg ggaagtcca tggggatcca gagcgatgtg 840
caggttgatg ccaattgcga aggggaatgc taccacagtg gagggactat aacaagcaga 900
ttgccttttc aaaacatcaa tagcagagca gttggcaaat gcccaagata tgtaaaaacg 960
gaaagtttat tattggcaac tgggatgaag aacgttcccc aaccttcaa aaaaaggaaa 1020
aaaagaggcc tgtttggcgc tatagcaggg ttattgaaa atggttggga aggtctggtc 1080

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gacgggtgggt acggttttcag gcatcagaat gcacaaggag aaggaactgc agcagactac 1140
aaaagcacc accggaagt tgatcagata accggaagt taaatagact cattgagaaa 1200
accaaccagc aatttgagct aatagataat gaattcactg aggttgaaaa gcagattggc 1260
aatttaatta actggacca agactccatc acagaagtat ggtcttaca tgctgaactt 1320
cttgtggcaa tggaaaacca gcacactatt gatttggctg attcagagat gaacaagctg 1380
tatgagcgag tgaggaaaca attaagggaa aatgctgaag aggatggcac tggttgcttt 1440
gaaatttttc ataaatgtga cgatgattgt atggctagta taaggaacaa tacttatgat 1500
cacagcaaat acagagaaga agcgatgcaa aatagaatac aaattgaccc agtcaaatg 1560
agtagtggct acaaatgtgt gatactttgg tttagcttcg gggcatcatg ctttttgctt 1620
cttgccattg caatgggctt tgttttcata tgtgtgaaga acggaacat gcggtgcact 1680
atttgtatat aa 1692

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SEQ ID NO: 49      moltype = DNA length = 1266
FEATURE           Location/Qualifiers
misc_feature      1..1266
                  note = Envelope; RRV
source            1..1266
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 49
agtgtaacag agcacttta tgtgtataag gctactagac catacctagc acattgcgcc 60
gattgcgggg acgggtactt ctgctatagc ccagttgcta tcgaggagat ccgagatgag 120
gcgtctgatg gcgtgcttaa gctccaagtc tccgcccata taggtctgga caaggcaggc 180
acccacgccc acacgaagct ccgatatatg gctggtcatg atgttcagga atctaagaga 240
gattccctga ggggtgtcac gtcgcagcgt tgcctccatc atgggacgat gggacacttc 300
atcgtcgcac actgtccacc aggcgactac ctcaagggtt cggttcgagga cgcagattcg 360
cacgtgaagg catgtaaggt ccaatacaag cacaatccat tgccggtggg tagagagaag 420
ttcgtggtta gaccacactt tggcgtagag ctgcatgca cctcatacca gctgacaacg 480
gctccaccgc acgaggagat tgacatgcat acaccgccag atataccgga tcgcaccctg 540
ctatcacaga cggcgggcaa cgtcaaaaata acagcaggcg gcaggactat cagggtacaac 600
tgtacctgcg gccgtgacaa cgtaggcact accagtactg acaagaccat caacacatgc 660
aagattgacc aatgccatgc tgcgtcacc agccatgaca aatggcaat tacctctcca 720
tttgttccca gggctgatca gacagctagg aaaggcaagg tacacgttcc gttccctctg 780
actaacgtca cctgccaggt gccgttggct cgagcgcggg atgccaccta tggtaagaag 840
gaggtgaccc tgagattaca ccagatcat ccgacgctct tctcctatag gatttttaga 900
gccgaaccgc acccgtaga ggaatgggtt gacaagttct ctgagcgcat catcccagtg 960
acggaagaag ggattgagta ccagtggggc aacaaccgcg cgtctgcctt gtgggcgcaa 1020
ctgacgacgc agggcaaac ccattggctgg ccacatgaaa tcattcagta ctattatgga 1080
ctataccccc ccgcacatct tgcgcagta tccggggcga gtctgatggc cctcctaact 1140
ctggcggcca catgctgcat gctggccacc gcgaggagaa agtgccaac accgtacgcc 1200
ctgacgcccag gagcggtggt accgttgaca ctgggggctgc tttgctgcgc accgagggcg 1260
aatgca 1266

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SEQ ID NO: 50      moltype = DNA length = 1938
FEATURE           Location/Qualifiers
misc_feature      1..1938
                  note = Envelope; MLV 10A1
source            1..1938
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 50
atggaaggtc cagcgttttc aaaacccctt aaagataaga ttaaccctgt gaagtcctta 60
atggctcatg gggctctatt aagagtaggg atggcagaga gcccccatca ggtctttaat 120
gtaacctgga gagtcaccaa cctgatgact gggcgtagcc ccaatgccac ctccctttta 180
ggaactgtac aagatgcctt cccaagatta tattttgatc tatgtgatct ggtcggagaa 240
gagtgaggac cttcagacca ggaaccatat gtcgggtatg gctgcaata ccccgagggg 300
agaaagcgga ccgggacttt tgacttttac gtgtgccctg ggcataccgt aaaatcgggg 360
tgtggggggc caagagaggg ctactgtggt gaatgggggt gtgaaaccac cggacaggct 420
tactggaagc ccacatcatc atggggacct atctccctta agcgcggtaa caccctctgg 480
gacacgggat gctccaaaat ggcttgtggc ccctgctacg acctctccaa agtatccaat 540
tccttccaag gggctactcg agggggcaga tgcaaccctc tagtccatga attcactgat 600
gcaggaaaaa aggctaattg ggacggggcc aaatcgtggg gactgagact gtaccggaca 660
ggaacagata ctattaccat gttctccctg acccgccagg tccctcaatat agggccccgc 720
atccccattg ggcctaactc cgtgatcact ggtcaactac cccctcccg acccgtgcag 780
atcaggctcc accggcctcc tcagcctcct cctacaggcg cagcctctat agtccctgag 840
actgccccac cttctcaaca acctgggacg ggagacaggc tgctaaacct ggtagaagga 900
gcctatcagg cgcttaacct caccaatccc gacaagacc aagaatgttg gctgtgctta 960
gtgtcgggac ctcttatta cgaaggagta gcggtcgtgg gcacttata caatcattct 1020
accgccccgg ccagctgtac ggccacttcc caacataagc ttaccctatc tgaagtgaca 1080
ggacaggggc tatgcatggg agcactacct aaaactcacc aggccttatg taacaccacc 1140
caaatgtccc gctcagatc ctactacct gcagcaccgc ctggaacaat gtgggcttgt 1200
agcactggat tgactccctg cttgtccacc acgatgctca atctaaccac agactattgt 1260
gtattagtgt agctctggcc cagaataatt taccactccc ccgattatat gtatggtcag 1320
cttgaacagc gtaacaaata taagagggag ccagatcgt tgaccctggc cttctgcta 1380
ggaggattaa ccattgggag gattgcagct ggaataggga cggggaccac tgcctaatc 1440
aaaacccagc agtttgagca gcttcacgcc gctatccaga cagacctcaa cgaagtogaa 1500
aaatcaatta ccaacctaga aaagtcaact acctcgttgt ctgaagtagt cctacagaa 1560
cgaagagccc tagatttgct ctctctaaaa gagggaggtc tctgcgcagc cctaaaagaa 1620
gaatgttgtt tttatgcaga ccacacggga ctagtgcagc acagcatggc caaactaagg 1680

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gaaaggctta atcagagaca aaaactattt gagtcaggcc aaggttgggt cgaagggcag 1740
ttaaatagat cccctgggtt taccacotta atctccacca tcatgggacc tctaatagta 1800
ctcttactga tcttactctt tggaccctgc attctcaato gatttggtcca atttggttaa 1860
gacaggatct cagtgggtcca ggctctgggt ttgactcaac aatatcacca gctaaaaact 1920
atagagtacg agccatga                                     1938

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SEQ ID NO: 51      moltype = DNA length = 2030
FEATURE            Location/Qualifiers
misc_feature       1..2030
                    note = Envelope; Ebola
source             1..2030
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 51
atgggtgtta caggaatatt gcagttacct cgtgatcgat tcaagaggac atcattcttt 60
ctttgggtta ttatctcttt ccaagaaca ttttccatcc cacttggagt catccacaat 120
agcacattac aggttagtga tgtcgacaaa ctggtttgcc gtgacaaact gtcacccaca 180
aatcaattga gatcagttgg actgaatctc gaagggaatg gagtggcaac tgacgtgcca 240
tctgcaacta aaagatgggg cttcaggtcc ggtgtccac caaaggtggt caattatgaa 300
gctggtgaat gggctgaaaa ctgctacaat cttgaaatca aaaaacctga cgggagtggag 360
tgtctaccag cagcgccaga cgggattcgg ggcttcccc gggtgccgga tgtgcacaaa 420
gtatcaggaa cgggaccgtg tgcggagac tttgccttcc acaaaggagg tgctttcttc 480
ctgtatgacc gaactgtctc cacagttatc taccgaggaa cgactttcgc tgaaggtgtc 540
gttgcatctc tgatactgcc ccaagctaag aaggacttct tcagctcaca ccccttgaga 600
gagccggtca atgcaacgga ggaccctgct agtggctact attctaccac aattagatat 660
caagctaccg gttttggaac caatggagaa gagtatattg tcgaggttga caatttgacc 720
tacgtccaac ttgaatcaag attcacacca cagtttctgc tccagctgaa tgagacaata 780
tatacaagtg ggaaggagg caataccacg ggaactaa tttggaagg caaccocgaa 840
attgatacaa caatcgggga gtgggccttc tgggaaacta aaaaacctc actagaaaaa 900
tctgcagtg agagttgtct ttcacagctg tatcaaacag agccaaaaac atcagtggtc 960
agagtccggc gcgaacttct tccgaccag ggaccaacac acaactgaa gaccacaaa 1020
tcatggcttc agaaaatttc tctgcaatgg ttcaagtga cagtcaggaa agggagagtg 1080
cagtgctgcc tctgacaacc ctggccacaa tctccacgag tcctcaaccc cccacaacca 1140
aaccaggtcg ggaacaacgc accacaata caccgtgta taaacttgac atctctgagg 1200
caactcaagt tgaacaacat caccgcagaa cagacaacga cagcacagcc tccgacactc 1260
ccccgcgcc gaccgcagcc ggacccttaa aagcagagaa cacaacacg agcaagggtg 1320
ccgacctctc ggaccccgcc accacaacaa gtcccaaaaa ccacagcgag accgctggca 1380
acaacaacac tcatcaccaa gataccggag aagagagtg cagcagcggg aagctaggct 1440
taattacca tactattgct ggagtcgag gactgatcac aggcgggagg agagctcgaa 1500
gagaagcaat tgtaattgct caaccacaa gcaaccctaa tttacattac tggactactc 1560
aggtatgaag tgtgcaatc ggactggcct ggataccata tttcgggcca gcagccgagg 1620
gaatttacc agaggggctg atgcacaatc aagatgggtt aatctgtggg ttgagacagc 1680
tggtcaacga gacgactcaa gctcttcaac tgttctgag agccacaacc gagctacgca 1740
ccttttcaat cctcaaccgt aaggcaattg atttcttgc gcagcgatg ggccggcacat 1800
gccacattt gggaccggac tgtgtatcg aaccacatga ttggaccaag aacataacag 1860
acaaaattga tcaagattat catgattttg ttgataaaac ccttcgggac caggggggaca 1920
atgacaattg gtggacagga tggagacaat ggataccggc aggtatttga gttacaggcg 1980
ttataattgc agttatcgct ttattctgta tatgcaaat tgtcttttag 2030

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SEQ ID NO: 52      moltype = DNA length = 237
FEATURE            Location/Qualifiers
misc_feature       1..237
                    note = Polymerase III shRNA promoters; U6 promoter
source             1..237
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 52
tttccatga ttccttcata ttgtcatata cgatacaagg ctgttagaga gataattgga 60
attaattga ctgtaaacac aaagatatta gtacaaaata cgtgacgtag aaagtaataa 120
tttcttgggt agtttgagct tttaaaatta tgttttaaaa tggactatca tatgcttacc 180
gtaaactgaa agtatttcga tttcttggct ttatatatct tgtggaaagg acgaaac 237

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SEQ ID NO: 53      moltype = DNA length = 243
FEATURE            Location/Qualifiers
misc_feature       1..243
                    note = Polymerase III shRNA promoters; 7SK promoter
source             1..243
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 53
ctgcagttat tagcatgccc caccatctg caaggcattc tggatagtg caaaacagcc 60
ggaaatcaag tccgtttatc tcaaaactta gcattttggg aataaatgat atttgctatg 120
ctgggttaaa tagattttag ttaaaatttc tctggaagct ctagtacgat aagcaacttg 180
acctaagtgt aaagttgaga tttccttcag gtttatatag cttgtgcgcc gcctggctac 240
ctc                                     243

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SEQ ID NO: 54      moltype = DNA length = 21
FEATURE            Location/Qualifiers
misc_feature       1..21

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source	note = FDPS target sequence #1 1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 54		
gtcctggagt acaatgcat t		21
SEQ ID NO: 55	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = FDPS target sequence #2	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 55		
gcaggatttc gttcagcact t		21
SEQ ID NO: 56	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = FDPS target sequence #3	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 56		
gccatgtaca tggcaggaat t		21
SEQ ID NO: 57	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = FDPS target sequence #4	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 57		
gcagaaggag gctgagaaag t		21
SEQ ID NO: 58	moltype = DNA length = 49	
FEATURE	Location/Qualifiers	
misc_feature	1..49	
	note = Non-targeting sequence	
source	1..49 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 58		
gccgctttgt aggatagagc tgcgagctcta tcctacaaag cggcttttt		49
SEQ ID NO: 59	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Forward primer	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 59		
aggaattgat ggcgagaagg		20
SEQ ID NO: 60	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
misc_feature	1..22	
	note = Reverse primer	
source	1..22 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 60		
cccaaagagg tcaaggtaat ca		22
SEQ ID NO: 61	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
misc_feature	1..18	
	note = Forward primer	
source	1..18 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 61		
agcgcggtca cagcttca		18
SEQ ID NO: 62	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	

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misc_feature      1..20
                  note = Reverse primer
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 62
ggcgacgtag cacagcttct                                20

SEQ ID NO: 63      moltype = DNA length = 130
FEATURE            Location/Qualifiers
misc_feature        1..130
                  note = Left Inverted Terminal Repeat (Left ITR)
source              1..130
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 63
cctgcaggca gctgcgcgct cgctcgctca ctgaggccgc cggggcgctg ggcgacctt  60
ggtcgccccg cctcagtgag cgagcgagcg cgcagagagg gaggggccaa ctccatcact 120
aggggttcct                                           130

SEQ ID NO: 64      moltype = DNA length = 151
FEATURE            Location/Qualifiers
misc_feature        1..151
                  note = Right Inverted Terminal Repeat (Right ITR)
source              1..151
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 64
gagcgggcgc aggaaccctt agtgatggag ttggccactc cctctctgcg cgctcgctcg  60
ctcactgagg ccggggcgacc aaaggctcgcc cgacgcccgg gctttgcccc ggcggcctca 120
gtgagcgagc gagcgcgagc ctgcctgcag g                                           151

SEQ ID NO: 65      moltype = DNA length = 1227
FEATURE            Location/Qualifiers
misc_feature        1..1227
                  note = RRE/rabbit poly A beta globin
source              1..1227
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 65
tctagaagga gctttgttcc ttgggttctt gggagcagca ggaagcacta tggggcgagc  60
gtcaatgacg ctgacggtac aggccagaca attattgtct ggtatagtgc agcagcagaa 120
caatttgctg agggctattg aggcgcaaca gcatctgttg caactcacag tctggggcat 180
caagcagctc caggcaagaa tcctggctgt ggaaagatac ctaaaggatc aacagctcct 240
agatcttttt cctctgcca aaaattatgg ggacatcatg aagccccctg agcatctgac 300
ttctggctaa taaaggaat ttattttcat tgcaatagtg tgttggaaat ttttgtgtct 360
ctcactcgga aggacatag ggagggcaaa tcatttaaaa catcagaatg agtatttggg 420
ttagagtttg gcaacatag ccatatgctg gctgccatga acaaaggctg ctataaagag 480
gtcatcagta tatgaaacag cccctgctg tccattcctt attccataga aaagccttga 540
cttgaggtta gatttttttt atattttgtt ttgtgttatt tttttcttta acatccctaa 600
aattttcctt acatgtttta ctagccagat ttttctcct cctctgacta ctcccagtc 660
tagctgtccc tcttctctta tgaagatccc tcgacctgca gcccaagctt ggcgtaatca 720
tggatcatag tgtttctgtg gtgaaattgt tatccgctca caattccaca caacatacga 780
gccggaagca taaagtgtta agcctggggt gcctaataag tgagctaact cacattaatt 840
gcgttgcgct cactgccgcg ttccagtcg ggaaacctgt cgtgccagcg gatccgcatc 900
tcaattagtc agcaaccata gtcccgcccc taactccgcc catcccgccc ctaactccgc 960
ccagttccgc ccattctccg ccccatggct gactaatttt ttttatttat gcagagggcg 1020
agggcgctcg ggcctctgag ctattccaga agtagtgagg aggccttttt ggaggcctag 1080
gcttttgcaa aaagctaact tgtttattgc agcttataat gggtacaaat aaagcaatag 1140
catcacaaat ttcacaaata aagcattttt ttcactgcat tctagtgtg gtttgtccaa 1200
actcatcaat gtatcttata acccggg                                           1227

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What is claimed is:

1. A pharmaceutical combination, comprising:
 - (a) a bisphosphonate drug; and
 - (b) a lentiviral particle comprising an envelope protein and a therapeutic vector comprising:
 - an EF-1 alpha promoter and
 - at least one encoded shRNA or microRNA that, when expressed, inhibits production of farnesyl diphosphate synthase (FDPS)
 wherein the bisphosphonate drug and lentiviral particle are administered together or separately,
 wherein the at least one encoded shRNA or microRNA comprises a sequence having at least 80 percent identity with SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO:

3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.

2. The pharmaceutical combination of claim 1, wherein the at least one encoded shRNA or microRNA activates a GD T cell.

3. The pharmaceutical combination of claim 1, wherein the at least one encoded shRNA or microRNA activates a GD T cell.

4. The pharmaceutical combination of claim 1, wherein the envelope protein is capable of infecting a cancer cell.

5. The pharmaceutical combination of claim 1, wherein the bisphosphonate drug comprises zoledronic acid.

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6. A pharmaceutical combination, comprising:
 (a) a bisphosphonate drug; and
 (b) a lentiviral particle comprising an envelope protein and a therapeutic vector comprising:
 at least one encoded shRNA or microRNA that, when expressed, inhibits production of farnesyl diphosphate synthase (FDPS) and activates a GD T cell wherein the bisphosphonate drug and lentiviral particle are administered together or separately,
 wherein the at least one encoded shRNA or microRNA comprises a sequence having at least 80 percent identity with SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.
7. The pharmaceutical combination of claim 6, wherein the envelope protein is capable of infecting a cancer cell.
8. The pharmaceutical combination of claim 6, wherein the bisphosphonate drug comprises zoledronic acid.
9. A method of treating a cancer in a subject in need thereof, using an immunotherapy-based composition, the method comprising:
 (a) administering a therapeutically-effective amount of an bisphosphonate drug to the subject; and
 (b) administering a therapeutically-effective amount of the immunotherapy-based composition to the subject, wherein the immunotherapy-based composition comprises a lentiviral particle, the lentiviral particle comprising an envelope protein and a therapeutic vector comprising:
 an EF-1 alpha promoter; and,
 at least one encoded shRNA or microRNA that, when expressed, inhibits production of farnesyl diphosphate synthase (FDPS),
 wherein the at least one encoded shRNA or microRNA comprises a sequence having at least 80 percent identity with SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.

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10. The method of claim 9, wherein the at least one encoded shRNA or microRNA activates a GD T cell.
11. The method of claim 9, wherein the at least one encoded shRNA or microRNA activates a GD T cell.
12. The method of claim 9, wherein the cancer is selected from one or more of a carcinoma, a leukemia, a lymphoma, a sarcoma, a myeloma, a mesothelioma, a mixed type, or mixtures thereof.
13. The method of claim 9, wherein the bisphosphonate drug comprises zoledronic acid.
14. A method of treating a cancer in a subject in need thereof, using an immunotherapy-based composition, the method comprising:
 (a) administering a therapeutically-effective amount of an bisphosphonate drug to the subject; and
 (b) administering a therapeutically-effective amount of the immunotherapy-based composition to the subject, wherein the immunotherapy-based composition comprises a lentiviral particle, the lentiviral particle comprising an envelope protein and a therapeutic vector comprising:
 at least one encoded shRNA or microRNA that, when expressed, inhibits production of farnesyl diphosphate synthase (FDPS) and activates a GD T cell, wherein the at least one encoded shRNA or microRNA comprises a sequence having at least 80 percent identity with SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, or SEQ ID NO: 10.
15. The method of claim 14, wherein the cancer is selected from one or more of a carcinoma, a leukemia, a lymphoma, a sarcoma, a myeloma, a mesothelioma, a mixed type, or mixtures thereof.
16. The method of claim 14, wherein the bisphosphonate drug comprises zoledronic acid.

* * * * *